Differential Equation

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Chapter - 4 Differential Equations

4.1 Introduction

Definition: An equation involving derivations of one or more dependent variable with respect to one or more independent variable is called differential equation.

For example:

(i) $\frac{dx}{dt}+\frac{dy}{dt}=\sin t$

(ii) $y=\sqrt{x}\frac{dy}{dx}+\frac{k}{dy/dx}$

(iii) $\left(\frac{d^3y}{dx^3}\right)^{2/3}+\left(\frac{d^3y}{dx^3}\right)^{3/2}=0$

(iv) $\frac{\partial^2 y}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial x^3}=0$

- **Ordinary Differential Equation:** A differential equation involving derivatives with respect to a single independent variable is known as ordinary differential equation.
  In the above equation (i), (ii) & (iii) are ordinary differential equation.
- **Partial Differential Equation:** A differential equation involving partial derivation with respect to more than one independent variable is known as partial differential equation.
- **Order of a Differential Equation:** The order of the highest derivative involved in a differential equation is called the order of the differential equation.
- **Degree of a Differential Equation:** The degree of a differential equation is the exponent of the highest derivative which occurs in it after making the derivatives free for radicals and fractions.

Example: Find the order and degree of the following differential equations:

(i) $\frac{d^2y}{dx^2}=\frac{y^{1/2}}{x^{1/3}\ 1+x^{2/3}}$

(ii) $y=\sqrt{x}\frac{dy}{dx}+\frac{k}{dy/dx}$

(iii) $\left(\frac{d^3y}{dx^3}\right)^{2/3}+\left(\frac{d^3y}{dx^3}\right)^{3/2}=0$

(iv) $\sin\left(\frac{d^2y}{dx^2}\right)+x=y$

Solution:(i) $\frac{d^2y}{dx^2}=\frac{y^{1/2}}{x^{1/3}\ 1+x^{2/3}}$

Here, the order is 2 and degree is 1.

(ii) $y=\sqrt{x}\frac{dy}{dx}+\frac{k}{dy/dx}$

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Or, $y\frac{dy}{dx}=\sqrt{x}\left(\frac{dy}{dx}\right)^2+k$

Here, the order is 1 and degree is 2.

(iii) $\left(\frac{d^2y}{dx^3}\right)^{2/3}+\left(\frac{d^3y}{dx^3}\right)^{3/2}=0$

Or, $\left(\frac{d^3y}{dx^3}\right)^{2/3}=-\left(\frac{d^3y}{dx^3}\right)^{3/2}$

Making the exponent of derivative free from fractions, we get

$\left(\frac{d^3y}{dx^3}\right)^{4}=-\left(\frac{d^3y}{dx^3}\right)^{9}$

Here, the order is 3 and degree is 9.

(iv) $\sin\left(\frac{d^2y}{dx^2}\right)+x=y$

Or, $\frac{d^2y}{dx^2}-\frac{1}{3!}\left(\frac{d^2y}{dx^2}\right)^3+\frac{1}{5!}\left(\frac{d^2y}{dx^2}\right)^5-...+x=y$

So, the order is 2 and degree cannot be determined.

Note: The order of a differential equation always exists but the degree may or may not exist.

4.2 <u>Linear differential equation:</u>

A differential equation is called linear if,

(i)Every dependent variable and every derivatives involved should be in first degree only and

(ii) No product of dependent variables and/or derivative occurs.

A differential equation which is not linear is known as non-linear differential equation.

- **Formation of differential equation:** Let the family of curves having *n*-parameters is,

  $F\ (x, y, c_1, c_2, ..., c_n) = 0$ …(i)

  where $c_1, c_2, ..., c_n$ are *n* arbitrary constants.

  Now, differentiating (i), '*n*' times so as the get '*n*' additional equations containing the '*n*' arbitrary constants and derivatives. Then eliminating all '*n*' arbitrary constants from the $(n + 1)$ equations [obtained by successive differentiation and the given equation], we get an equation involving a derivative of *n*th order and this equation is the differential equation of the given family of curve (i).

  Thus the obtained differential equation of the family of curves (i) is of form

  $\phi\ (x, y, y_1, y_2, ..., y_n) = 0$ …(ii)

  where, $y_1, y_2, ..., y_n$ are the derivatives of order 1, 2, …, *n* respectively.

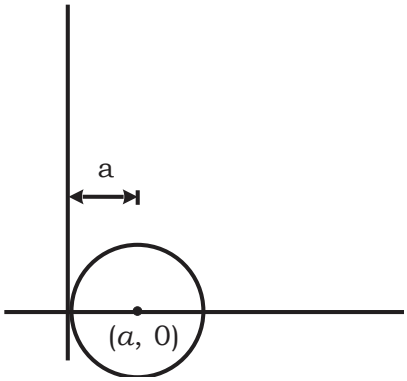
Note: The number of independent parameter in a family of curves is the order of the differential equation by the given family of curve.

Example: Find the differential equation representing the family of circles touching *y*-axis at the origin.

Solution: The general equation of family of circles, with centre (g, h) and radius *r* is

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$$x-g\ ^{2}+\ y-h\ ^{2}=r^{2} \qquad \ldots(i)$$



Since, it is given that the given family of circle touches the *y*-axis at origin so the given circles must have centre at *x*-axis. Let $a,0$ is the centre of the given circle, hence comparing with (i), we get,

$g = a$, $h = 0$ and $r = a$

Thus the equation of the given family of circles is

$$x-a\ ^{2}+y^{2}=a^{2}$$

Or, $x^{2}+y^{2}-2ax=0 \qquad \ldots(ii)$

Differentiating (ii), w.r.t *x*, we get

$$2x+2y\frac{dy}{dx}-2a=0$$

Or, $2a=2x+2y\frac{dy}{dx} \qquad \ldots(iii)$

Now, from (ii) and (iii), we get the required differential equation—

$$x^{2}+y^{2}-2x^{2}-2xy\frac{dy}{dx}=0$$

Or, $x^{2}+2xy\frac{dy}{dx}-y^{2}=0$

Or, $\frac{dy}{dx}=\frac{y^{2}-x^{2}}{2xy}$

Since, in the above example, equation (ii) has only one arbitrary constant (a), so we differentiate equation (ii) only one time.

Example: Find the third order differential equation whose solution is the 3-parameter family of curves defined by

$x^{2}+y^{2}+2ax+2by+c=0,$

where *a*, *b*, *c* are parameters.

Solution: Given, $x^{2}+y^{2}+2ax+2by+c=0 \qquad \ldots(i)$

Differentiating (i) three times in succession, we have

$x+yy^{1}+a+by^{1}=0, \qquad \ldots(ii)$

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$1+\left[y^1\right]^2+yy^2+by^2=0,$ …(iii)

and $3y^1y^2+yy^3+by^3=0.$ …(iv)

We now eliminate *a*, *b*, *c* from (i), (ii), (iii) and (iv). To do so, we simply eliminate *b* from (iii) and (iv).

Multiplying both sides of (iii) by y^3 and (iv) by y^2 and subtracting, we have

$y^3+y^3\left[y^1\right]^2+yy^2y^3-3y^1\left[y^2\right]^2-yy^2y^3=0$

Or, $\left[1+\left(y^1\right)^2\right]y^3-3y^1\left(y^2\right)^2=0.$

Example: Find the differential equation of the family of curves $y=e^x\left(A\cos x+B\sin x\right)$, where A and B are arbitrary constants.

Solution: Given that, $y=e^x\left(A\cos x+B\sin x\right)$ …(i)

Differentiating (i) with respect to *x*, we get

$y'=e^x\left(-A\sin x+B\cos x\right)+e^x\left(A\cos x+B\sin x\right)$

Or, $y'=e^x\left(-A\sin x+B\cos x\right)+y,$ using (i) …(ii)

Differentiating (ii) again with respect to *x*, we get

$y''=-e^x\left(A\cos x+B\sin x\right)+e^x\left(-A\sin x+B\cos x\right)+y',$ …(iii)

Now from (ii), $e^x\left(-A\sin x+B\sin x\right)=y'-y.$ …(iv)

$y''=-y+y'-y+y'$ or $y''-2y'+2y=0.$

Example: By eliminating the constants *a* and *b* obtain the differential equation for which $xy=ae^x+be^{-x}+x^2$ is a solution.

Solution: Given that $xy=ae^x+be^{-x}+x^2$ …(i)

Differentiating (i) with respect to *x*, we get

$xy'+y=ae^x-be^{-x}+2x$ …(ii)

Differentiating (ii) again with respect to *x*, we get

$xy''+y'+y'=ae^x+be^{-x}+2$

or $xy''+2y'=\left(xy-x^2\right)+2,$ using (i)

or $xy''+2y'-xy+x^2-2=0.$

Example: Find the differential equation of all circles of radius *a*.

Solution: The equation of all circles of radius *a* is

$(x-h)^2+(y-k)^2=a^2,$ …(i)

where *h* and *k*, the coordinates of the centre of the circle, are to to be taken as arbitrary constants.

Differentiating (i) with respect to *x*, we get

$(x-h)+(y-k)y'=0.$ …(ii)

Differentiating (ii) again with respect to *x*, we get

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$$1+\ y'^{\,2}+\ y-k\ \ y''=0\ \ \text{or}\ \ y-k=\frac{1+\ y'^{\,2}}{y''} \qquad \ldots(\text{iii})$$

Putting this value of $y - k$ in (ii), we get

$$x-h=-\ \ y-k\ \ y'=\frac{1+\ y'^{\,2}\ \ y'}{y''} \qquad \ldots(\text{iv})$$

Using (iii) and (iv), (i) gives

$$\frac{1+\ y'^{\,2\ \ 2}\ \ y'^{\,2}}{y''^{\,2}}=\frac{1+\ y'^{\,2\ \ 2}}{y''^{\,2}} \quad \text{or} \quad 1+\ y'^{\,2\ \ 3}=a^2\ \ y''^{\,2}.$$

- **Solution of differential equation:** Any relation between the dependent and independent variables, when substituted in the differential equation, reduces it to an identity is called a solution or integral of the differential equation.

  Let $\phi\ \ x, y, y_1, y_2, ..., y_n\ \ = 0$ …(i)

  be an *n*th order differential equation, then

  (a) A solution of (i) having '*n*' independent arbitrary constant is called a **general solution' or complete primitive of** eqn.(i)

  (b) A solution of (i) obtained from a general solution of (i) by taking particular values to one or more of the '*n*' independent arbitrary constant is called a **particular solution of** eqn. **(i).**

  (c) A solution of (i) which cannot be obtained from any general solution of (i) by any choice of the '*n*' independent arbitrary constants is called a **singular solution of** eqn. **(i).**

  e.g.: Consider the differential equation

  $$\left(\frac{dy}{dx}\right)^2-4y=0 \qquad \ldots(\text{i})$$

  Here, $y=\ \ x+c\ ^{\,2}$ is a general solution of (i).

  Taking, $c = 1$, $y=\ \ x+1\ ^{\,2}$ is a particular solution of (i)

  whereas, $y = 0$ is singular solution of (i) at it satisfies (i) and cannot be obtained by any choice of '*c*' of the general solution.

4.2.1 EQUATION OF FIRST ORDER AND FIRST DEGREE

The differential equation of the form

$$F\left(x, y, \frac{dy}{dx}\right)=0$$

Or, $\frac{dy}{dx}=+\ \ x, y$

Or, $M\ \ x, y\ \ dx+N\ \ x, y\ \ dy=0$

Represent differential equation of firs order and first degree.

4.3 Methods of solving differential equation of first order and first degree:

4.3.1 Separation of Variables:

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If a differential equation of the first order and first degree is of the form

$f_1(x)\,dx + f_2(y)\,dy = 0$ …(i)

Where $f_1(x)$ and $f_2(y)$ are either constant or function of only x and y respectively, the we say that the variables are separable in the given differential equation.

The differential equation (i) is being solved by integrating and adding as arbitrary constant of integration to one of the both sides, thus the solution is

$$\int f_1(x)\,dx + \int f_2(y)\,dy = c$$

Or, $\phi_1(x) + \phi_2(y) = c$, where '$c$' is an arbitrary constant.

Example: Solve $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$.

Solution: For separating variables, we re-write given equation as

$\frac{dy}{dx} = e^{-y}(e^x + x^2)$ or $e^y dy = (x^2 + e^x)$

Integrating, $e^y = \frac{1}{3}x^3 + e^x + c.$

Example: Solve $y - x\frac{dy}{dx} = a\left(y^2 + \frac{dy}{dx}\right)$.

Solution: The given equation can be re-written as

$$a + x\frac{dy}{dx} = y - ay \text{ or } \frac{dx}{x+a} = \frac{dy}{y(1-ay)}$$

or $\frac{dx}{x+a} = \left[\frac{a}{1-ay} + \frac{1}{y}\right]dy.$

Integrating, $\log(x+a) = -\log(1-ay) + \log y + \log c,$

or, $\log(x+a) = \log\left(\frac{cy}{1-ay}\right)$ or $x + a = \frac{cy}{1-ay}$

or, $(x+a)(1-ay) = cy$, which is the required solution.

Example: Solve $3e^x \tan y\,dx + (1-e^x)\sec^2 y\,dy = 0.$

Solution: Separating the variables, we get

$\frac{3e^x}{1-e^x}dx + \frac{\sec^2 y}{\tan y}dy = 0.$

Integrating, $-3\log(1-e^x) + \log(\tan y) = \log c.$

or $\log(\tan y) = \log(1-e^x)^{-3} + \log c,$ or $\tan y = c(1-e^x)^{-3}$.

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Example: If $\frac{dy}{dx} = e^{x+y}$ and it is given that for $x = 1,\ y = 1$; find y when $x = -1$.

Solution: Given equation is $e^{-y}dy = e^{x}dx$

Integrating it, $-e^{-y} = e^{x} + c.$ …(i)

Putting $x = 1$, $y = 1$ in $1 - e^{-1} = e + c$ so that $c = -e^{-1} - e.$

Hence (i) becomes $-e^{-y} = e^{x} - e^{-1} - e.$ …(ii)

Putting $x = -1$ in (ii), we obtain

$-e^{-y} = e^{-1} - e^{-1} - e$ so that $y = -1$.

4.3.2 Equation reducible to separation of variables:

The equation of the form

$$\frac{dy}{dx} = f\ \ ax + by + c \qquad \ldots(i)$$

Can be reduced to an equation in which variable can be separated putting $ax + by + c = v$, we get

$$\frac{dy}{dx} = \frac{1}{b}\left[\frac{dv}{dx} - a\right]$$

And, the equation (i) reduces to

$$\frac{1}{b}\left[\frac{dv}{dx} - a\right] = f\ \ v$$

Or, $\frac{dv}{dx} = a + b\ \ f\ \ v$

Or, $\frac{dv}{a + b + \ v} = dx$

Integrating both sides, we get

$$\int \frac{dv}{a + b\ \ f\ \ v} = \int dx + c$$

Or, $\phi\ \ v\ = x + c$

Or, $\phi\ \ ax + by + c\ = x + c$, where '$c$' is the constant of integration.

Example: Solve $\frac{dy}{dx} = \ 4x + y + 1^{\ 2}$.

Solution: Let $4x + y + 1 = v.$ …(i)

Differentiating (i) with respect to x, we get

$4 + \frac{dy}{dx} = \frac{dv}{dx}$ or $\frac{dy}{dx} = \frac{dv}{dx} - 4.$ …(ii)

Using (i) ad (ii), the given equation becomes

$\frac{dv}{dx} - 4 = v^{2}$ or $\frac{dv}{dx} = 4 + v^{2}.$

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Now separating variables x and v, $dx = \frac{dv}{4+v^2}$.

Integrating, $x + c\left(\frac{1}{2}\right)\tan^{-1}\left(\frac{v}{2}\right)$, where c is an arbitrary constant.

or, $2x + 2c = \tan^{-1}\left(\frac{v}{2}\right)$ or $v = 2\tan\ 2x + 2c$

or, $4x + y + 1 = 2\tan\ 2x + 2c$, using 1

Example: Solve $\frac{dy}{dx} = \sec\ x + y$ or $\cos\ x + y\ dy = dx.$

Solution: Let $x + y = v,$

So that $\frac{dy}{dx} = \left(\frac{dv}{dx}\right) - 1.$ …(i)

Using (i), the given equation becomes

$\left(\frac{dv}{dx}\right) - 1 = \sec v$ or $\frac{dv}{dx} = 1 + \frac{1}{\cos v}$

or $dx = \frac{\cos v}{1+\cos v}dv = \frac{2\cos^2\frac{1}{2}v - 1}{1 + 2\cos^2\frac{1}{2}v - 1}dv$ or $dx = \left(1 - \frac{1}{2}\sec^2\frac{1}{2}v\right)dv.$

Integrating, $x + c = v - \tan\frac{1}{2}v$ or $y - \tan\frac{1}{2}\ x + y\ = c.$

Example: Solve $x + y\ dx - dy\ = dx + dy.$

Solution: Re-writing the given equation, we get

$x + y - 1\ dx = \ x + y + 1\ dy$ or $\frac{dy}{dx} = \frac{x+y-1}{x+y+1}$ ...(i)

Let $x + y = v.$ …(ii)

So that as usual $\frac{dy}{dx} = \frac{dv}{dx} - 1.$ …(iii)

Using (ii) and (iii), (i) becomes

$\frac{dv}{dx} - 1 = \frac{v-1}{v+1}$ or $\frac{dv}{dx} = \frac{2v}{v+1}$ or $2dx = \left(1 + \frac{1}{v}\right)dv.$

$\therefore$ Integrating, $2x + c = v + \log v$ or $x - y + c = \log\ x + y\ .$

4.3.3 Homogenous Equations:

A differential equation of the form

$\frac{dy}{dx} = f\ y/x$ …(i)

is paid to be homogenous equation of first order and first degree.

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Putting $y/x = v$, i.e. $y = vx$ …(ii)

Differentiating (ii) with respect to x, we get

$$\frac{dy}{dx} = v + x\frac{dv}{dx} \quad \text{…(iii)}$$

Using (ii) and (iii) in (i), we get

$$v + x\frac{dv}{dx} = f\ v$$

Or, $x\frac{dv}{dx} = f\ v\ - v$

Now, separating the variable x and v, we get

$$\frac{dx}{x} = \frac{dv}{f\ v\ - v}$$

Integrating both sides we get

$\log x + c = \int \frac{dv}{f\ v\ - v}$, where c is an arbitrary constant.

After integrating, we replace v by y/x

Example: Solve $x^3 + 3xy^2\ dx + y^3 + 3x^2 y\ dy = 0.$

Solution: $$\frac{dy}{dx} = -\frac{x^3 + 3xy^2}{y^3 + 3x^2 y} = -\frac{1 + 3\left(\frac{y}{x}\right)^2}{\left(\frac{y}{x}\right)^3 + 3\left(\frac{y}{x}\right)}. \quad \text{….(i)}$$

Take $\frac{y}{x} = v$ *i.e.* $y = vx.$

$$\therefore\ \frac{dy}{dx} = v + x\frac{dv}{dx}. \quad \text{…(ii)}$$

From (i) and (ii), $v + x\frac{dv}{dx} = -\frac{1 + 3v^2}{v^2 + 3v}$

or, $x\frac{dv}{dx} = -\frac{1 + 3v^2}{v^2 + 3v} - v = -\frac{v^4 + 6v^2 + 1}{v^3 + 3v}$ or $4\frac{dx}{x} = -\frac{4v^3 + 12v}{v^4 + 6v^2 + 1}dv.$

Integrating, $4\log x = -\ v^4 + 6v^2 + 1\ + \log c$

or $\log x^4 = \log\left[\frac{c}{v^4 + 6v^2 + 1}\right]$ *i.e.* $x^4\ v^4 + 6v^2 + 1\ = c$

or $y^4 + 6x^2 y^2 + x^4 = c$ or $x^2 + y^2\ ^2 + 4x^2 y^2 = c,$ as $\frac{y}{x} = v$

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Example: Solve $\frac{dy}{dx}=\frac{y}{x}+\tan\frac{y}{x}$.

Solution: Since the R.H.S. of the given equation is function of $\frac{y}{x}$ alone,

we conclude that it must be a homogeneous equation.

Take $\frac{y}{x}=v$, *i.e.* $y=vx \ \therefore \ \frac{dy}{dx}=v+x\frac{dv}{dx}$.

Using (i), the given equation becomes

$v+x\frac{dv}{dx}=v+\tan v$ or $\frac{dx}{x}=\frac{\cos v}{\sin v}dv$.

Integrating, $\log x=\log\sin v+\log c$

or $x=c\sin v$ *i.e.* $x=c\sin\left(\frac{v}{x}\right)$.

Example: Solve $x\cos\left(\frac{y}{x}\right)\ ydx+xdy\ =y\sin\left(\frac{y}{x}\right)\ xdy-ydx$ …(i)

or $\left(x\cos\frac{y}{x}+y\sin\frac{y}{x}\right)y-\left(y\sin\frac{y}{x}-x\cos\frac{y}{x}\right)x\frac{dy}{dx}=0.$ …(ii)

Solution: Rewriting (i), we get (ii). So (i) and (ii) are the same equations.

From (ii), $\frac{dy}{dx}=\frac{\left\{x\cos\left(\frac{y}{x}\right)+y\sin\left(\frac{y}{x}\right)\right\}y}{\left\{y\sin\left(\frac{y}{x}\right)-x\cos\left(\frac{y}{x}\right)\right\}x}$

or $\frac{dy}{dx}=\frac{\left[\cos\left(\frac{y}{x}\right)+\left(\frac{y}{x}\right)\sin\left(\frac{y}{x}\right)\right]\left(\frac{y}{x}\right)}{\left[\left(\frac{y}{x}\right)\sin\left(\frac{y}{x}\right)-\cos\left(\frac{y}{x}\right)\right]}$. …(iii)

Take $\frac{y}{x}=v$ *i.e.* $y=vx \ \therefore \ \frac{dy}{dx}=v+x\frac{dv}{dx}$.

Using, (iv), (iii) becomes $v+x\frac{dv}{dx}=\frac{v\ \cos v+v\sin v}{v\sin v-\cos v}$

or $x\frac{dv}{dx}=\frac{v\ \cos v+v\sin v}{v\sin v-\cos v}-v=\frac{2v\cos v}{v\sin v-\cos v}$

or $2\frac{dc}{x}=\frac{v\sin v-\cos v}{v\cos v}dv=\left[\frac{\sin v}{\cos v}-\frac{1}{v}\right]dv.$

Integrating, $2\log x=-\log\cos v-\log v+\log c$

or $\log x^2=\log\left(\frac{c}{v}\cos v\right)$

$\therefore \ x^2v\cos v=c$ or $xy\cos\left(\frac{y}{x}\right)=c.$ $\left[\because \ v=\frac{y}{x}\right]$

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Example: Solve $x\left(\frac{dy}{dx}\right) = y \ \log y - \log x + 1$.

Solution: Given $\frac{dy}{dx} = \frac{y}{x}\left(\log\frac{y}{x} + 1\right)$ …(i)

Putting $\frac{y}{x} = v$ or $y = xv$, we have $\frac{dy}{dx} = v + x\frac{dv}{dx}$. …(ii)

From (i) and (ii), $v = x\frac{dv}{dx} = v \ \log v + 1$ or $\frac{dx}{x} = \frac{dv}{v\log v}$.

Integrating, $\log x + \log c = \log \log v$ or $xc = \log v$.

$\therefore v = e^{xc}$ or $\left(\frac{y}{x}\right) = e^{xc}$ or $y = xe^{xc}$.

4.3.4 Equation reducible to homogenous form:

The differential equation of the form

$$\frac{dy}{dx} = \frac{ax + by + c}{a_1x + b_1y + c_1} \quad \text{…(i)}$$

can be reduced to homogeneous form as discussed.

CASE 1 when $\frac{a}{a_1} \neq \frac{b}{b_1}$

Take, $x = X + h$ and $y = Y + k$ …(ii)

Where, X and Y are new variable and h and k are constant to be so choose that the resulting equation becomes homogenous in X and Y.

Differentiating (ii), we get

$dx = dx$ and $dy = dy$

i.e. $\frac{dy}{dx} = \frac{dy}{d}$ …(iii)

Using (ii) and (iii) in (i), we get

$$\frac{dy}{dx} = \frac{a\ X + h\ + b\ Y + k\ + c}{a_1\ X + h\ + b_1\ Y + k\ + c_1}$$

Or, $\frac{dy}{dx} = \frac{aX + bY + \ ah + bk + c}{a_1X + b_1Y + \ a_1h + b_1k + c_1}$ …(iv)

To make (iv) homogenous, we choose h and k such that

$ah + bk + c = 0$ and $a_1h + b_1k + c_1 = 0$ …(v)

solving (v), we get

$h = \frac{bc_1 - b_1c}{ab_1 - a_1b}$ and $k = \frac{ca_1 - c_1a}{ab_1 - a_1b}$

Since, $ab_1 - a_1b \neq 0$ as $\frac{a}{a_1} \neq \frac{b}{b_1}$ so h and k will exist

Now, with condition (v), equation (iv) reduce to

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$$\frac{dy}{dx}=\frac{ax+by}{a_1x+b_1y}=\frac{a+b\ y/x}{a_1+b_1\ y/x}$$

which is a homogenous equation in X and Y and after solving it we replace X and Y by $x + h$ and $y + k$ where h and k has been derived earlier.

Example: Solve $\frac{dy}{dx}=\frac{x+2y-3}{2x+y-3}$.

Solution: Take $x=X+h,\ y=Y+k,$ so that $\frac{dy}{dx}=\frac{dY}{dX}$. …(i)

$\therefore$ Given equation becomes $\frac{dY}{dX}=\frac{X+2Y+\ h+2k-3}{2X+Y+\ 2h+k-3}$. …(ii)

Choose h, k so that $h+2k-3=0$ and $2h+k-3=0.$ …(iii)

Solving (iii) we get $h=1,\ k=1$ so that from (i), we have

$X=x-1, Y=y-1$ …(iv)

Using (iii) in (ii), we get

$$\frac{dY}{dX}=\frac{X+2Y}{2X+Y}=\frac{1+\left(\frac{2Y}{X}\right)}{2+\left(\frac{Y}{X}\right)}$$ …(v)

Take $\frac{Y}{X}=v,$ *i.e.*$Y=vX$. So $\frac{dY}{dX}=v+X\frac{dv}{dX}$. …(vi)

From (v) and (vi),

$$v+X\frac{dv}{dX}=\frac{1+dv}{2+v} \text{ or } X\frac{dv}{dX}=\frac{1+2v}{2+v}-v=\frac{1-v^2}{2+v}$$

$$\text{or } \frac{dX}{X}=\frac{2+v\ dv}{1-v\ \ 1+v}=\left[\frac{1}{2}\left(\frac{1}{1+v}\right)+\frac{3}{2}\left(\frac{1}{1-v}\right)\right]dv.$$

[Resolving into partial fractions]

Integrating, $\log X+\log c=\left(\frac{1}{2}\right)\left[\log\ 1+v\ -3\log\ 1-v\ \right]$

$$\text{or } 2\log(cX)=\log\frac{1+v}{1+v^{\ 3}} \text{ or } X^2c^2=\frac{1+v}{1-v^{\ 3}}$$

$$\text{or } X^2c^2\left(1-\frac{Y}{X}\right)^3=1+\frac{Y}{X}, \text{ as } v=\frac{Y}{X}$$

or $c^2\ \ X-Y\ ^3=X+Y$ or $c^2\ \ x-1-\ y-1\ \ ^2=x-1+y-1$, by (iv)

or $c'\ x-y\ ^2=\ x+y-2\ ,$ taking $c'\ x-y\ ^2=\ x+y-2\ ,$

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Example: Solve $(2x^2+3y^2-7)\,x\,dx-(3x^2+2y^2-8)\,y\,dy=0$

Solution: Given $(2x^2+3y^2-7)\,x\,dx-(3x^2+2y^2-8)\,y\,dy=0$ …(i)

Let $x^2=u$ and $y^2=v$ so that

$2x\,dx=du$ and $2y\,dy=dv.$ …(ii)

From (i) and (ii), $(2u+3v-7)\,du-(3u+2v-8)\,dv=0$

or $\frac{dv}{du}=\frac{2u+3v-7}{3u+2v-8}.$ …(iii)

Taking $u=U+h,\ v=V+k$ so that $\frac{dv}{du}=\frac{dV}{dU},$ …(iv)

The given equation becomes $\frac{dV}{dU}=\frac{2U+3Y+(2h+3k-7)}{3U+2V+(3u+2k-8)}.$ …(v)

Choose h, k so that $2h+3k-7=0$ and $3h+2k-8=0.$ …(vi)

Solving (iii), we get $h = 2, k = 1$ so that from (iv), we have

$U=u-2$ and $V=v-1.$

or $U=x^2-2$ and $V=y^2-1,$ by (ii) …(vii)

and (v) becomes $\frac{dV}{dU}=\frac{2U+3V}{3U+2V}=\frac{2+3\left(\frac{V}{U}\right)}{3+2\left(\frac{V}{U}\right)}.$ …(viii)

Take $\frac{V}{U}=w,$ *i.e.*$V=wU.$ So $\frac{dV}{dU}=w+U\frac{dw}{dU}.$ …(ix)

From (viii) and (ix), $w+U\frac{dw}{dU}=\frac{2+3w}{3+2w}$ or $U\frac{dw}{dU}=\frac{2(1-w^2)}{3+2w}$

or $\frac{2dU}{U}=\frac{3+2w}{1-w^2}dw=\left[\frac{3}{1-w^2}-\frac{-2w}{1-w^2}\right]dw.$

Integrating, $2\log U=\frac{3}{2}\log U=\frac{3}{2}\log\frac{l+w}{l-w}-\log(l-w^2)+\frac{1}{2}\log c$

or $4\log U=3\log\left(\frac{1+w}{1-w}\right)-2\log(1-w^2)+\frac{1}{2}\log c\,\backslash$

or $\log\frac{U^4}{c}=\log\left(\frac{1+w}{1-w}\right)^3-\log(1-w^2)^2$

or $\log\frac{U^4}{c}=\left[\log\left(\frac{1+w}{1-w}\right)^3\cdot\frac{1}{(1-w^2)^2}\right]$

or $\frac{U^4}{c}=\frac{(1+w)^3}{(1-w)^5(1+w)^2}$ or $(1-w)^5U^4=c(1+w)$

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or $\left(1-\frac{V}{U}\right)^5 U^4 = c\left(1+\frac{V}{U}\right)$ or $U-V^{\ 5} = C\ U+V$

or $x^2-y^2-1^{\ 5} = c\ x^2+y^2-3$, by (vii).

CASE 2 when $\frac{a}{a_1}=\frac{b}{b_1}=\lambda$ say

This implies $a=\lambda a_1$ and $b=\lambda b_1$

Then the equation

$$\frac{dy}{dx}=\frac{ax+by+c}{a_1x+b_1y+c_1} \quad \ldots(i)$$

reduce in the form

$$\frac{dy}{dx}=\frac{\lambda\ a_1x+b_1y\ +c}{a_1x+b_1y+c_1} \quad \ldots(ii)$$

Now, putting $a_1x+b_1y=v$, so $\frac{dy}{dx}=\frac{1}{b_1}\left[\frac{dv}{dx}-a_1\right]$ in (ii), we get

$$\frac{1}{b_1}\left[\frac{dv}{dx}-a_1\right]=\frac{\lambda v+c}{v+c_1}$$

Or, $\frac{dv}{dx}=\frac{b_1\ \lambda v+c}{v+c_1}+a_1$ …(iii)

Now, the equation (iii) can be solved by the method of separation of variable and replacing to v by a_1x+b_1y in obtained solution, we will get the required solution.

Example: Solve $\frac{dy}{dx}=\frac{4x+6y+5}{3y+2x+4}$.

Solution: The given equation may be re-written as

$$\frac{dy}{dx}=\frac{2\ 2x+3y\ +5}{2x+3y\ +4}. \quad \ldots(i)$$

$\therefore$ We take $2x+3y=v$. …(ii)

Differentiating, $2+3\ \frac{dy}{dx}=\frac{dv}{dx}$ or $\frac{dy}{dx}=\frac{1}{3}\left(\frac{dv}{dx}-2\right)$. …(iii)

Using (ii) and (iii), (i) gives

$$\frac{1}{3}\left(\frac{dv}{dx}-2\right)=\frac{2v+5}{v+4} \text{ or } \frac{dv}{dx}=\frac{3\ 2v+5}{v+4}+2=\frac{8v+23}{v+4}$$

$$\text{or } \frac{dx}{dv}=\frac{v+4}{dv+23}=\frac{\frac{1}{8}\ 8v+23\ +\left(4-\frac{23}{8}\right)}{8v+23}=\left[\frac{1}{8}+\frac{9}{8\ 8v+23}\right]$$

Separating variables, $dx=\left[\frac{1}{8}+\frac{9}{8\ 8v+23}\right]dv$.

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Integrating, $x+c=\left(\frac{v}{8}\right)+\left(\frac{9}{64}\right)\log\ 8v+23$

or $\quad 8x+8c=2c+3y+\left(\frac{9}{8}\right)\log\ 16x+24y+23$

[using (ii) and multiplying by 8]

or $3y-6x+\left(\frac{9}{8}\right)\log\ 16x+24y+23\ =8c$

or $y-2x+\left(\frac{3}{8}\right)\log\ 16x+24y+23\ =c'$,

where $c'\left(=\frac{8c}{3}\right)$ is an arbitrary constant.

CASE 3 when $a=b_1$ and $b=a_1$

Or $a=-b_1$ and $b=-a_1$

Then the given differential equation

$$\frac{dy}{dx}=\frac{ax+by+c}{a_1x+b_1y+c_1} \qquad \ldots(iv)$$

can be solved by applying componendo and dividendo.

Example: Solve the differential equation

$$\frac{dy}{dx}=\frac{x+2y+8}{2x+y+7}$$

Here, $a=b_1=1$ and $b=a_1=2$, so applying componendo and dividendo, we get

$$\frac{dy+dx}{dy-dx}=\frac{3x+3y+15}{-x+y+1}$$

Or, $\frac{dy+dx}{3\ x+y+5}=\frac{dy-dx}{-x+y+1}$

Or, integrating both sides, we get

$$\frac{1}{3}\log\ x+y+z\ =\log\ y-x+1\ +\log c$$

Or, $x+y+5=c_1\ \ y-x+1\ ^3$, where c_1 is the arbitrary constant.

4.3.5 <u>Exact differential equation:</u>

A differential equation which can be obtained by differentiating its primate (solution) without any elimination and reduction is said **'Exact'.**

e.g.: The equation $xdy+ydx=0$ is exact as it is obtained from its solution $xy= c$

In other words, the differential equation of the form

$M\ \ x,\ y\ \ dx+N\ \ x,y\ \ dy=0$

is called exact if there exists a function $f\ \ x,\ y\ \ $ **of *x* and *y* such that**

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$d\left[f\ \ x,\ y\ \right] = M\ \ x,\ y\ \ dx + N\ \ x,\ y\ \ dy$

Necessary and sufficient condition for a differential equation

$Mdx + Ndy = 0$

Where M and N are function of *x* and *y*, to be exact is

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$

Working rule:

(a) Compare the given differential equation with the form $Mdx + Ndy = 0$ and find out M and N.

The find $\frac{\partial M}{\partial x}$ and $\frac{\partial N}{\partial y}$. If $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$, we say to given equation is exact.

(b) Find $\int Mdx$; treating y as constant

(c) Find $\int Ndy$; treating x as constant and omit those terms which have been obtained in integral $\int Mdx$.

(d) The complete solution of the given differential equation is

$\int Mdx + \int Ndy = c$

Omitting the terms obtained in $\int Mdx$

Example: Solve $y\sin 2x\ dx - \ 1 + y^2 + \cos^2 x\ \ dy = 0$

Solution: Comparing the given equation with $M\ dx + N\ dy = 0,$ we get

$M = y\sin 2x$ and $N = -\ 1 + y^2 + \cos^2 x$.

$M = x + \frac{a^2 y}{x^2 + y^2}$ and $N = y - \frac{a^2 x}{x^2 + y^2}.$

So $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$ Hence the given equation is exact.

Now $\int M\ dx = \int y\sin 2x\, dx = -\frac{1}{2} y\cos 2x$ …(i)

and $\int N\ dy = \int\ 1 + y^2 + \cos^2 x\ \ dy = -y - \frac{y^3}{3} - y\cos^2 x$

[Treating x as constant]

$= -\frac{1}{3} y^3 - y\ 1 + \cos^2 x$

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$$= -\frac{y^3}{3} - y\left(1 + \frac{1+\cos 2x}{2}\right)$$

$$= -\frac{y^3}{3} - \frac{1}{2}y\ \ 3 + \cos 2x$$

$$= -\frac{y^3}{3} - \frac{3}{2}y - \frac{1}{2}y\cos 2x$$

$$= -\frac{y^3}{3} - \frac{3}{2}y \qquad \text{...(ii)}$$

[Omitting $-\frac{1}{2}y\cos 2x$ which has been already obtained in $\int M\,dx$ *i.e.* in (i) above]

Hence the required solution from (i) and (ii) is $-\frac{1}{2}y\cos 2x - \frac{1}{3}y^3 - \frac{3}{2}y = C_1$, or $y\cos 2x + \frac{2}{3}y^3 + 3y = C$, where $C = -2C_1$ is an arbitrary constant.

Example: Solve $ax + hy + g\ \ dx + \ hx + by + f\ \ dy = 0.$

or $\frac{dy}{dx} + \frac{ax + hy + g}{hx + by + f} = 0.$

Solution: Here $M = ax + hy + g,\ N = hx + by + f.$

So $\frac{\partial M}{\partial y} = h = \frac{\partial N}{\partial x}$ and equation is exact.

Now $\int M\,dx = \int\ ax + hy + g\ \ dx = \left(\frac{1}{2}\right)ax^2 + hxy + gx$ …(i)

[Treating y as constant]

and $\int N\,dy = \int\ hx + by + f\ \ dy = hxy + \left(\frac{1}{2}\right)by^2 + fy$

[Treating x as constant]

$= \left(\frac{1}{2}\right)by^2 + fy.$ …(ii)

[Omitting hxy which already occurs in $\int M\ \ dx$].

There-fore the required solution from (i) and (ii) is

$\left(\frac{1}{2}\right)ax^2 + hxy + gx + \left(\frac{1}{2}\right)by^2 + fy = C$

or $ax^2 + 2hxy + by^2 + 2gx + 2fy = C',$

where $C' = \ 2C$ is an arbitrary constant.

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Example: Solve $x\,dx + y\,dy = \dfrac{a^2\ x\,dy - y\,dx}{x^2 + y^2}$ …(i)

2nd Form $x^3 + xy^2 + a^2 y\ dx + \ y^3 + yx^2 - a^2 x\ dy = 0$ …(ii)

3rd Form $x\,dx + y\,dy\ \ x^2 + y^2\ = a^2\ x\,dy - y\,dx$. …(iii)

Solution: On dividing by $x^2 + y^2$, (ii) as well as (iii) will reduce to (i). So let us solve (i) which can be rewritten in standard form $M\,dx + N\,dy = 0$ as given below

$$\left\{x + \frac{a^2 y}{x^2 + y^2}\right\}dx + \left(y - \frac{a^2 x}{x^2 + y^2}\right)dy = 0 \quad \text{…(iv)}$$

Thus, here $M = x + \dfrac{a^2 y}{x^2 + y^2}$ and $N = y - \dfrac{a^2 x}{x^2 + y^2}$.

$$\therefore \quad \frac{\partial M}{\partial y} = a^2 \frac{1.\ x^2 + y^2\ - y.2y}{x^2 + y^2\ ^2} = a^2 \frac{x^2 - y^2}{x^2 + y^2\ ^2}$$

and $\dfrac{\partial N}{\partial x} = -a^2 \dfrac{1.\ x^2 + y^2\ - x.2x}{x^2 + y^2\ ^2} = -a^2 \dfrac{y^2 - x^2}{x^2 + y^2\ ^2} = \dfrac{a^2\ x^2 - y^2}{x^2 + y^2\ ^2}$.

$\therefore$ Here, $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$, hence (i) is an exact equation.

Now $\int M\ dx = \int \left\{x + \dfrac{ay^2}{x^2 + y^2}\right\}dx = \dfrac{x^2}{2} + a^2 y.\dfrac{1}{y}\tan^{-1}\left(\dfrac{x}{y}\right)$

[Treating y as constant]

$$= \frac{1}{2}x^2 + a^2 \tan^{-1}\left(\frac{x}{y}\right) \quad \text{…(v)}$$

and $\int N\,dx = \int \left\{y - \dfrac{a^2 x}{x^2 + y^2}\right\}dy = \dfrac{y^2}{2} - a^2 x.\dfrac{1}{x}\tan^{-1}\left(\dfrac{y}{x}\right)$.

[Treating x as constant]

$$= \frac{1}{2}y^2 - a^2 \tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}y^2 - a^2 \cot^{-1}\left(\frac{x}{y}\right)$$

$$= \frac{1}{2}y^2 - a^2\left[\frac{1}{2}\pi - \tan^{-1}\left(\frac{x}{y}\right)\right] = \frac{1}{2}y^2 - \frac{1}{2}\pi a^2 \ldots\text{(vi)}$$

[Omitting $\tan^{-1}\left(\dfrac{x}{y}\right)$ which already occurs in $\int M\,dx$]

Hence the solution from (v) and (vi) is

$$\frac{1}{2}x^2 + a^2 \tan^{-1}\left(\frac{x}{y}\right) + \frac{1}{2}y^2 - \frac{1}{2}a^2\pi = C$$

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or $x^2 + 2a^2 \tan^{-1}\left(\frac{x}{y}\right) + y^2 = C'$

where $C' = 2C + a^2\pi$ is an arbitrary constant.

4.3.6 Integrating factor:

If a differential equation of the form $Mdx + Ndy = 0$ is not exact, it can always be made exact by multiplying by some function of *x* and *y*. Such multiplier is called as integrating factor.

A differential equation of the form $Mdx + Ndy = 0$ posses an **infinite number of integrating factor.**

There is no general method for finding the integrating factor. Here we discuss some specific rules to determine an integrating factor.

Rule 1: By inspection: Often as I.F. (integrating factor) of given equation of the form $Mdx + Ndy$ can be formed out by inspection.

Here, by rearranging the terms of the given equation and/or by dividing by a suitable function of *x* and *y*, the equation. Thus obtained will contain several parts inferable easily.

Example: Solve $y\,dx - x\,dy + (1 + x^2)\,dx + x^2 \sin y dy = 0$

Solution: Dividing each term of the given equation by x^2, we get

$$\frac{y\,dx - x\,dy}{x^2} + \frac{1 + x^2}{x^2} + dx + \sin y\,dy = 0$$

or $$-\frac{x\,dy - y\,dy}{x^2} + \left(\frac{1}{x^2} + 1\right)dx + \sin y\,dy = 0$$

or $$-d\left(\frac{y}{x}\right) + \left(1 + \frac{1}{x^2}\right)dx + \sin y\,dy = 0.$$

Integrating $$-\left(\frac{y}{x}\right) + x - \left(\frac{1}{x}\right) - \cos y = c$$

or $-y + x^2 - 1 - x\cos y = cx$, where *c* is an arbitrary constant.

Example: Solve $y\sin 2x\,dx = (1 + y^2 + \cos^2 x)\,dy$.

Solution: Re-writing $-2y\sin x\cos x\,dx + \cos^2 x\,dy + (1 + y^2)\,dy = 0$

or $d(y\cos^2 x + (1 + y')\,dy = 0.$

Integrating, $y\cos^2 x + y + \frac{y^3}{3} = c.$

Example: Solve $(x^3 + xy^2 + a^2 y)\,dx + (y^3 + yx^2 - a^2 x)\,dy = 0$

Solution: Re-writing the given equation, we have

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$x\ (x^2+y^2)\ dx + y\ (x^2+y^2)\ dy + a^2\ (y\,dx - x\,dy) = 0$

or $x\,dx + y\,dy + a^2 \dfrac{y\,dx - x\,dy}{x^2+y^2} = 0$

or $x\,dx + y\,dy + a^2 d\left(\tan^{-1}\dfrac{x}{y}\right) = 0.$

Integrating, $\left(\dfrac{1}{2}\right)x^2 + \left(\dfrac{1}{2}\right)y^2 + a^2 \tan^{-1}\left(\dfrac{x}{y}\right) = \dfrac{1}{2}c$

or $x^2 + y^2 + 2a^2 \tan^{-1}\left(\dfrac{x}{y}\right) = c.$

Rule 2: If the given equation $M\,dx + N\,dy = 0$ is homogenous and $Mx + Ny \neq 0$ then $\dfrac{1}{Mx+Ny}$ is an integrating factor of the differential equation.

Example: Solve $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0.$

Solution: Given $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0.$ …(i)

Clearly (i) is a homogeneous differential equation. Comparing (i) with $Mdx + Ndy = 0,$ we have

$M = x^2y - 2xy^2$ and $N = -(x^3 - 3x^2y)$. …(ii)

$\therefore\ \ Mx + Ny = x\,(x^2y - 2xy^2) - y\,(x^3 - 3x^2y) = x^2y^2 \neq 0,$

Showing that I.F. of (i) = $\dfrac{1}{Mx+Ny} = \dfrac{1}{x^2y^2}$.

On multiplying (i) by $\dfrac{1}{x^2y^2}$, we have

$\left(\dfrac{1}{y} - \dfrac{2}{x}\right)dx - \left(\dfrac{x}{y^2} - \dfrac{3}{y}\right)dy = 0,$

which must now be an exact equation. So, the required solution by the usual rule is

$\left(\dfrac{x}{y}\right) - 2\log x + 3\log y = c$, c being an arbitrary constant.

Rule 3: If the differential equation $Mdx + Ndy = 0$ is of the form

$f_1(xy)\,ydx + f_2(xy)\,xdy = 0$ …(i)

then, $\dfrac{1}{Mx - Ny}$ is an integrating factor, provided $Mx - Ny \neq 0$

Example: Solve $(xy\sin xy + \cos xy)\,y\,dx + (xy\sin xy - \cos xy)\,x\,dy = 0.$

Solution: Given $(xy\sin xy + \cos xy)\,y\,dx + (xy\sin xy - \cos xy)\,x\,dy = 0.$ …(i)

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Comparing (i) with $Mdx + Ndy = 0$, we have

$M = y\ xy\sin xy + \cos xy$ and $N = x\ xy\sin xy - \cos xy$, …(ii)

Showing that (i) is of the form $f_1\ xy\ y\,dx + f_2\ xy\ x\,dy = 0.$

Again, $Mx - Ny = xy\ xy\sin xy + \cos xy\ - xy\ xy\sin xy - \cos xy$

or $Mx - Ny = 2xy\cos xy \neq 0,$

showing that I.F. of (i) = $\dfrac{1}{Mx - Ny} = \dfrac{1}{2xy\cos xy}.$

On multiplying (i) by $\dfrac{1}{2xy\cos xy}$, we have

$$\frac{1}{2}\left(y\tan xy + \frac{1}{x}\right)dx + \frac{1}{2}\left(x\tan xy - \frac{1}{y}\right)dy = 0$$

which must be exact and so by the usual rule, the required solution is

$$\frac{1}{2}\ \log\sec xy + \log x\ - \left(\frac{1}{2}\right)\log y = \left(\frac{1}{2}\right)\log c$$

or $\log\sec\ xy + \log\left(\dfrac{x}{y}\right) = \log c$ or $\left(\dfrac{x}{y}\right)\sec\ xy = c.$

Rule 4: If the differential equation is of the form $Mdx + Ndy = 0$ and $\dfrac{1}{N} = \left(\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}\right) = \phi\ x$

where $\phi\ x$ is the function of x only then

$\text{I.F.} = e^{\int \phi\ x\ dx}$

Example: Solve $x^2 + y^2 + x\ dx + xy\ dy = 0.$

Solution: Given $x^2 + y^2 + x\ dx + xy\ dy = 0.$ …(i)

Comparing (i) with $Mdx + Ndy = 0$, we have

$M = x^2 = y^2 + x$ and $N = xy.$

Here $\dfrac{\partial M}{\partial y} = 2y$ and $\dfrac{\partial N}{\partial x} = y.$ …(ii)

$$\therefore \frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{xy}\ 2y - y\ = \frac{1}{x},$$

which is a function of x alone.

$\therefore$ I.F. of (i) $= e^{\int 1/x\ dx} = e^{\log x} = x.$

Multiplying (i) by x, we have $x^3 + xy^2 + x^2\ dx + x^2 y\,dy = 0,$

Which must be exact equation and so its solution is

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$$\left(\frac{x^4}{4}\right)+\frac{x^2y^2}{2}+\left(\frac{x^3}{3}\right)=\frac{c}{12} \quad \text{or} \quad 3x^4+6x^2y^2+4x^3=c.$$

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Rule 5: If the differential equation is of the form $Mdx + Ndy = 0$ and $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \psi\ y$

where $\psi\ y$ is the function of y only then

$\text{I.F.} = e^{\int \psi\ y\ dy}$

Example: $2xy^4e^y + 2xy^3 + y\ dx + x^2y^4e^y - x^2y^2 - 3x\ dy = 0.$ …(i)

Solution: Comparing (i) with $Mdx + Ndy = 0$, we get

$M = 2xy^4e^y + 2xy^3 + y$ and $N = x^2y^4e^y - x^2y^2 - 3x.$

Here $\frac{\partial M}{\partial y} = 8xy^3e^y + 2xy^4e^y + 6xy^2 + 1,$

$\frac{\partial N}{\partial x} = 2xy^4e^y - 2xy^2 - 3.$

$\therefore \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -4\ 2xy^3e^y + 1\ = -\frac{4}{y}\ 2xy^4e^y + 2xy^3 + y\ = \frac{-4M}{y}$

$\Rightarrow \quad \frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = -\frac{4}{y},$ which is a function of y alone.

$\Rightarrow$ I.F. of (i) $= e^{\int\left(\frac{-4}{y}\right)dy} = e^{-4\log y} = \left(\frac{1}{y^4}\right).$

Multiplying (i) by $\frac{1}{y^4}$, we have

$\left\{2xe^y + \left(\frac{2x}{y}\right) + \left(\frac{1}{y^3}\right)\right\}dx + \left\{x^2e^y - \left(\frac{x^2}{y^2}\right) - 3\left(\frac{x}{y^4}\right)\right\}dy = 0,$

which must be exact and so by usual method its solution is

$x^2e^y + \left(\frac{x^2}{y}\right) + \left(\frac{x}{y^3}\right) = c.$

Rule 6: If the given equation $Mdx + Ndy = 0$ is of the form $x^\alpha y^\beta\ Mydx + nxdy\ = 0$ …(i)

Then its integrating factor is

$x^{k\ m\ -1-\alpha}\ y^{k\ n\ -1-\beta}$, where k is any scalar.

If the differential equation is of the form

$x^\alpha y^\beta\ mydx + nxdy\ + x^{\alpha_1}y^{\beta_1}\ m_1ydx + n_1xdy\ = 0$ …(ii)

then by Rule (vi) $x^{k\ m\ -1-\alpha}\ y^{k\ n\ -1-\beta}$ and $x^{k_1\ m_1\ -1-\alpha_1}\ y^{k_1\ n_1\ -1-\beta_1}$ are I.F. for 1st and 2nd term of L.H.S. of equation (ii).

The above mentioned the factors will be identical if we choose k and k_1 such that

$km - 1 - \alpha = k_1m_1 - 1 - \alpha_1$ …(iii)

$kn - 1 - \beta = k_1n_1 - 1 - \beta_1$ …(iv)

Solving (iii) and (iv), we evaluate values of k and k_1. Substituti9ng these values in any one of the factors, we obtained an I.F. for (ii).

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Also, for the differential equation given in (ii), we can assume $x^h y^k$ be an integrating factor and then we can find out the value of *h* and *k*.

Example: Solve $y^2 + 2x^2 y\ dx + 2x^3 - xy\ dy = 0$.

Solution: The given equation can be rewritten as

$y^2\ dx - xy\ dy + 2x^2 y\ dx + 2x^3\ dy = 0$

or $y\ y\ dx - x\ dy + x^2\ 2y\,dx + 2x\,dy = 0.$

So let $x^h y^k$ be an I.F. of the given equation.

Multiplying the given equation by $x^h y^k$, we have

$$x^h y^{k+2} + 2x^{h+2} y^{k+1}\ dx + 2x^{h+3} y^k - x^{h+1} y^{k+1}\ dy = 0, \quad \ldots(i)$$

which must be exact. For (i), we have

$M = x^h y^{k+2} + 2x^{h+2} y^{k+1},\ N = 2x^{h+3} y^k - x^{h+1} y^{k+1}.$

Since (i) is exact, we must have $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

i.e., $k+2\ x^h y^{k+1} + 2\ k+1\ x^{h+2} y^k = 2\ h+3\ x^{h+2} y^k - h+1\ x^h y^{k+1}$

Now equating the coefficients of $x^h y^{k+1}$ and $x^{h+2} y^k$, we get

$k+2 = -\ h+1$ and $2\ k+1 = 2\ h+3$,

i.e., $h + k = -3$ and $h - k = -2.$ …(ii)

Solving system (ii), we get $h = -\left(\frac{5}{2}\right)$ and $k = -\left(\frac{1}{2}\right)$

$\therefore$ I.F. $= x^{-5/2} y^{-1/2}.$

Multiplying the given equation by I.F. $x^{-5/2} y^{-1/2}$, we get

$$x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2}\ dx + 2x^{1/2} y^{-1/2} + x^{-3/2} y^{1/2}\ dy = 0$$

which must be exact. For this new equation, as usual solution is $-\left(\frac{2}{3}\right) x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2} = c.$

Example: Given that the differential equation $2x^2 y^2 + y\ dx - x^3 y - 3x\ d = 0$ has an I.F. of the form $x^h y^k$, find its general solution.

Solution: Given $2x^2 y^2 + y\ dx - x^3 y - 3x\ d = 0$ …(1)

Multiplying both sides of (1) by, I.F. $x^h y^k$, we get

$$2x^{h+2} y^{k+2} + x^h y^{k+1}\ dx + 3x^{h+1} y^k - x^{h+3} y^{k+1}\ dy = 0 \quad \ldots(2)$$

For (2) to be exact, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

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$$\Rightarrow \quad 2\ (k+2)\ x^{h+2}y^{k+1} + (k+1)\ x^h y^k$$
$$= 3\ (h+1)\ x^h y^k - (h+3)\ x^{h+2}y^{k+1}$$
$$\Rightarrow \quad 2\ (k+2) = -(h+3) \quad \text{and} \quad (k+1) = 3\ (h+1)$$
$$\Rightarrow \quad h + 2k = -7 \quad \text{and} \quad 3h - k = -2$$
$$\Rightarrow \quad h = \frac{-11}{7} \quad \text{and} \quad k = -\frac{19}{7}.$$

So an I.F. of (1) is $x^{-11/7}y^{-19/7}$. Multiplying (1) by $x^{-11/7}y^{-19/7}$, we have

$$2x^{3/7}y^{-5/7} + x^{-11/7}y^{-12/7}\ dx + 3x^{-4/7}y^{-19/7} - x^{-19/7}y^{-12/7}\ dy = 0,$$

Which must be exact. Hence, as usual, the required solution is

$$\frac{2x^{10/7}y^{-5/7}}{\frac{10}{7}} + \frac{x^{-4/7}y^{-12/7}}{\left(\frac{-4}{7}\right)} = \frac{7c}{20} \quad \text{or} \quad 4x^{10/7}y^{-5/7} - 5x^{-4/7}y^{-12/7} = c.$$

4.3.7 Linear differential equation:

A first order differential equation of the form

$$\frac{dy}{dx} + Py = \theta \qquad \ldots(i)$$

Where P and Q are either constant or function of x only is known as a linear differential equation of first order.

Method of solving

(i) Write the given equation in the form

$$\frac{dy}{dx} + Py = \theta$$

(ii) Find the I.F. by

$$\text{I.F.} = e^{\int P dx}$$

(iii) The required solution is

$$y \times \text{I.F.} = \int Q(\text{I.F.})dx + c$$

Example: Solve $x\cos x\left(\frac{dy}{dx}\right) + y\ (x\sin x + \cos x) = 1.$

Solution: Re-writing given equation, we have

$$\frac{dy}{dx} + \left(\tan x + \frac{1}{x}\right)y = \frac{\sec x}{x}$$

I.F. = $e^{\int\left(\frac{\tan x+1}{x}\right)dx} e^{\log\sec x + \log x} = e^{\log x\sec x} = x\sec x.$

Hence the required solution is

$$yx\sec x = \int \sec^2 x\ dx + c \quad \text{or} \quad yx\sec x = \tan x + c$$

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Example: Integrate $1+x^2\left(\frac{dy}{dx}\right)+2xy-4x^2=0$ .Obtain equation the curve satisfying this equation and passing through the origin.

Solution: Re-writing given equation, $\frac{dy}{dx}+\frac{2x}{1+x^2}=\frac{4x^2}{1+x^2}$.

Here $\int P\,dx=\int\frac{2x}{1+x}\,dx=\log\,1+x^2$

So I.F. = $e^{\int P\,dx}=\,1+x^2$.

Hence the required solution is

$y\,1+x^2\,=\int\frac{4x^2}{1+x^2}.\,1+x^2\,\,dx+c$

or $y\,1+x^2\,=\left(\frac{4}{3}\right)x^3+c.$

Example: Solve $x+2y^3\left(\frac{dy}{dx}\right)=y.$

Solution: Here it is possible to put the equation in form $\frac{dx}{dy}+P_1x=Q_1.$

Thus, we have $\frac{dx}{dy}=\frac{x+2y^3}{y}$ *i.e.* $\frac{dx}{dy}-\frac{1}{t}x=2y^2$ …(1)

In (1), $\int P_1\,dy=-\int\left(\frac{1}{y}\right)dy=-\log y\quad\therefore\ \text{I.F.}=e^{-\log y}=\frac{1}{y}.$

Hence the required solution is $\frac{x}{y}=\int 2y^3.\left(\frac{1}{y}\right)dy+c.$

or $\frac{x}{y}=y^2+c,$ where c is an arbitrary constant.

Example: Solve $1+y^2\,\,dx=\,\tan^{-1}y-x\,\,dy.$

Solution: Here, $\frac{dx}{dy}+\frac{x}{1+y^2}=\frac{\tan^{-1}y}{1+y^2}.$

So, $\int P_1\,dy=\int\frac{1}{x+y^2}dy=\tan^{-1}y\quad\therefore\quad\text{I.F.}=e^{\int P_1\,dy}=e^{\tan^{-1}y}.$

Hence the required solution is

$xe^{\tan^{-1}}y=\int e^{\tan^{-1}y}.\frac{\tan^{-1}y}{1+y^2}dy=c.$

or $xe^{\tan^{-1}}y=\int e^t.\,t\,\,dt+c\qquad\left[\text{Put }\tan^{-1}y=t\ \therefore\frac{dy}{1+y^2}=dt\right]$

or $xe^{\tan^{-1}}y=te^t-e^t+c\qquad$ [Integrating by parts]

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or $xe^{\tan^{-1}y} = e^{\tan^{-1}y}\ \tan^{-1}y - 1\ + c$

or $x = \tan^{-1}y - 1 + c\ e^{-\tan^{-1}y}$, dividing by $e^{\tan^{-1}y}$.

Example: Solve $x = \ 1-x^2\ \ dy + \ 2x^2y - y - ax^3\ \ dx = 0.$

Solution: Re-writing the given equation, we have

$$x = \ 1-x^2\ \ \frac{dy}{dx} + y\ \ 2x^2 - 1\ = ax^3 \quad \text{or} \quad \frac{dy}{dx} + \frac{2x^2-1}{x\ 1-x^2}\,y = \frac{ax^2}{1-x^2}. \qquad \ldots(1)$$

Comparing (1) with $\frac{dy}{dx} + Py = Q,$ we have

$$P = \frac{2x^2-1}{x\ 1-x^{\ 2}} = -\frac{1}{x} - \frac{1}{2\ x+1} - \frac{1}{2\ x-1} \quad \text{and} \quad Q = \frac{ax^2}{1-x^2} \qquad \ldots(2)$$

$$\int P\,dx = -\int\left[\frac{1}{x} + \frac{1}{2\ x+1} + \frac{1}{2\ x-1}\right]dx$$

$$= -\left[\log x + \frac{1}{2}\log\ \ x+1\ + \frac{1}{2}\log\ \ x-1\ \right]$$

$$= -\left[\log x + \frac{1}{2}\log\ \ x^2-1\ \right] = -\log\left[x\sqrt{x^2-1}\ \right] = \log\left[x\sqrt{x^2-1}\ \right]^{-1}$$

$\therefore$ Integrating factor $= e^{\int P\,dx} = e^{\log\left[x\sqrt{x^2-1}\ \right]^{-1}} = \left[x\sqrt{x^2-1}\ \right]^{-1}$

Hence the solution of (1) is given by

$$y\,\text{I.F.} = \int Q(\text{I.F.})\,dx + c$$

$$\text{or} \quad \frac{y}{x\sqrt{x^2-1}} = \int \frac{ax^2}{1-x^2} \times \frac{1}{\sqrt{x^2-1}}\,dx + c = c - a\int \frac{x\,dx}{x^2-1^{\ 3/2}}$$

$$= c - \frac{a}{2}\int \frac{dt}{t^{3/2}}, \text{ putting } x^2 - 1 = t$$

$$= c - \frac{a}{2}\left[\frac{t^{-1/2}}{-1/2}\right] = c + \frac{a}{\sqrt{t}} = c + \frac{a}{\sqrt{x^2-1}}$$

or $y = cx\sqrt{x^2-1}\ + ax.$

4.3.8 Equation reducible to linear form (Bernoulli's Equation) :

A differential equation of the form

$$f'\ y\ \frac{dy}{dx} + P\left[f\ \ y\ \right] = Q \qquad \ldots(i)$$

Where P and Q are either constant or function of x only can be reduced to the linear form as follows

Let $f\ \ y\ = v$ $\qquad \ldots(ii)$

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Differentiating (ii), we get

$$f'\ y\ \frac{dy}{dx} = \frac{dv}{dx} \qquad \ldots(iii)$$

Using (ii) and (iii) in (i), we get

$$\frac{dv}{dx} + Pv = Q \qquad \ldots(iv)$$

Bernoulli's Equation: An equation of the form

$$\frac{dy}{dx} + Py = Qy^n \qquad \ldots(i)$$

Where P and Q are constant or function of *x* only and '*n*' is constant except *o* and *l*, is called Bernaulli's Equation.

This is a special case of equations reducible to linear form.

Method of solving :

Divide both side of (i) by y^n, we get

$$y^{-n}\frac{dy}{dx} + Py^{-n+1} = Q \qquad \ldots(ii)$$

Putting $y^{-n+1} = v$ and so $1-n\ y^{-n}\frac{dy}{dx} = \frac{dv}{dx}$

Or, $y^{-n}\frac{dy}{dx} = \frac{1}{1-n}\frac{dv}{dx}$

Thus, equation (ii) reduces

$$\frac{1}{1-n}\frac{dv}{dx} = Pv = Q$$

Or, $\frac{dv}{dx} + 1-n\ Pv = 1-n\ Q$

Hence, I.F. $= e^{1-n\int Pdx}$ and the required solution will be

$$v \times \text{I.F.} = 1-n \int Q \times \text{I.F.}\ dx + c$$

Or, $\boxed{y^{1-n} \times \text{I.F.} = 1-n \int Q \times \text{I.F.}\ dx + c}$

Example: Solve $\left(\frac{dy}{dx}\right) = e^{x-y}\ e^x - e^y$.

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Solution: Re-writing $\left(\frac{dy}{dx}\right) = e^{2x}.e^{-y} - e^{x}$ *or* $\frac{dy}{dx} + e^{x} = e^{2x}.e^{-y}$.

Now dividing by $e^{-y}, e^{y}\frac{dy}{dx} + e^{x}.e^{y} = e^{2x}$.

Putting $e^{y} = v$ so that $e^{y}\frac{dy}{dx} = \frac{dv}{dx}$, we get $\frac{dv}{dx} + e^{x}v = e^{2x}$.

Its I.F. $= e^{\int P\,dx} = e^{\int e^{x}\,dx} = e^{e^{x}}$ and the solution is

$v.e^{e^{x}} = \int e^{2x}.e^{e^{x}}\,dx + c = \int e^{x}.e^{e^{x}}.e^{x}dx + c$

$= \int te^{t}dt + c$ [Put $e^{x} = t$ so that $e^{x}dx = dt$]

$= \int t.e^{t} - \int 1.e^{t}dt + c$ [Integrating by parts]

$= t.e^{t} - e^{t} + c = e^{t}\ t - 1\ + c$

i.e. $e^{y}e^{e^{x}} = e^{e^{x}}\ e^{x} - 1\ + c$ [$\because v = e^{y}$ and $t = e^{x}$]

or $e^{y} = e^{x} - 1 + ce^{-e^{x}}$. [Dividing by $e^{e^{x}}$]

Example: Solve $\frac{dz}{dx} + \frac{z}{x}\log z = \frac{z}{x^{2}}.\ \log z^{\ 2}$.

Solution: Here we have *z* in place of *y* and so the method of solution will remain similar. Dividing by $z\ \log z^{\ 2}$, we get

$$\frac{1}{z\ \log z^{\ 2}}\frac{dz}{dx} = \frac{1}{x}\frac{1}{\log z} = \frac{1}{x^{2}}. \qquad \ldots(1)$$

Putting $\frac{1}{\log z} = v$ so that $\frac{-1}{\log z^{\ 2}}\frac{dz}{dx} = \frac{dv}{dx}$, (1) becomes

$$-\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x^{2}} \text{ or } \frac{dv}{dx} - \frac{1}{x}v = -\frac{1}{x^{2}}, \qquad \ldots(2)$$

Whose I.F.= $e^{-\int 1/x\,dx} = e^{-\log x} = \frac{1}{x}$ and so solution is

$$\frac{v}{x} = \int\left(-\frac{1}{x^{3}}\right)dx + c = \frac{1}{2x^{2}} + c \text{ or } \frac{1}{x\ \log z} = \frac{1}{2x^{2}} + c.$$

Example: $x\left(\frac{dy}{dx}\right) + y\log y = xye^{x}$.

Solution: Dividing by *xy*, the given equation reduces to

$$\frac{1}{y}\frac{dy}{dx} + \frac{1}{x}\log y = e^{x}. \qquad \ldots(1)$$

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Let log$y = v$ so that $\frac{1}{y}\frac{dy}{dx} = \frac{dv}{dx}$. …(2)

Using (2), (1) gives $\frac{dv}{dx} + \frac{1}{x}v = e^x$. …(3)

Comparing (3) with $\frac{dv}{dx} + Pv = Q$, we have

$P = \frac{1}{x}$ and $Q = e^x$.

$\int P\,dx = \int \frac{1}{x}dx = \log x$ so I.F. $e^{\int P\,dx} = e^{\log x} = x$.

Hence solution of (3) is

$v.(\text{I.F.}) = \int Q.\ \text{I.F.}\ dx$ or $vx = \int x e^x\ dx$

or $vx = xe^x - \int e^x\,dx = xe^x - e^x + c$

or $x\log y = e^x\ (x-1) + c$, by (2)

Example: Solve $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$, given $y = 1$ when $x = 1$.

Solution: $2xy\frac{dy}{dx} = x^2 + y^2 + 1$

or $2xy\frac{dy}{dx} - y^2 = 1 + x^2$

Dividing by x, $2y\frac{dy}{dx} - \frac{1}{x}y^2 = \left(\frac{1}{x} + x\right)$ …(1)

Putting $y^2 = v$ so that $2y\left(\frac{dy}{dx}\right) = \frac{dv}{dx}$, …(2)

(1) gives $\frac{dv}{dx} - \frac{1}{x}v = \frac{1}{x} + x$. …(3)

Comparing (3) with $\left(\frac{dv}{dx}\right) + Pv = Q$, here

$P = -\frac{1}{x}$ and $Q = x + \left(\frac{1}{x}\right)$

$\therefore \int P\ dx = -\int \frac{1}{x}\ dx = -\log x$

so I.F. $= e^{\int P\,dx} e^{\log x^{-1}} = x^{-1}$.

and solution of (3) is given by

v I.F. $= \int Q$ I.F. $dx + c$

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or $y^2x^{-1} = \int\left(x+\frac{1}{x}\right)x^{-1}dx = \int\ 1+x^{-2}\ \ dx$

or $y^2x^{-1} = x - x^{-1} + c$

or $y^2 = \ c + x\ \ x - 1.$ …(5)

Given $y = 1$ where $x = 1$. So (5) gives $1 = c + 1 - 1$ or $c = 1$.

Hence, from (5), the required solution is $y^2 = x\ \ x+1\ -1.$

Examples based on Bernaulli's equation:

Example: Solve $\frac{dy}{dx} + y = y^2 \log x.$

Solution: Re-writing, $y^{-2}\frac{dy}{dx} + \frac{1}{x}y^{-1} = \frac{1}{x}\log x.$ …(1)

Putting $y^{-1} = v$ so that $-y^{-2}\frac{dy}{dx} = \frac{dv}{dx}$ and (1) gives

$-\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x}\log x$

or $\frac{dv}{dx} - \frac{1}{x}v = -\frac{1}{x}\log x.$ …(2)

I.F. of (2) $e^{-\int 1/x\ dx} = e^{-\log x} = x^{-1} = \frac{1}{x}.$

So solution of (2) is $vx^{-1} = -\int x^{-2}\log x\,dx + c$

or $y^{-1}x^{-1} = -\left[\log x \times \frac{x^{-1}}{-1} - \int \frac{1}{x}\times\frac{x^{-1}}{-1}dx\right] + c$

or $\frac{1}{y} = \log x + x + 1 + cx.$

Example: Solve $x^3y^2 + xy\ \ dx = dy$.

Solution: Re-writing the given equation, we have

$\frac{dy}{dx} = x^3y^3 + xy$

or $y^{-2}\frac{dy}{dx} - xy^{-1} = x^3.$ …(1)

Putting $y^{-1} = v$ so that $-y^{-2}\left(\frac{dy}{dx}\right) = \frac{dv}{dx},$

(1) gives $-\left(\frac{dv}{dx}\right) - xv = x^3$

or $\left(\frac{dv}{dx}\right) + xv = -x^3,$

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which is linear whose I.F, $= e^{\int x dx} = e^{x^2/2}$.

$\therefore$ Solution is $v e^{x^2/2} = -\int x^3 e^{x^2/2} dx = -\int x^2 e^{x^2/2} x dx$ …(2)

Putting $\frac{x^2}{2} = t$ so that $xdx = dt$, (2) gives

$v e^{x^2/2} = -2\int te^t \, dt = -2\left[te - \int te^t \, dt\right] + c$

$= -2\ te^t - e^t \ + c = -2e^t \ t - 1 \ + c$

$y^{-1} e^{x^2/2} = -2e^{x^2/2}\left[\left(\frac{x^2}{2}\right) - 1\right] + c$ or $y^{-1} = \ 2 - x^2 \ + ce^{-x^2/2}$.

Example: Solve $\frac{dy}{dx} = \ x^2 y^3 + xy \ = 1$.

Solution: Here we proceed as explained in Remark of Article 2.13.

Rewriting, $\frac{dx}{dy} = x^2 y^3 + xy$

or $x^{-2}\frac{dx}{dy} - x^{-1} y = y^3$. …(1)

Putting, $x^{-1} = v$, $\therefore -x^{-2}\frac{dx}{dy} = \frac{dv}{dy}$, (1) becomes

$-\frac{dv}{dy} - yv = y^3$, *i.e.* $\frac{dv}{dy} + yv = -y^3$,

which is linear in v and y and I.F. $= e^{\int y\, dy} = e^{x^2/2}$. So solution is

$v.e^{y^2/2} = -\int y^3 e^{y^2/2} dy + c$

$= -2\int te^t dt + c$ [Put $y^2/2 = t \quad \therefore \ ydy = dt$]

$= -2\left[t.e^t - \int 1.e^t dt + c\right] = -2\ te^t - e^t \ + c.$

or $x^{-1} e^{y^2/2} = -2e^{y^2/2}\left(\frac{y^2}{2} - 1\right) + c \qquad \left[\because v = x^{-1}\right]$

$\frac{1}{x} = 2 - y^2 + ce^{-y^2/2}$.

4.4. Orthogonal trajectory

Trajectory: A curve which cuts every member of a given family and curves in accordance with some given rule is called trajectory of the given family of curves.

- If a curve cuts every member of given family of curves of an angle $\alpha \neq 90°$ is called an ***Oblique Trajectory*** or $\alpha -$ ***Trajectory***.

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- If a curve cuts every member of the given family of curves at right angle *i.e.* $\alpha = 90^\circ$ is called ***Orthogonal Trajectory***.

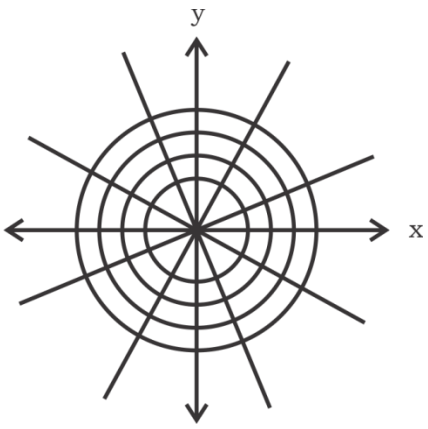
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For example:

Consider two family of curves

$y = mx$ …(1)

and $x^2 + y^2 = a^2$ …(2)



where *m* and *a* are parameters. In the figure, we see that every line given by (1) cuts the family of concentric circles given by (ii) at right angle.

So, $y = mx$ is the orthogonal trajectory of the family of circles $x^2 + y^2 = a^2$.

- When each member of a given family of curves intersects all other member orthogonally, then the given family of curves is known as ***Self Orthogonal.***
- Method for finding the Orthogonal trajectories in Cartesian coordinate

  Let the given family of curve is $f\ (x, y, c\ ) = 0$ …(1)

Step 1 Find the differential equation of (1)

$\phi\left(x, y, \frac{dy}{dx}\right) = 0$ [say] …(2)

Step 2 For orthogonal trajectories replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ *i.e.* differential equation of orthogonal trajectory is

$\psi\left(x, y, \frac{dx}{dy}\right) = 0$ [say] …(3)

Step 3 Solving the differential equation (3), we get some function $h\ (x, y, k\ ) = 0$, where *k* is an arbitrary constant which is the required orthogonal family of curves or Orthogonal trajectory.

Example: Find the orthogonal trajectory of the family of curve $x^2 + y^2 = a^2$

Solution: Give family of curve is $x^2 + y^2 = a^2$ …(1)

First we find the differential equation of (1)

Differentiating with respect to *x*, we get

$2x + 2y\frac{dy}{dx} = 0$

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$\Rightarrow x + y\frac{dy}{dx} = 0$ …(2)

Second, we replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, then we get from (2)

$x - y\frac{dx}{dy} = 0$

$\Rightarrow xdy - ydx = 0$ …(3)

Now, for the required family of orthogonal family of curve, we solve (3)

From (3), we have $\frac{dy}{y} = \frac{dx}{x}$

Integrating, we get $\log y = \log x + \log c$, where $\log c$ is constant of integration

or $\boxed{y = cx}$

so, the orthogonal trajectory is $y = cx$.

Example: Find the orthogonal trajectory of $y = cx^2$.

Solution: The given family of curve is $y = cx^2$ …(1)

Differentiating with respect to *x*, we get

$\frac{dy}{dx} = 2cx$ …(2)

From (1),

$c = \frac{y}{x^2}$

so, (2) becomes

$\frac{dy}{dx} = \frac{2y}{x}$ …(3)

Now, for orthogonal trajectory we replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, so (3) becomes

$-\frac{dx}{dy} = \frac{2y}{x}$

$\Rightarrow xdx + 2ydy = 0$ …(4)

Integrating (4), we get

$\frac{x^2}{2} + y^2 = c^2$, where *c* is constant of integration

$\Rightarrow x^2 + 2y^2 = 2c^2$

or, it can be written as

$\Rightarrow x^2 + 2y^2 = R^2$ where R is arbitrary constant and $R^2 = 2c^2$.

- Method for finding the orthogonal trajectories in Polar coordinate

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Let the given family of cardiod is $f\ \ r,\theta,c\ =0\ \ \ldots(1)$

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Step 1 Find the differential equation of (1)

$$\phi\left(r,\theta,\frac{dr}{d\theta}\right)=0 \qquad \ldots(2)$$

Step 2 For orthogonal trajectory replace $\frac{dr}{d\theta}$ by $-r^2\frac{d\theta}{dr}$ *i.e.* differential equation of orthogonal trajectory is

$$\psi\left(r,\theta,\frac{d\theta}{dr}\right)=0 \qquad \ldots(3)$$

Step 3 Solving the differential equation (3), we get some function $h\ r,\theta,k\ =0$, where k is an arbitrary constant which is the required orthogonal trajectory.

Example: Find the orthogonal trajectory of cardiod of $r=a\ 1-\cos\theta$.

Solution: The given family of curve is $r=a\ 1-\cos\theta$ …(1)

Differentiating with respect to θ, we get

$$\frac{dr}{d\theta}=a\sin\theta$$

or $a=coec\theta\frac{dr}{d\theta}$

Putting it in (1), we get

$$r=\text{cosec}\,\theta\frac{dr}{d\theta}\ 1-\cos\theta$$

or $r=\frac{dr}{d\theta}\left(\frac{1-\cos\theta}{\sin\theta}\right)$ …(2)

For orthogonal trajectory, we replace $\frac{dr}{d\theta}$ by $-r^2\frac{d\theta}{dr}$ in (2) and get

$$r=-r^2\frac{d\theta}{dr}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

or, $\frac{dr}{r}=-\frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}.\cos\frac{\theta}{2}}d\theta$

or $\frac{dr}{r}=-\tan\frac{\theta}{2}d\theta$

or $\frac{dr}{r}+\tan\frac{\theta}{2}d\theta=0$ …(3)

Integrating (3), we get

$\ln r+2\ln\left(\sec\frac{\theta}{2}\right)=\ln c,$, where $\ln c$ is constant of integration.

or, $r\sec^2\frac{\theta}{2}=c$

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or $r = c\ \cos^2\frac{\theta}{2}$

or $r = c\ \ 1+\cos\theta\ \ \ r = c\ \ 1+\cos\theta$

4.5 <u>Higher order differential equation</u>:

LINEAR DIFFERENTIAL EQUATION WITH CONSTANT CO-EFFICIENTS:

A differential equation of the form

$$a_0\frac{d^n y}{dx^n}+a_1\frac{d^{n-1}y}{dx^{n-1}}+a_2\frac{d^{n-2}y}{dx^{n-2}}+...+a_{n-1}\frac{dy}{dx}+a_n y = x \qquad \ldots(1)$$

where, $a_0 \neq 0,\ \ a_0, a_1, ...a_n \in \mathbb{R}$ and x is a constant or function of x only, is called linear differential equation with constant co-efficient of order n.

Let $D^n\ \ y\ = \frac{d^n y}{dx^n}$

[D is an operator and the index of D indicates the number of times the operator of differentiate has been carried out. Also the negative index of D is equivalent to an integration. *i.e.* $D^{-1}x = \int xdx$ and $D^{-2}x = \int\ \int xdx\ \ dx$ and so on]

Then the differential equation (1) can be re-written as

$$a_0D^n+a_1D^{n-1}+a_2D^{n-2}+...+a_{n-1}D+a_n\ \ y = x\ \ \ a_0D^n+a_1D^{n-1}+a_2D^{n-2}+...+a_{n-1}D+a_n\ \ y = x$$

or $f\ \ D\ \ y = x \qquad \ldots(2)$

- **The complete primitive or general solution of (1) or (2) is**
  $y = C.F. + P.I.$ **[*C.F.*→complementary function and P.I. → Particular Integral]**
- **Complementary function is the solution of equation**
  $f\ \ D\ \ y = 0$
- **Particular integral** $= \frac{1}{f\ \ D}X.$

<u>Working Rule for *C.F.* :</u>

For the C.F. we have to find solution of the equation

$f\ \ D\ \ y = 0$

We construct an **auxiliary equation**

$f\ \ D\ \ = 0$

for solving the auxiliary equation we shall get n roots and according to the nature of roots we can write the complementary function of $f\ \ D\ \ y = x$ as per the table given below.

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S. No.	Roots of $f(D)=0$	C.F.
1.	If all the roots are distinct.	$C.F.=c_1e^{m_1x}+c_2e^{m_2x}+\ldots+c_ne^{m_nx}$
2.	If any root m_k is repeated r times and root are different.	$C.F.=\left(c_1+c_2x+c_3x^2+\ldots+c_rx^{r-1}e^{m_kx}\right)+c_{r+1}e^{m_1x}+c_{r+2}e^{m_2x}$
3.	If the roots are complex say, $\alpha\pm i\beta$	$C.F.=c_1e^{(\alpha+i\beta)x}+c_2e^{(\alpha-i\beta)x}$ $=e^{\alpha x}\left(c_1\cos\beta x+c_2\sin\beta x\right)$ or $C.F.=c_1e^{\alpha x}\cos\left(\beta x+c_2\right)$ or $C.F.=c_1e^{\alpha x}\sin\left(\beta x+c_2\right)$
4.	If the complex roots $\alpha+i\beta$ repeated two times	$C.F.=e^{\alpha x}\left[\left(c_1+c_2\right)\cos\beta x+\left(c_3+c_4x\right)\sin\beta x\right]$
5.	If roots surds say, $\alpha\pm\sqrt{\beta}$	$C.F.=e^{\alpha x}\left[c_1\cos h\sqrt{\beta}x+c_2\sin h\sqrt{\beta}x\right]$

Example: Solve $D^4-81\ y=0$

Solution: Here auxiliary equation is $D^4-81=0$. …(1)

$\because D^4-81=\ D^2+9\ \ D^2-9\ =\ D+3i\ \ D-3i\ \ D+3\ \ D-3$.

Hence (1) gives $D=\pm 3i,\ 3,\ -3$.

Now the part of C.F. corresponding to unequal roots 3, –3 is $c_1e^{3x}+c_2e^{-3x}$.

Again the part of C.F. corresponding to imaginary root $\pm 3i$ is given by ($\because$ here $\alpha=0$ and $\beta=3$).

Hence the general solution is $y=c_1e^{3x}+c_2e^{-3x}+c_3\cos 3x+c_4\sin 3x,$ where c_1,c_2,c_3 and c_4 are arbitrary constants.

Example: Solve $\frac{d^3y}{dx^3}-4\frac{d^2y}{dx^2}+5\frac{dy}{dx}-2y=0.$

Solution: If $D=\frac{d}{dx}$, the given equation in symbolic form is given by $D^3-4D^2+5D-2\ \ y=0.$

$\therefore$ Auxiliary equation is $D^3-4D^2+5D-2\ \ y=0.$ …(1)

$D^3-4D^2+5D-2=D^2\ \ D-1\ -3D\ \ D-1\ +2\ \ D-1$

$=\ D-1\ \ D^2-3D+2\ =\ D-1\ \ D-1\ \ D-2$.

$\therefore$ The roots of (1) are 1, 1, 2 and required solution is $y=\ c_1+c_2x\ e^x+c_3e^{2x}$, c_1, c_2 c_3 are arbitrary constants.

Method of finding P.I.:

The given differential equation is $f\ \ D\ \ y=X$.

Then, $P.I.=\frac{1}{f\ \ D}X.[\frac{1}{f\ \ D}$ is the inverse operator of $f\ \ D\ ]$

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1. General Method:

General method of $f\ D$ can be resolved in linear factors, the

$$P.I. = \frac{1}{f\ D} X$$

$$= \frac{1}{D-\alpha_1\ \ D-\alpha_2\ \ ...\ D-\alpha_n} X$$

$$= \left[\frac{A_1}{D-\alpha_1} + \frac{A_2}{D-\alpha_2} + ... + \frac{A_n}{D-\alpha_n}\right] X$$

This *P.I.* can be find out by the help of the given two formulas.

(i) $\frac{1}{D-\alpha} X = e^{\alpha x} \int X\, e^{-\alpha x} dx$ and

(ii) $\frac{1}{D+\alpha} X = e^{-\alpha x} \int X\, e^{-\alpha x} dx.$

Example: Solve $D^2 + a^2\ \ y = \tan\ ax.$

Solution: Here auxiliary equation is $D^2 + a^2 = 0.$ So $D = \pm\ ia.$

$\therefore\ C.F. = c_1 \cos ax + c_2 \sin ax.$

Now $P.I. = \frac{1}{D^2 + a^2} \tan ax = \frac{1}{D+ai\ \ D-ai} \tan ax$

$$\therefore P.I. = \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right] \tan ax. \qquad \ldots(1)$$

Now, $\frac{1}{D-ia} \tan ax = e^{iax} \int e^{-iax} \tan ax\ dx$

$= e^{iax} \int \cos ax - i \sin ax\ \frac{\sin ax}{\cos ax} dx,$ by Euler's theorem

$$= e^{iax} \int \left[\sin ax - i \frac{1-\cos^2 ax}{\cos ax}\right] dx$$

$$= e^{iax} \int \left[\sin\ ax - i\ \sec\ ax - \cos\ ax\ \right] dx$$

$$= e^{iax} \left[-\frac{\cos ax}{a} - \frac{i}{a} \log \tan\left(\frac{\pi}{4} + \frac{ax}{2}\right) + i. \frac{\sin ax}{a}\right]$$

$$= -\frac{e^{iax}}{a}\left[\cos ax - i \sin ax\ + i \log \tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right]$$

$$= -\frac{e^{iax}}{a}\left[e^{-iax} + i \log \tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right],$$ by Euler's theorem

Thus, $\frac{1}{D-ia} \tan ax = -\frac{1}{a}\left[1 + ie^{iax} \log \tan\left(\frac{\pi}{4} + \frac{ax}{2}\right)\right].$ $\qquad \ldots(2)$

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Replacing i by $-i$ in (2), we get

$$\frac{1}{D-ia}\tan ax = -\frac{1}{a}\left[1-ie^{-iax}\log\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right]. \qquad \ldots(3)$$

$\therefore$ From (1), (2) and (3), we get

$$P.I. = \frac{1}{2ia}\left\{-\frac{i}{a}\ e^{iax}+e^{-iax}\ \log\tan\left(\frac{\pi}{4}+\frac{ax}{2}\right)\right\}$$

$$= -\left(\frac{1}{a^2}\right)\cos ax\log\tan\left(\frac{1}{4}\pi+\frac{1}{2}ax\right).$$

$\therefore$ Solution is $y = C.F.+P.I.$

or $y = c_1\cos ax + c_2\sin ax - \left(\frac{1}{a^2}\right)\cos ax\log\tan\left(\frac{1}{4}\pi+\frac{1}{2}ax\right).$

Example: Solve $D^2+a^2\ \ y=\sec ax.$

Solution: Here auxiliary equation is $D^2+a^2=0.$ So $D=\pm ia.$

$\therefore\ C.F. = c_1\cos ax + c_2\sin ax$

$$P.I. = \frac{1}{D^2-a^2}\sec ax = \frac{1}{2ia}\left[\frac{1}{D-ia}-\frac{1}{D+ia}\right]\sec ax. \qquad \ldots(1)$$

Now, $\frac{1}{D-ia}\sec ax = e^{iax}\int e^{-iax}\sec ax\ dx$

$$= e^{iax}\int \cos ax - i\sin ax\ .\frac{1}{\cos ax}dx$$

$$= e^{iax}\int 1-i\tan ax\ \ dx$$

$$= e^{iax}\left\{x+\left(\frac{i}{a}\right)\log\cos ax\right\}. \qquad \ldots(2)$$

Replacing i by $-I$ in (2), we get

$$\frac{1}{Dia}\sec ax = e^{-iax}\left[x-\left(\frac{i}{a}\right)\log\cos ax\right]. \qquad \ldots(3)$$

$\therefore$ From (1), (2) and (3), we get

$$P.I. = \left(\frac{1}{2ia}\right)\left[e^{iax}\left\{x+\left(\frac{i}{a}\right)\log\cos ax\right\}-e^{iax}\left\{x-\left(\frac{i}{a}\right)\log\cos ax\right\}\right]$$

$$= x\frac{e^{iax}-e^{-iax}}{2ia}+\frac{1}{a^2}\ \log\cos ax\ .\frac{e^{iax}+e^{-iax}}{2}$$

$$= \left(\frac{x}{a}\right)\sin ax+\left(\frac{1}{a^2}\right)\cos ax\log\cos ax$$

$\therefore$ Solution is $y = C.F.+P.I.$

i.e., $y = c_1\cos ax + c_2\sin ax + \left(\frac{x}{a}\right)\sin ax + \left(\frac{1}{a^2}\right)\cos ax\log\cos ax.$

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2. Shortcut Method:

Here we discuss different short methods of finding P.I. which are much shorter than the general methods.

Rule I : When X is of the form e^{ax}

$$P.I. = \frac{1}{f(D)} e^{ax} = \frac{e^{ax}}{f(a)}, \text{ where } f(a) \neq 0$$

When $f(a) = 0$

$$P.I. = \frac{1}{f(D)} e^{ax}$$

$$= \frac{1}{\phi(D)(D-a)^r} e^{ax}$$

$$= \frac{1}{\phi(a)} \cdot \frac{1}{(D-a)^r} e^{ax}$$

$$= \frac{1}{\phi(a)} \cdot \frac{x^r}{r!} e^{ax}$$

Example: Solve $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^x$.

Solution: The given equation in symbolic form is

$(D^2 - 3D + 2)\, y = e^x$, whose auxiliary equation is

$D^2 - 3D + 2 = 0$ *i.e.* $(D-1)(D-2) = 0$. So $D = 1, 2$.

$\therefore C.F.$ is $c_1 e^x + c_2 e^x$.

$$P.I. = \frac{1}{(D-1)(D-2)} e^x = \frac{1}{D-1} \frac{1}{1-2} e^x \text{ (Using formula I)}$$

$$= -\frac{x}{1!} e^x \text{ (by formula II)}$$

$\therefore$ Solution is $y = c_1 e^x + c_2 e^{2x} - xe^x$.

Example: Solve $(D^3 - 5D^2 + 7D - 3)\, y = e^{2x} \cosh x$.

Solution: The auxiliary equation is $D^3 - 5D^2 + 7D - 3 = 0$.

or $(D^2)(D-1) - 4D(D-1) + 3(D-1) = 0$

or $(D-1)(D^2 - 4D + 3) = 0$

or $(D-1)(D-1)(D-3) = 0$.

$\therefore D = 1, 1, 3$ and $C.F. = (c_1 + c_2 x)\, e^x + c_3 e^{2x}$.

$$P.I. = \frac{1}{(D-1)^2 (D-3)} e^{2x} \cosh x = \frac{1}{(D-1)^2 (D-3)} e^{2x} \frac{e^x + e^{-x}}{2}$$

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$$= \frac{1}{D-1^{\,2}\;\; D-3}\left(\frac{1}{2}e^{3x}+\frac{1}{2}e^{x}\right).$$

Now *P.I.* for $\frac{1}{2}e^{x} = \frac{1}{2}\frac{1}{D-1^{\,2}\;\; D-3}e^{x}$

$= \frac{1}{2}\frac{1}{D-1^{\,2}}\frac{1}{1-3}e^{x}$ (Using formula 1)

$= \frac{1}{2\;\;-2}\frac{1}{D-1^{\,2}}e^{x} = \frac{1}{4}\times\frac{x^2}{2!}e^{x}$ (by formula 2)

P.I. corresponding to $\frac{1}{2}e^{3x} = \frac{1}{2}\frac{1}{D-1\;\; D-1^{\,2}}e^{3x}$

$= \frac{1}{2\;\; D-3}\frac{1}{4}e^{3x} = \frac{1}{8}\cdot\frac{x}{1!}e^{3x} = \frac{x}{8}e^{3x}$

$\therefore$ Solution is. $y = \; c_1 + c_2\; x\;\; e^{x} + c_3 e^{2x} - \left(\frac{1}{8}\right)x^2 e^{x} + \left(\frac{1}{8}\right)xe^{3x}$

Example: Solve $\left(\frac{d^3y}{dx^3}\right) - y = \; e^{x}+1^{\,2}$

Solution: Re-writing the given equation in symbolic form, we get

$$D^3-1\;\; y = e^{2x}+2e^{x}+1, \qquad \ldots(1)$$

Whose auxiliary equation is $D^3-1=0$

or $\;\; D-1\;\;\; D^2+D+1\;\; = 0$

so that $D=1, D=\frac{-1\pm i\sqrt{3}}{2}$

$\therefore C.F = c_1 e^{x} + e\left(-\frac{1}{2}\right)\left[c_2\cos\left(\frac{\sqrt{3}}{2}\right)x + c_3\sin\left(\frac{\sqrt{3}}{2}\right)x\right]. \therefore C.F = c_1 e^{x}$

P.I. corresponding to $1 = \frac{1}{D^3-1}1 = \frac{1}{D^3-1}e^{0.x} = \frac{1}{-1}e^{0.x} = -1,$

P.I. corresponding to $e^{2x} = \frac{1}{D^3-1}e^{2x} = \frac{1}{8-1}e^{2x} = \frac{1}{7}e^{2x}$

and *P.I.* corresponding to

$$2e^{x}\;\; 2\frac{1}{D-1}\cdot\frac{1}{D^2+D+1}e^{x} = 2\frac{1}{D-1\,3}\frac{1}{3}e^{x}$$

$= \frac{2}{3}\frac{x}{1!}e^{x} = \frac{2x}{3}e^{x}.$

$\therefore$ Required solution is given by

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$$y = c_1 e^x + e^{\left(-\frac{1}{2}\right)x}\left[c_2 \cos\left(\frac{\sqrt{3}}{2}\right)x + c_3 \sin\left(\frac{\sqrt{3}}{2}\right)x\right] - 1$$

$$+\left(\frac{1}{7}\right)e^{2x} + \left(\frac{2x}{3}\right)e^x$$

Rule II: When X is cos *ax* or sin *ax*

$$P.I. = \frac{1}{f\ D}\cos ax = \frac{1}{\phi\ D^2}\cos ax = \frac{1}{\phi\ -a^2}\cos ax; \phi\ -a^2\ \neq 0$$

when, $\phi\ -a^2\ = 0$, then

$$\frac{1}{D^2 + a^2}\sin ax = \frac{-x}{2a}\cos ax, and$$

$$\frac{1}{D^2 + a^2}\sin ax = \frac{x}{2a}\sin ax.$$

Example: Solve $\left(\frac{d^4 y}{dx^4}\right) - m^4\ y = \sin mx$

Solution: Given $D^4 - m^4\ \ y = \sin mx, D = \frac{d}{dx}$ …(1)

whose auxiliary equation is $D^4 - m^4 = 0$ giving $D = \pm m, \pm im$

so, $C.F = c_1 e^{mx} + c_2 e^{-mx} + c_3 \cos mx + c_4 \sin mx$

$$P.I. = \frac{1}{D^2 + m^2}\frac{1}{D^2 - m^2}\sin mx = \frac{1}{D^2 + m^2}\frac{1}{-2x^2}\sin mx$$

$$= \frac{1}{-2m^2}\frac{1}{D^2 + m^2}\sin mx = -\frac{1}{2m^2} \times \left(-\frac{x}{2m}\cos mx\right) = \frac{x}{4m^3}\cos mx$$

So solution is $y = c_1 e^{mx} + c_2 e^{-mx} + c_3 \cos mx + c_4 \sin mx + \left(\frac{x}{4m^3}\right)\cos mx$

Example: Find the solution of $\left(\frac{d^2 y}{dx^2}\right) + 4y = 8\cos 2x$ given that $y = 0$ and $\frac{dy}{dx} = 0,$ when $x = 0$

Solution: Re-writing the given equation in symbolic form, we have

$D^2 + 4\ \ y = 8\cos\ 2x.$ …(1)

Also given that when $x = 0, y = 0$ …(2)

And when $x = 0, \frac{dy}{dx} = 0.$ …(3)

The auxiliary equation of (1) is

$D^2 + 4 = 0$ so that $D = \pm 2i$

So $C.F = c_1 \cos 2x + c_2 \sin 2x.$

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Also, $P.I. = 8\frac{1}{D^2+2^2}\cos 2x = 8.\frac{x}{2\cdot 2}\sin 2x = 2x\sin 2x$

Hence the general solution of (1) is

$y = c_1\cos 2x + c_2\sin 2x + 2x\sin 2x.$ …(4)

Putting $x = 0$ and $y = 0$ (due to (2)), (4) yields $c_1 = 0$. Then (4) reduces to

$y = c_2\sin 2x + 2x\sin 2x = c_2 + 2x\ \sin 2x$ …(5)

From (5), $\frac{dy}{dx} = 2\sin 2x + 2\ c_2 + 2x\ \cos 2x.$ …(6)

Putting x = 0 and $\frac{dy}{dx} = 0$ (due to (3)), (6) yields $0 = 2c_2 \Rightarrow c_2 = 0.$

Hence from (5), the required solution is $y = 2x\sin 2x$

Example: Solve $D^2 + a^2\ \ y = \sin ax$

Solution: Here the auxiliary equation is $D^2 + a^2 = 0$

i.e. $D = \pm ia$ so $C.F = c_1\cos ax + c_2\cos ax$

$P.I. = \frac{1}{D^2+a^2}\sin ax = \frac{-x}{2a}\cos ax.$

$\therefore$ Solution is $y = c_1\cos ax + c_2\sin a\,x - \left(\frac{x}{2a}\right)\cos ax.$

Example: Solve $D^2 + 4\ \ y = \sin^2 x,$

Solution: Here auxiliary equation is $D^2 + 4 = 0$ so that $D = \pm 2i.$

Hence $C.F. = \cos 2x + c_2\sin 2x.$

Also

$P.I. = \frac{1}{D^2+4}\sin^2 x = \frac{1}{2}\frac{1}{D^2+4}\ 1 - \cos 2x$

$= \frac{1}{2}\left[\frac{1}{D^2+4}e^{0.x} - \frac{1}{D^2+4}\cos 2x\right]$

$= \frac{1}{2}\left[\frac{1}{0+4}e^{0.x} - \left(\frac{x}{2\times 2}\right)\sin 2x\right] = \frac{1}{8} - \frac{1}{8}x\sin 2x.$

$\therefore$ Solution is $y = c_1\cos 2x + c_2\ \sin 2x + \left(\frac{1}{8}\right) - \left(\frac{x}{8}\right)\sin 2x.$

Example: Solve $D^3 + a^2\,D\ \ y = \sin ax$

Solution: Here the auxiliary equation is

$D^3 + a^2 D = 0$

or $D\ \ D^2 + a\ \ = 0$

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$\therefore D = 0 \pm ia,$

and $C.F. = c_1 e^{0.x} + c_2 \cos ax + c_3 \sin ax = c_1 + c_2 \cos ax + c_3 \sin ax$

$$P.I. = \frac{1}{D^2 + a^2 \ D} \sin ax = \frac{1}{D^2 + a^2} \left[\frac{1}{D} \sin ax \right]$$

$$= \frac{1}{D^2 + a^2} \left(-\frac{\cos ax}{a} \right) \qquad \left[\because \frac{1}{D} \text{means integration} \right]$$

$$= -\frac{1}{a} \times \frac{1}{D^2 + a^2} \cos ax = -\frac{1}{a} \times \frac{x}{2a} \sin ax$$

So solution is $y = c_1 + c_2 \cos ax + c_3 \sin ax - \left(\frac{x}{2a^2} \right) \sin ax.$

Rule III: When X is x^M or a polynomial of degree M.

$$P.I. = \frac{1}{f \ d} x^M$$

$$= \frac{1}{aD^r \left[1 \pm F \ D \right]^n} x^M \quad \text{[Where } D^r \text{ is the lowest degree term from}$$

$f \ D$ and a is any constant]

$$\frac{1}{aD^r} \left[1 \pm F \ D \right]^{-n} x^M$$

$$\boxed{1 + x^{\ n} = 1 + nx + \frac{1 \ n-1}{2!} x^2 + \frac{1 \ n-1 \ \ n-2}{3!} x^3 + \ldots}$$

Example: Solve $D^3 - D^2 - 6D \ \ y = x^2 + 1.$

Solution: The auxiliary equation is $D \ \ D^2 - D - 6 \ = 0$

or $D \ \ D - 3 \ \ \ D + 2 \ = 0$

$\therefore D = 0, 3, -2$

and $C.F. = c_1 e^{0.x} + c_2 e^{3x} + c_3 e^{-2x} = c_1 + c_2 e^{3x} + c_3 e^{-2x}$

$$P.I. = \frac{1}{D^3 - D^2 - 6D} \ \ x^2 + 1 = \frac{1}{-6D \left(1 + \frac{D}{6} - \frac{D^2}{6} \right)} \ \ x^2 + 1$$

$$= -\frac{1}{6D \left[1 + \left(\frac{D}{6} - \frac{D^2}{6} \right) \right]^{-1}} \ \ x^2 + 1$$

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$$=-\frac{1}{6D}\left[1-\left(\frac{D}{6}-\frac{D^2}{6}\right)+\left(\frac{D}{6}-\frac{D^2}{6}\right)^2-...\right]\ x^2+1$$

$$=-\frac{1}{6D}\left[1-\frac{D}{6}+\frac{7}{36}D^2+...\right]\ x^2+1$$

$$=-\frac{1}{6D}\left[x^2+1-\frac{1}{6}D\ x^2+1\ +\frac{7}{36}D^2\ \ x^2+1\ +...\right]$$

$$=-\frac{1}{6D}\left[x^2+1-\frac{1}{6}\ 2x+0\ +\frac{7}{36}\ 2+0\ +...\right]$$

$$=-\frac{1}{6D}\left[x^2-\frac{1}{3}x+\frac{25}{18}\right]=-\frac{1}{6}\left[\frac{x^3}{3}-\frac{x^2}{6}+\frac{25}{18}x\right].$$

Hence the required solution is $v=C.F.+P.I.$

i.e. $y=c_1+c_2e^{3x}+c_3{}^{-2x}-\left(\frac{1}{18}\right)\left(x^3-\frac{x^2}{2}+\frac{25x}{6}\right).$

Example: Solve $D^3+8\ y=x^4+2x+1.$

Solution: The auxiliary equation is $D+2\ \ D^2-2D+4\ =0$

so $D=-2,1\pm i\sqrt{3}$

$\therefore C.F.=c_1e^{-2x}+c_2e^{x}\ \cos x\sqrt{3}+c_3$

$$P.I.=\frac{1}{D^3+8}\ x^4+2x+1=\frac{1}{8\left(1+\frac{1}{8}D^3\right)}\ x^4+2x+1$$

$$=\frac{1}{8}\left(1+\frac{1}{8}D^3\right)^{-1}\ x^4+2x+1=\frac{1}{8}\left(1-\frac{1}{8}D^3+...\right)\ x^4+2x+1$$

$$=\frac{1}{8}\left[\ x^4+2x+1\ -\frac{1}{8}D^3\ x^4+2x+1\ \right]$$

$$=\frac{1}{8}\left[\ x^4+2x+1\ -3x\right]=\frac{1}{8}\ x^4-x+1$$

$$\therefore y=c_1e^{-2x}+c_2e^{x}.\ \cos x\sqrt{3}+c_3\ +\left(\frac{1}{8}\right)\ x^4-x+1\ .$$

Rule IV: When $X=e^{ax}.V,$ where V is function of x.

$$P.I.=\frac{1}{f\ \ D}e^{ax}.V$$

$$=e^{ax}\frac{1}{f\ \ D+a}V.$$

Example: Solve $D^2+3D+2\ y=e^{2x}\sin x$

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Solution: Auxiliary equation $D^2+3D+2=0$ gives $D=-2,-1.$

So, we have $C.F.=c_1e^{-2x}+c_2e^{-x}$

$$P.I.=\frac{1}{D^2+3D+2}e^{2x}\sin x=e^{2x}\frac{1}{D+2\ ^2+3\ D+2\ +2}\sin x$$

$$=e^{2x}\frac{1}{D^2+7D+12}\sin x=e^{2x}\frac{1}{-1^2+7D+12}\sin x$$

$$=e^{2x}\frac{11-7D}{11-7D\ \ 11+7D}\sin x=e^{2x}\ 11-7D\ .\frac{1}{121-49D^2}\sin x$$

$$=e^{2x}\ 11-7D\ .\frac{1}{121+49}\sin x=\frac{e^{2x}}{170}\ 11\sin x-7\cos x\ .$$

$$\therefore y=c_1e^{2x}+c_2e^{-2x}+\left(\frac{1}{170}\right)e^{2x}\ 11\sin x-7\cos x\ .$$

Example: $D^2-2D+1\ y=x^2e^{3x}$

Solution: Here $A.E.=D^2-2D-2D+1=0\Rightarrow D=1,1.$

so $C.F.=\ c_1+c_2\ x\ e^x.$

$$P.I.=\frac{1}{D-1\ ^2}x^2e^{3x}=e^{3x}\frac{1}{D+3-1}x^2=\frac{1}{4}e^{3x}\frac{1}{\left(1+\frac{D}{2}\right)^2}x^2$$

$$=\frac{1}{4}e^{3x}\left(1+\frac{D}{2}\right)^{-2}x^2=\frac{1}{4}e^{3x}\left[1-2\left(\frac{D}{2}\right)+3\left(\frac{D}{2}\right)^2+...\right]x^2$$

$$=\frac{1}{4}e^{3x}\left[x^2-Dx^2+\frac{3}{4}D^2x^2\right]=\frac{1}{4}e^{3x}\left[x^2-2x+\frac{3}{2}\right].$$

$$\therefore y=\ c_1+c_2x\ e^x+\left(\frac{e^{3x}}{4}\right)\left(x^3-2x+\frac{3}{2}\right)$$

Example: Solve $D^2-1\ y=\cosh x\cos x$

Solution: Given $D^2-1\ y=\cosh x\cos x=\frac{1}{2}\ e^x+e^{-x}\ \cos x,\ \ldots(1)$

Here auxiliary equation is $D^2-1=0$, giving $D=1,-1.$

Hence $C.F.=c_1e^x+c_2e^{-x}.$

P. I. corresponding to $\left(\frac{1}{2}\right)e^x\cos x$

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$$= \frac{1}{2}\frac{1}{D^2-1}e^x \cos x = \frac{1}{2}e^x \frac{1}{D+1^{\,2}-1}\cos x$$

$$= \frac{1}{2}e^x \frac{1}{D^2+2D}\cos x = \frac{1}{2}e^x \frac{1}{-1+2D}\cos x$$

$$= \frac{e^x}{2}\ 2D+1\ .\frac{1}{2D+1\ \ 2D-1}\cos x$$

$$= \frac{e^x}{2}\ 2D+1\ \cdot\frac{1}{4D^2-1}\cos x = \frac{e^x}{2}\ 2D+1\ \frac{1}{-4-1}\cos x$$

$$= -\frac{e^x}{10}\ 2D+1\ \cos x = -\frac{e^x}{10}\ -2\sin x+\cos x\ .$$

P. I. corresponding to $\left(\frac{1}{2}\right)e^{-x}\cos x$

$$= \frac{1}{2}\frac{1}{D^2-1}e^{-x}\cos x = \frac{1}{2}e^{-x}\frac{1}{D-1^{\,2}-1}\cos x$$

$$= \frac{e^{-x}}{2}\frac{1}{D^2-2D}\cos x = \frac{e^{-x}}{2}\frac{1}{-1-2D}\cos x$$

$$= -\frac{e^{-x}}{2}\ 2D-1\ \frac{1}{2D-1\ \ 2D+1}\cos x$$

$$= -\frac{e^{-x}}{2}\ 2D-1\ \frac{1}{4D^2-1}\cos x = \frac{e^{-x}}{2}\ 2D-1\ \frac{1}{-4-1}\cos x$$

$$= \frac{e^{-x}}{2}\ 2D-1\ \cos x = \frac{e^{-x}}{10}\ -2\sin x-\cos x\ .$$

$$\therefore y = c_1 e^x + c_2 e^{-x} - \left(\frac{1}{10}\right)e^x\ -2\sin x-\cos x$$

$$+\left(\frac{1}{10}\right)e^{-x}\ -2\sin x-\cos x$$

Or $y = c_1 e^x + c_2 e^{-x} + \frac{2}{5}\sin x\left(\frac{e^x-e^{-x}}{2}\right) - \frac{1}{5}\cos x\left(\frac{e^x+e^{-x}}{2}\right)$

Or $y = c_1 e^x + c_2 e^{-x} + \left(\frac{2}{5}\right)\sin x\sinh x - \left(\frac{1}{5}\right)\cos x\cosh x.$

Example: Solve $D^4+D^2+1\ y = e^{-x/2}\cos\left(\frac{x\sqrt{3}}{2}\right).$

Solution: The auxiliary equation is $D^4+D^2+1\ =0$ *or* $D^2+1^{\,2}-D^2-0$

Or $D^2+D+1\ \ D^2-D+1\ =0$

so that D^2+D+1 or $D^2-D+1=0$

giving $D=\frac{-1\pm i\sqrt{3}}{2}, D=\frac{1\pm i\sqrt{3}}{2}$

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$$\therefore C.F. = e^{-x/2}\left[c_1 \cos\left(\frac{x\sqrt{3}}{2}\right) + c_2 \sin\left(\frac{x\sqrt{3}}{2}\right)\right]$$
$$+ e^{x/2}\left[c_3 \cos\left(\frac{x\sqrt{3}}{2}\right) + c_4 \sin\left(\frac{x\sqrt{3}}{2}\right)\right]$$

$$P.I. = \frac{1}{D^4 + D^2 + 1} e^{-x/2} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{\left(D - \frac{1}{2}\right)^4 + \left(D - \frac{1}{2}\right)^2 + 1} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{D^4 - 4D^3.\frac{1}{2} + 6D^2\left(\frac{1}{2}\right)^2 - 4D\left(\frac{1}{2}\right)^3 + \frac{1}{2}^4 + D^2 - D + \left(\frac{1}{4}\right) + 1} \times \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{D^4 - 2D^3 + \left(\frac{5}{2}\right)D^2 - \left(\frac{3}{2}\right)D + \left(\frac{21}{16}\right)} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{\left[D^2 + \left(\frac{3}{4}\right)\right]\left[D^2 - 2D + \left(\frac{7}{4}\right)\right]} \cos\left(\frac{x\sqrt{3}}{2}\right)$$ [Since D^r is zero when

$D^2 = -\frac{3}{4}$ so $D^2 + \left(\frac{3}{4}\right)$ must be factor of the denominator]

$$= e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{4}\right)} . \frac{1}{\left(\frac{-3}{4}\right) - 2D + \left(\frac{7}{4}\right)} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{4}\right)} \frac{1}{1 - 2D} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{4}\right)} . \frac{1 + 2D}{1 - 2D} . \frac{1}{1 + 2D} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{4}\right)} . (1 + 2D) . \frac{1}{1 - 4D^2} \cos\left(\frac{x\sqrt{3}}{2}\right)$$
$$= e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{2}\right)} . \frac{1 + 2D}{1 - 4\left(-\frac{3}{4}\right)} \cos\left(\frac{x\sqrt{3}}{2}\right)$$

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$$= \frac{1}{4} e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{4}\right)} . \; 1 + 2D \; . \cos\left(\frac{x\sqrt{3}}{2}\right)$$

$$= \frac{1}{4} e^{-x/2} \frac{1}{D^2 + \left(\frac{3}{4}\right)} \left[\cos\left(\frac{x\sqrt{3}}{2}\right) - \sqrt{3} \sin\left(\frac{x\sqrt{3}}{2}\right) \right]$$

$$= \frac{1}{4} e^{-x/2} \left[\frac{1}{D^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \cos\left(\frac{x\sqrt{3}}{2}\right) - \sqrt{3} \frac{1}{D^2 \left(\frac{\sqrt{3}}{2}\right)^2} \sin\left(\frac{x\sqrt{3}}{2}\right) \right]$$

$$= \frac{1}{4} e^{-x/2} \left[\frac{x}{2\left(\frac{\sqrt{3}}{2}\right)} \sin\left(\frac{x\sqrt{3}}{2}\right) + \sqrt{3} . \frac{x}{2\left(\frac{\sqrt{3}}{2}\right)} \cos\left(\frac{x\sqrt{3}}{2}\right) \right]$$ [Using Rule II]

$$= \frac{x}{4\sqrt{3}} e^{-x/2} \left[\sin\left(\frac{x\sqrt{3}}{2}\right) + \sqrt{3} \cos\left(\frac{x\sqrt{3}}{2}\right) \right].$$

$$\therefore y = e^{-x/2} \left[c_1 \cos\left(\frac{x\sqrt{3}}{3}\right) + c_2 \sin\left(\frac{x\sqrt{3}}{2}\right) \right] + e^{x/2} \left[c_3 \cos\left(\frac{x\sqrt{3}}{2}\right) + c_4 \sin\left(\frac{x\sqrt{3}}{2}\right) \right]$$

$$+ \left(\frac{x}{4}\sqrt{3}\right) e^{-x/2} \left[\sin\left(\frac{x\sqrt{3}}{2}\right) + \sqrt{3} \cos\left(\frac{x\sqrt{3}}{2}\right) \right]$$

Rule V: When $X = x^M . V,$ where V is $\cos ax$ or $\sin ax$

$$P.I. = \frac{1}{f\ D} x^M, \cos\ ax$$

$$= real\ part\ of \left[\frac{1}{f\ D} e^{iax} . x^M \right]$$

and,

$$P.I. = \frac{1}{f\ D} x^M \sin ax$$

$$= imaging\ part\ of \left[\frac{1}{f\ D} e^{iax} . x^M \right]$$

Example: Solve $D^2 + 2D + 1 \; y = x \cos x.$

Solution: A.E. is $D + 1^{\ 2} = 0,$ so $D = -1, -1.$

$\therefore$ As usual, $C.F. = \; c_1 + c_2 \; x \; e^x.$

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$$P.I. = \frac{1}{D+1^{\,2}} x\cos x = real\ part\ of\ \frac{1}{D+1^{\,2}} xe^{ix}. \qquad \ldots(1)$$

Now, $\frac{1}{D+1^{\,2}} x e^{ix} = e^{ix} \frac{1}{D+i+1} x = \frac{e^{inx}}{1+i^{\,2}} \frac{1}{\left[1+\frac{D}{1+i}\right]^2}.x$

$$= \frac{e^{ix}}{2i}\left[1+\frac{D}{1+i}\right]^{-2} x = \frac{e^{ix}}{2i}\left[1-\frac{2D}{1+i}+\ldots\right]x$$

$$= -\frac{1}{2}e^{ix}\left[x-\frac{1-i}{2}.2\right], as\ \frac{1}{1+i} = \frac{1-i}{1+i\ \ 1-i} = \frac{1-i}{2}$$

$$= -\frac{1}{2}ie^{ix}\left[\ x-1\ +i\right] = -\frac{1}{2}e^{ix}\left[\ x-1\ i-1\right]$$

$$= -\frac{1}{2}\ \cos x + i\sin x\ \left[\ x-1\ i-1\right]. \ldots(2)$$

From (1) and (2), $P.I. = -\frac{1}{2}\left[-\cos x - \ x-1\ \sin x\right]$.

$$\therefore y = \ c_1 + c_2\ x\ e^{-x} + \left(\frac{1}{2}\right)\left[\cos x + \ x-1\ \sin x\right].$$

Remark: The solution is of general nature. However, if $m = 1$ *i.e.*, $X=xV$, where V is any function we can find *P.I.* by another method with the help of the following formula:

$$\frac{1}{F\ \ D}\ x.D\ = x.\frac{1}{F\ \ D}V - \frac{F,\ D}{f\ \ D^{\,2}}V.$$

We now illustrate an application of this formula in following example.

Example: Solve $D^2 - 2D + 1\ \ y = x\sin x.$

Solution: A.E. is $D-1^{\,2} = 0,$ so $D = 1, 1.$

$$\therefore C.F. = \ c_1 + c_2\ x\ e^{x}$$

$$P.I. = \frac{1}{D^2-2D+1} x\sin x$$

$$= x\frac{1}{D^2-2D+1}\sin x - \frac{2D-2}{D^2-2D+1}\sin x$$

$$= x\frac{1}{-1-2D+1}\sin x - \frac{2D-2}{-1-2D+1}\sin x$$

$$= -\frac{1}{2}x\frac{1}{D}.\sin x - \frac{1}{2}.\frac{1}{D^2}\ \ D-1\ \sin x$$

$$= \frac{1}{2}x\cos x - \frac{1}{2}.\frac{1}{D^2}\ \cos x - \sin x\ = \frac{1}{2}x\cos x - \frac{1}{2}.\frac{1}{D}\ \sin x - \cos x$$

$$= \left(\frac{1}{2}\right)x\cos x - \left(\frac{1}{2}\right)\ -\cos x - \sin x.$$

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$\therefore y = (c_1 + c_2 x)e^x + \left(\frac{1}{2}\right)(x\cos x + \cos x + \sin x)$.

Example: Solve $(D^2 - 2D + 1)y = xe^x \sin x$ or $(D-1)^2 y = xe^x \sin x$.

Solution: A.E. is $(D-1)^2 = 0$ so D = 1, 1 and $C.F. = (c_1 + c_2 x)e^x$.

$$P.I. = \frac{1}{(D-1)^2} e^x (x\sin x) = e^x . \frac{1}{(D+1-1)^2} x\sin x$$

$$= e^x \frac{1}{D^2} x\sin x = e^x . \frac{1}{D} \int x\sin x\, dx$$

$$= e^x . \left(\frac{1}{D}\right) . \left[-x\cos x - \int 1.(-\cos x)\, dx\right]. \qquad \textit{Integrating by parts}$$

$$= e^x \int (\sin x - x\cos x)\, dx = e^x \left[-\cos x - (x\sin x - \int 1.\sin x\, dx)\right]$$

$$= e^x (-\cos x - x\sin x - \cos x) = -e^x (x\sin x + 2\cos x) .$$

$$\therefore y = (c_1 + c_2 x)e^x - e^x (x\sin x + 2\cos x).$$

Example: Solve $(D^4 + D^2 + 1)y = ax^2 + be^{-x}\sin 2x$.

Solution: Given $(D^4 + D^2 + 1)y = ax^2 + be^{-x}\sin 2x$

Here auxiliary equation is $D^4 + D^2 + 1 = 0$ or $(D^2 + 1)^2 - D^2 = 0$

or $(D^2 + D + 1)(D^2 - D + 1) = 0$

so that $D^2 + D + 1 = 0$ *or* $D^2 - D + 1 = 0$

giving $D = \frac{-1 \pm i\sqrt{3}}{2}, D = \frac{1 \pm i\sqrt{3}}{2}$

$$\therefore C.F. = e^{-x/2}\left[c_1 \cos\left(\frac{x\sqrt{3}}{2}\right) + c_2 \sin\left(\frac{x\sqrt{3}}{2}\right)\right]$$

$$+ e^{x/2}\left[c_3 \cos\left(\frac{x\sqrt{3}}{2}\right) + c_4 \sin\left(\frac{x\sqrt{3}}{2}\right)\right]$$

Now, *P. I.* corresponding to ax^2

$$= a\frac{1}{D^4 + D^2 + 1} x^2 = a\left[1 + (D^4 + D^2)\right]^{-1} x^2$$

$$= a\left[1 - D^2 + \ldots\right]x^2 = a(x^2 - 2)$$

Now, *P. I.* corresponding to $be^{-x}\sin 2x$

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$$= b\frac{1}{D^4+D^2+1}e^{-x}\sin 2x = be^{-x}\frac{1}{D-1^4+\ D-1^2+1}\sin 2x$$

$$= be^{-x}\frac{1}{D^4-4D^3+6D^2-4D+1+D^2-2D+1+1}\sin 2x$$

$$= be^{-x}\frac{1}{D^4-4D^3+7D^2-6D+3}\sin 2x$$

$$= be^{-x}\frac{1}{-2^{2\ 2}-4D\ -2^2\ +7\ -2^2\ -6D+3}\sin 2x$$

$$= be^{-x}\frac{1}{10D-9}\sin 2x = be^{-x}\ 10D+9\ .\frac{1}{100D^2-81}\sin 2x$$

$$= be^{-x}\ 10D+9\ \frac{1}{100\ \ -2^{\ 2}-81}\sin 2x$$

$$= -be^{-x}\ 10D+9\ \frac{1}{481}\sin 2x = -\frac{be^{-x}}{481}\ 20\cos 2x+9\sin 2x$$

$$\therefore y = e^{-x/2}\left[c_1\cos\left(\frac{x\sqrt{3}}{2}\right)+c_2\sin\left(\frac{x\sqrt{3}}{2}\right)\right]+e^{x/2}\left[c_3\cos\left(\frac{x\sqrt{3}}{2}\right)+c_4\sin\left(\frac{x\sqrt{3}}{2}\right)\right]$$

$$+a\ \ x^2-2\ -\left(\frac{b}{481}\right)e^{-x}\ \ 20\cos 2x+9\sin 2x$$

Example: Solve $D^2-1\ y = xe^x+\cos^2 x.$

Solution: Given: $D^2-1\ y = xe^x+\left(\frac{1}{2}\right)\ 1+\cos 2x\ .\quad \ldots(1)$

The auxiliary equation is $D^2-1=0$ so that $D=\pm 1.$

So, $C.F.=c_1e^x+c_2e^{-x}$

P.I. corresponding to xe^x

$$= \frac{1}{D^2-1}xe^x = e^x\frac{1}{D+1^{\ 2}-1}x = e^x\frac{1}{D^2+2D}x = \frac{e^x}{2}\frac{1}{D\left(1+\frac{D}{2}\right)}x$$

$$= \frac{e^x}{2}.\frac{1}{D}\left(1+\frac{D}{2}\right)^{-1}x = \frac{e^x}{2}.\frac{1}{D}\left(1-\frac{D}{2}+\ldots\right)x = \frac{e^x}{2}.\frac{1}{D}\left(x-\frac{1}{2}\right)$$

$$= \frac{e^x}{2}\left(\frac{x^2}{2}-\frac{x}{2}\right) = \frac{1}{4}e^x\ \ x^2-x\ ,$$

P.I. corresponding to $\frac{1}{2} = \frac{1}{2}\frac{1}{D^2-1}e^{0.x} = \frac{1}{2}.\frac{1}{-1}e^{0.x} = -\frac{1}{2}.$

and P. I. corresponding to $\left(\frac{1}{2}\right)\cos 2x$

$$= \frac{1}{2}\frac{1}{D^2-1}\cos 2x = \frac{1}{2}\frac{1}{-2^2-1}\cos 2x = -\frac{1}{10}\cos 2x$$

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$$\therefore y = c_1 e^{x} + c_2 e^{-x} + \left(\frac{1}{4}\right) e^{x} \left(x^2 - x \right) - \frac{1}{10} \cos 2x.$$

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4.6 Homogeneous linear equation or Cauchy - Euler equation

A linear differential equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^n \frac{d^{n-1} y}{dx^{n-1}} + ... + a_n y = X \quad \ldots(1)$$

$$i.e.,\ x^n D^n + a_1 x^{n-1} D^{n-1} + ... + a_n\ y = X \quad \ldots(2)$$

where $a_1, a_2, \ldots, a_n$ are constants and *X* is either constant or a function of *x* only is called **Cauchy-Euler Equation or homogeneous linear differential equation.**

Note: In such equation the exponent of *x* and the order of derivative in each term will be same.

Method to solve homogeneous linear differential equation

Step I: We take $x = e^z$ $x = e^z$

$$so,\ z = \log^x \ and \ \frac{dz}{dx} = \frac{1}{x} \quad \ldots(3)$$

$$and\ \frac{dy}{dx} = \frac{dy}{dz} . \frac{dz}{dx}$$

$$= \frac{dy}{dz}\left(\frac{1}{x}\right)$$

$$or\ x\frac{dy}{dx} = \frac{dy}{dz} \quad \ldots(4)$$

$$Here\ x\frac{d}{dx} = \frac{d}{dz} = D_1 \qquad say \quad \ldots(5)$$

$$Again,\ \frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dz}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dz}\left(\frac{dy}{dz}\right).\frac{dz}{dx} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{dy^2}{dz^2} \qquad [bt\ 3]$$

$$or\ x^2\frac{d^2 y}{dx^2} = \frac{d^2 y}{dz^2} - \frac{dy}{dz} = D_1^2 - D_1\ y$$

and similarly, we can get the following formula.

$$x\frac{dy}{dx} = D_1 y \qquad or \qquad xD = D_1$$

$$x^2\frac{d^2 y}{dx^2} = D_1\ D_1 - 1\ y \qquad or \qquad x^2 D^2 = D_1\ D_1 - 1$$

$$x^3\frac{d^3 y}{dx^3} = D_1\ D - 1\ D_1 - 2\ y \qquad or \qquad x^3 D^3 = D_1\ D_1 - 1\ D_1 - 2$$

$$- \qquad - \qquad - \qquad - \qquad -$$

$$- \qquad - \qquad - \qquad - \qquad -$$

$$x^n\frac{d^n y}{dx^n} = D_1\ D_1 - 1\ D_1 - 2\ ...\ D_1 - n + 1\ y \quad \ldots(6)$$

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Step II: Substituting the result obtained in (3) and (6) in the given equation (1) or (2), we get

$f\ D_1\ y = Z$ …(7)

Where z is constant or function of z only.

Step III: Now the equation (7) becomes the linear differential equation with constant co-efficient and earlier we had gone through this type of differential equation, so we can solve it by using the method of solving of linear differential equation with constant coefficient.

Example: Solve $x^2 D^2 - cD + 2\ y = x \log x.$

Solution: Let $x = e^z$ so that $z = \log x$ and let $D_1 = \frac{d}{dz}.$ …(1)

Then the given equation becomes

$\left[D_1\ D_1 - 1\ - D_1 + 2\right] y = ze^x \text{ or } D_1^2 - 2D_1 + 2\ y = ze^x,$

Which is linear differential equation with constant coefficients.

Its auxiliary equation is $D_1^2 - 2D_1 + 2 = 0. \text{ So } D_1 = \pm i.$

$\therefore C.F. = e^z\ c_1 \cos z + c_2 \sin z$

$= x\left[c_1 \cos\ \log x\ + c_2 \sin\ \log x\ \right], \text{ using } (1)$

$P.I. = \frac{1}{D_1^2 - 2D + 2} ze^z = e^z \frac{1}{D_1 + 1\ ^2 - 2\ D_1 + 1\ + 2} z$

$= e^z \frac{1}{D_1^2 + 1}.z = e^z\ 1 + D_1^2\ ^{-1}.z = e^z\ 1 - \ldots\ z$

$= e^z.z = x \log x, \text{ using } 1$

$\therefore y = x\left[c_1 \cos\ \log x\ + c_2 \sin\ \log x\ \right] + x \log x.$

Example: Solve $x^2 D^2\ y - 3x\,Dy + 5y = x^2 \sin \log x.$

Solution: Given $x^2 D^2 - 3xDy + 5\ y = x^2 \sin \log x.$ …(1)

Let $x = e^x$ so that $z = \log x$ and let $D_1 = \frac{d}{dz}.$ …(2)

Then $xD = D_1$ and $x^2 D^2 = D_1\ D_1 - 1$. …(3)

Using (2) and (3), (1) reduces to

$\left[D_1\ D_1 - 1\ - 3D_1 + 5\right] y = e^{2x} \sin z$

or $D_1^2 - 4D_1 + 5\ y = e^{2z} \sin z$ …(4)

A. E. of (4) is $D_1^2\ - 4D_1 + 5 = 0$ giving $D_1 = \frac{4 \pm \sqrt{16 - 20}}{2} = 2 \pm i.$

$\therefore C.F. = e^{2z}\ c_1 \cos z + c_2 \sin z\ = x^2\ c_1 \cos \log s + x_2 \sin \log x\ , by\ 2$

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$$P.I.=\frac{1}{D_1^2-4D_1+5}e^{2z}\sin z=e^{2z}\frac{1}{D_1+2^{\ 2}-4\ D_1+2\ +5}\sin z$$

$$=e^{2z}\frac{1}{D_1^2+1}\sin z=e^{2z}\left(-\frac{z}{2}\cos z\right)=-\frac{x^2}{2}\log x\cos\log x, by\ \ 2$$

$$\therefore y=x^2\ \ c_1\cos\log x+c_2\sin\log x\ -\left(\frac{1}{2}\right)x^2\log x\cos\log x.$$

Example: *Solve* $x^3\ \ d^3y/dx^3\ +2x^2\ \ d^2y/dx^2\ +2y=10\ \ x+1/x$

Solution: Given $x^3\ D^3+2x^2D^2+2\ =10\ \ x+x^{-1}\ ,D=d/dx$ …(1)

Let $x=e^z$ so that $z=\log x$ and let $D_1=d/dz.$ …(2)

Then, $xD=D_1, x^2\ D^2=D_1\ \ D_1-1\ , x^3\ D^3=D_1\ \ D_1-1\ \ \ D_1-2\ .$ …(3)

Using (2) and (3), (1) reduces to

$$\left[D_1\ \ D_1-1\ \ \ D_1-2\ +2D_1\ \ D_1-1\ +2\right]y=10\ \ e^z+e^{-z}$$

Or $D_1^3-D_1^2+2\ \ y=10e^z+10e^{-z}$ …(4)

A.E. of (4) is $D_1^3-D_1^2+2=0$

Or $D_1+1\ \ \ D_1^2-2D_1+2\ =0$ giving $D_1=-1,\pm i.$

$$\therefore C.F.=c_1e^{-z}+e^z\ \ c_1\cos z+c_2\sin z$$

$$=c_1\ x^{-1}+x\ \ c_1\cos\log x+c_2\sin\log x$$

P.I. corresponding to

$$10e^z=10\frac{1}{D_1+1\ \ \ D_1^2-2D_1+2}e^z=10\frac{1}{2\ 1-2+2}e^z=5x, by\ \ 2$$

and P.I. corresponding to $10e^{-z}$

$$=10\frac{1}{D_1+1\ \ \ D_1^2-2D_1+2}e^{-z}=10\frac{1}{D_1+1}.\frac{1}{1+2+2}e^{-z}$$

$$=2\frac{1}{D_1+1}e^{-z}.1=2e^{-z}\frac{1}{D_1-1+1}.1=2e^{-z}\frac{1}{D_1}.1$$

$$2e^{-z}.z=2x^{-1}\log x, by\ \ 2$$

$$\therefore y=c_1\ x^{-1}+x\ \ c_1\cos\log x+c_2\sin\log x\ +5x+2x^{-1}\log x.$$

Example: *Solve* $x^2\frac{d^2y}{dx^2}-3x\frac{dy}{dx}+y=\frac{\log x\sin\log x+1}{x}$

Solution: Given $x^2\ D^2-3xD+1\ \ y=x^{\ 1}1+\log x\sin\log x\ .$ …(1)

Let $x=e^z$ so that $z=\log x$ and let $D_1=d/dz.$ …(2)

Then $xD=D_1$ and $x^2\ D^2=D_1\ \ D_1-1\ .$ …(3)

Using (2) and (3), (1) reduces to

$$\left[D_1\ \ D_1-1\ -3D_1+1\right]y=e^z\ \ 1+z\sin z$$

Or $D_1^2-4D_1+1\ \ y=e^{-z}+e^{-z}\ z\sin z.$ …(4)

A.E. of (4) is $D_1^2-4D_1+1=0$ so that $D_1=2\pm\sqrt{3}.$

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$\therefore C.F. = e^{2z}\left[c_1 \cosh\ \sqrt{3}z\ + c_2 \sinh\ \sqrt{3}z\ \right]$

$= x^2\left[c_1 \cosh\ \sqrt{3}\log x\ + c_2 \sinh\ \sqrt{3}\log x\ \right]$ by (2)

P.I. corresponding to e^{-z}

$= \frac{1}{D_1^2 - 4D_1 + 1}e^{-z} = \frac{1}{1+4+1}e^{-z}\frac{1}{6}x^{-1}, = \frac{1}{D_1^2}$ by (2)

and P.I. corresponding to $e^{-z}z\sin z$

$= \frac{1}{D_1^2 - 4D_1 + 1}e^{-z}\ z\sin z\ = e^{-z}\frac{1}{D_1 - 1\ ^2 - 4\ D_1 - 1\ + 1}z\sin z$

$e^{-z}\frac{1}{D_1^2 - 6D_1 + 6}z\sin z$

$= e^{-z}\left[z\frac{1}{D_1^2 - 6D_1 + 6}\sin z - \frac{2D_1 - 6}{D_1^2 - 6D_1 + 6\ ^2}\sin z\right]$

[Proceeding like Example: 2. Of Art. 4.19, Chapter 4]

$= e^{-z}\left[z\frac{1}{-1-6D_1+6}\sin z - \ 2D_1 - 6\ \frac{1}{-1-6D_1+6\ ^2}\sin z\right]$

$= e^{-z}\left[z\frac{1}{5-6D_1}\sin z - \frac{2D_1 - 6}{5-6D_1\ ^2}\sin z\right]$

$= e^{-z}\left[z\frac{5+6D}{25-36D_1^2}\sin z - \ 2D_1 - 6\ \frac{1}{25-60D_1+36D_1^2}\sin z\right]$

$e^{-z} = \left[z\ 5+6D_1\ \frac{1}{25+36}\sin z - \ 2D_1 - 6\ \frac{1}{25-60D_1-36}\sin z\right]$

$= e^{-z}\left[\frac{z}{61}\ 5+6D_1\ \sin z + \ 2D_1 - 6\ \frac{1}{11+60D_1}\sin z\right]$

$e^{-z}\left[\frac{z}{61}\ 5\sin z + 6\cos z\ + \frac{120\,D_1^2 - 382D_1 + 66}{-3600-121}\sin z\right]$

$= e^{-z}\left[\frac{z}{61}\ 5\sin z + 6\cos z\ + \frac{120D_1^2 - 382D_1 + 66}{-3600-121}\sin z\right]$

$= e^{-z}\left[\frac{z}{61}\ 5\sin z + \cos z\ + \frac{120\ -\sin z\ -382\cos z + 66\sin z}{3721}\right]$

$\therefore y = x^2\left[c_1 \cosh\ \sqrt{3}\log x\ + c_2 \sinh\ \sqrt{3}\log x\ + 1/\ 6x\ \right]$

$+ \frac{1}{x}\left(\frac{\log x}{61}\ 5\sin\log x + 6\cos\log x\ + \frac{54\sin\log x + 382\cos\log x}{3721}\right)$

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EQUATION REDUCIBLE TO HOMOGENEOUS FORM

A differential equation of the type

$$a+bx\ ^{n}\frac{d^n y}{dx^n}+a_1\ a+bx\ ^{n-1}\frac{d^{n-1}y}{dx^{n-1}}+..+a_{n-1}\ a+bx\ \frac{dy}{dx}+a_n y=X, \qquad \ldots(1)$$

where a, b, a_1, a_2..a_n are constants and X is a function of x alone, can be reduced to homogeneous form by putting $a+bx=\upsilon$

Now $a+bx=\upsilon$ gives $b=d\upsilon/dx.$ …(2)

$$\therefore \frac{dy}{dx}=\frac{dy}{d\upsilon}.\frac{d\upsilon}{dx}=b\frac{dy}{d\upsilon}, \text{ using (2).}$$

$$\therefore \frac{d}{dx}=b\frac{d}{d\upsilon} \qquad \ldots(3)$$

$$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=b\frac{d}{d\upsilon}\left(b\frac{dy}{d\upsilon}\right)=b^2\frac{d^2y}{d\upsilon^2}, \text{using (3).}$$

Similarly, $\frac{d^n y}{dx^n}=b^n\frac{d^n y}{d\upsilon^n}.$

Putting these in (1) and dividing by b^n, we get

$$\upsilon^n\frac{d^n y}{d\upsilon^n}+\frac{a_1}{b}\upsilon^{n-1}\frac{d^{n-1}y}{d\upsilon^{n-1}}+...+\frac{a_n}{b^n}y=\frac{V}{b^n}, \qquad \ldots(4)$$

which is a **standard homogeneous equation**. Here V is the value of X in terms of v by using (2). Equation (4) is now solved as usual. Finally v is replaced by $ax+b$ to get the required solution.

Example: *Solve* $\left[\ 3x+2\ ^{2}D^2+3\ 3x+2\ D-36\right]y=3x^2+4x+1.$

Solution: Let $3x+2=\upsilon$ So $dy/dx=3\ dy/d\upsilon$ an

$d^2y/dx^2=3^2\ \ d^2y/d\upsilon^2$.

Hence $\upsilon^2\,3^2\left(\frac{d^2y}{d\upsilon^2}\right)+3\upsilon.3\left(\frac{dy}{d\upsilon}\right)-36y=3\left(\frac{\upsilon-2}{3}\right)^2+4\left(\frac{\upsilon-2}{3}\right)+1$

Or $\upsilon^2\frac{d^2y}{d\upsilon^2}+\upsilon\frac{dy}{d\upsilon}-4y=\frac{1}{27}\ \upsilon^2-1$. …(1)

which is homogenous in y and v. Here we have v in place of x as usual.

So let $\upsilon=e^z$ or $z=\log\upsilon$ and $D_1=d/dz.$ Then (1) gives

$\left[D_1\ D_1-1\ +D_1-4\right]y=\ e^{2z}-1\ /27$ *i.e.* $D_1^2-4\ y=\ e^{2z}-1\ /27$

Here A.E. is $D_1^2-4=0.\ \therefore D_1=\pm 2$ and hence

$C.F.=c_1e^{2z}+c_2e^{-2z}=c_1\upsilon^2+c_2\upsilon^{-2} \qquad \left[\because \upsilon=e^z\right]$

$i.e\,C.F.=c_1\ 3x+2\ ^{2}+c_2\ 3x+2\ ^{-2}\ \ \because \upsilon=3x+2$

$$P.I.=\frac{1}{27}\frac{1}{D_1^2-4}\ e^{2z}-1=\frac{1}{27}\left[\frac{1}{D_1-2\ \ D_1+2}e^{2z}-\frac{1}{D_1^2-4}e^{0.z}\right]$$

$$=\frac{1}{27}\left[\frac{1}{D_1-2}\frac{1}{2+2}e^{2z}-\frac{1}{0-4}e^{0.z}\right]=\frac{1}{27\times 4}\left[\frac{1}{D_1-2}e^{2z}+1\right]$$

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$$=\frac{1}{108}\left[\frac{z}{1!}e^{2z}+1\right]=\frac{1}{108}\ ze^{2z}+1$$

$$=1/108\ \left[v^2\log v+1\right]=1/108\ \left[\ 3x+2\ ^2\log\ 3x+2\ +1\right]$$

$\therefore$ Required general solution is

$$y=c_1\ 3x+2\ ^2+c_2\ 3x+2\ ^{-2}+1/108\ \left[\ 3x+2\ ^2\log\ 3x+2\ +1\right]$$

Example: *Solve :* $1+x\ ^2\frac{d^2y}{dx^2}+\ 1+x\ \frac{dy}{dx}+y=4\cos\log\ 1+x$

Solution: Let $1+x=v$ So $dy/dx=dx=dy/dv, d^2y/dx^2=d^2y/dv^2$

$$\therefore v^2\frac{d^2y}{dv^2}+v\frac{dy}{dv}+y=4\cos\log v \qquad \ldots(1)$$

Now put $v=e^z$ *i.e.* $z=\log v$ and let $D_1=d/dz$. So (1) gives

$$\left[D_1\ D_1-1\ +D_1+1\right]y=4ocs\,z \text{ or } D_1^2+1\ y=4\cos z.$$

Auxiliary equation $D_1^2+1=0$ So $D_1=\pm i.$

$$\therefore C.F.=c_1\cos z+c_2\sin z=c_1\cos\log v+c_2\sin\log v$$

$$=c_1\cos\log\ 1+x\ +c_2\sin\log\ 1+x\ .$$

Now $P.I.=4\frac{1}{D_1^2+1}\cos z=4.\frac{z}{2\times1}\sin z$

$$=2\log\ 1+x\ \sin\log\ 1+x\ \ asz=\log v=\log\ 1+x$$

$$\therefore y=c_1\cos\log\ 1+x\ +c_2\sin\log\ 1+x\ +2\log\ 1+x\ \sin\log\ 1+x$$

Example: *Solve :* $x+1\ ^2y_2+\ x+1\ y_1=\ 2x+3\ \ 2x+4$

Solution: Given $x+1\ ^2\frac{d^2y}{dx^2}+\ x+1\ \frac{dy}{dx}=4x^2+14x+12.$ …(1)

Let $1+x=v$ so that $dy/dx=dy/dv$ and $d^2y/dx^2=d^2y/dv^2$ …(2)

Then (1) reduces to $v^2\frac{d^2y}{dv^2}+v\frac{dy}{dv}=4\ v=1\ ^2+14\ v-1\ +12$

Or $\left(v^2\frac{d}{dv^2}+v\frac{d}{dv}\right)y=4v^2+6v+2.$ …(3)

Now, put $v=e^z$ *i.e* $z=\log v$ and let $D_1=d/dz.$ …(4)

Then (3) reduces to $\left[D_1\ D_1-1\ +D_1\right]y=4e^{2z}+6e^z+2$

Or $D_1^2y=4e^{2z}+6e^z+2.$ …(5)

A.E. of (5) is $D_1^2=0$ giving $D_1=0,0$

$$\therefore C.F.=\ c_1+c_2z\ \ e^{0.z}=c_1+c_2z=c_1+c_2\log v,\ \text{by (4)}$$

$$=c_1+c_2\log\ 1+x\ ,by\ \ 2$$

$$P.I.\ \frac{1}{D_1^2}\ 4e^{2z}+6e^z+2\ =\frac{1}{D_1}\ 2e^{2z}+6e^z+2z$$

$$=e^{2z}+6e^z+z^2=v^2+6v+\ \log v\ ^2\ \text{by (4)}$$

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$= 1+x^{2}+6\ 1+x\ +\left[\log\ 1+x\ \right]^{2}$, by (2)

$= x^{2}+8x+7+\left[\log\ 1+x\ \right]^{2}$

$\therefore y=c_1+c_2\log\ 1+x\ +x^{2}+8x+\left[\log\ 1+x\ \right]^{2}$

Or $y=c_1+c_2\log\ 1+x\ +x^{2}+8x+\left[\log\ 1+x\ \right]^{2}$

where $c_1=c_1+7$ is another arbitrary constant.

4.7 Ordinary simultaneous differential equations

Here we discuss the ordinary differential equations involving two or more dependent variables.
For the determination of dependent variable, there must be as many equations as there are dependent variables.

Method of Solving

Step – I By the process of elimination, no obtain an equation involving one dependent variable Ninth one independent variable.

Step – II After solving the derived equation, me get the solution and substituting it we get the other dependent variable.

The method can be easily understand by seeing the gives example

Example: *Solve* $dx/dt\ -7x+y=0$ …(i)

$dy/dt\ -2x-5y=0$ …(ii)

First method (symbolic method with use of D).

Step I Writing *D* for d/dt the given equations *(i)* and *(ii)* can be rewritten in the symbolic form as follows:
(D – 7) x + y = 0 …(1)
And -2 x + (d - 5) y = 0 …(2)

Step II We now eliminate *x* (say) as follows. Multiplying (1) by 2 and operating (2) by (D - 7), we get
2 (D - 7) x + 2y = 0 …(3)
-2 (D - 7) x + (D - 7) (D - 5) y = 0 …(4)
Adding (3) and (4), [(D - 7) (D - 5) + 2] y = 0
i.e (D^2 – 12 D + 37) y = 0,
which is linear equation with constant coefficients.
its. A.E. (auxiliary equation) is D^2 – 12 D + 37 = 0.
Solving for $D, D=6\pm i$,

$\therefore \quad y=e^{6t}\ c_1\cos t+c_2\sin t\ \ldots(5)$

Step III We now try to get *x* by using (5). In this connection remember that we must avoid integration to get *x*. Thus if we use (1) to get *x*, then after putting value of *y* we have to integrate for getting *x*. Hence we must use (2) because this will not involve any subsequent integration to obtain *x*. Now from (5), differentiating w.r.t. 't', we get

$Dy=6e^{6t}\ c_1\cos t+c_2\sin t\ +e^{6t}\ -c_1\sin t+c_2\cos t$

Or $Dy=e^{6t}\left[\ 6c_1+c_2\ \cos t+\ 6c_2-c_1\ \sin t\right]$

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Substituting the values of *y* and *Dy* in (2), we have

$2x = Dy - 5y = e^{6t}\left[\ c_1 + c_2\ \cos t + \ 6c_2 - c_1\ \sin t - 5\ c_1 \cos t + c_2 \sin t\ \right]$

Or $\quad 2x = e^{6t}\left[\ c_1 + c_2\ \cos t + \ c_2 - c_1\ \sin t\right]$

Or $\quad x = \ 1/2\ e^{6t}\left[\ c_1 + c_2\ \cos t + \ c_2 - c_1\ \sin t\right].$ …(6)

Thus (5) and (6) together give the required solution.

Remark: We can also eliminate *y* first (as we did to eliminate *x*) and then obtain *x*. This value of *x* can be put in (1) to get the desired value of *y*.

Alternative Method :

(Use of differentiation): We shall now use differentiation to solve the given equations.
To eliminate *x*, we differentiate *(ii)* w.r.t *t* and obtain

$\frac{d^2y}{dt^2} - 2\frac{dx}{dt} - 5\frac{dy}{dt} = 0.$ *...(iii)*

Now $\frac{dx}{dt} = 7x - y$ [from *(i)*]

$= \frac{7}{2}\left(\frac{dy}{dt} = 5y\right) - y$ [Putting value of *x* from *(ii)*]

$= \frac{7}{2}\frac{dy}{dt} - \frac{37y}{2}$

Putting this value of dx/dt in *(iii)*, we obtain

$\frac{d^2y}{dt^2} - 7\frac{dy}{dt} + 37y - 5\frac{dy}{dt} = 0$ or $D^2 - 12D + 37\ y = 0$

Now get *y* as done in first method. In fact repeat the whole method after this step. Thus we get the same values of *x* and *y* as in first method. Thus the general solution is given by (5) and (6).

Remarks: This method will be used when found very necessary. In almost all problems we shall use the first method.

Important Note:
Generally *t* will be the independent variable and *x* and *y* will be dependent variables. In some problems any other variable, *x* say, will be given as the independent variable and *y* and *z* as the dependent variables. This point should be noted carefully while doing any problem.

Example: *Solve:* $d^2x/dt^2 - 3x - 4y = 0, d^2y/dt^2 + x + y = 0$

Solution: Writing *D* **for** *d/dt*, the given equations become

$D^2 - 3\ x - 4y = 0$ …(1)

And $x + \ D^2 + 1\ y = 0$ …(2)

Eliminating *y*, $\left[\ D^2 + 1\ \ D^2 - 3\ + 4\right]x = 0$ Or $D^2 - 1^{\ 2}\ y = 0$ …(3)

A.E. is $D^2 - 1^{\ 2} = 0 \quad \therefore \quad D = 1,1,-1,-1.$

Hence the Solution: of (3) is $x = \ c_1 + c_2 t\ e^t + \ c_3 + c_4 t\ e^{-t},$ …(4)

Where c_1, c_2, $c_3$and c_4 are arbitrary constants. Then form (4), we get

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$Dx = c_2 e^t + (c_1 + c_2 t)\, e^t + c_4 e^{-t} - (c_3 + c_4 t)\, e^{-t}$

$= (c_1 + c_2 + c_2 t)\, e^t + (c_4 - c_3 - c_4 t)\, e^{-t}$

Or $D^2 x = (c_1 + 2c_2 + c_2 t)\, e^t - (2c_4 - c_3 - c_4 t)\, e^{-t}$ …(5)

But from (1), $4y = D^2 x - 3x$.

Hence using (4) and (5), (6) becomes …(6)

$4y = (c_1 + c_2 + 2c_2 t)\, e^t - (2c_4 - c_3 - c_4 t)\, e^{-t}$

$-3\left[(c_1 + c_2 t)\, e^t + (c_3 + c_4 t)\, e^{-t}\right]$

Or $4y = (2c_2 - c_1 - c_2 t)\, e^t - (2c_4 + c_3 + c_4 t)\, e^{-t}$

Or $y = (1/2)(c_2 - c_1 - c_2 t)\, e^t - (1/2)(c_4 + c_3 + c_4 t)\, e^{-t}$ …(7)

The required solution is given by (4) and (7).

Example: Solve the following simultaneous equations:

(i) $\frac{dy}{dx} + y = z + e^x, \frac{dz}{dx} + z = y + e^x$

(ii) $\frac{dx}{dt} + x = y + e^t, \frac{dy}{dt} + y = x + e^t$.

Solution: (i) Writing D for d/dx, the given equations become

$(D+1)\, y - z = e^x$ …(1)

And $-y + (D+1)\, z = e^x$ …(2)

Operating (1) by $(D+1)$, $z = (D+1)\, e^x$. …(3)

Adding (2) and (3), we get

$\left[(D+1)^2 - 1\right] y = e^x + e^x + e^x$. $\quad \because D = d/dx$

Or $D(D+2)\, y = 3e^x \therefore A.E.\ D(D+2) = 0$ gives $D = 0, -2$

$\therefore C.F. = C_1 + C_2 e^{-2x}$

Now P.I. $= \frac{1}{D(D+2)} e^x = e^x$

$\therefore \quad y = c_1 + c_2 e^{-2x} + e^x$. …(4)

From (4), $Dy = dy/dx = -2c_2 e^{-2x} + e^x$. …(5)

$\therefore$ From (1), $z = Dy + y - e^x$

$= -2c_2 e^{-2x} + e^x + c_1 + c_2 e^{-2x} + e^x - e^x$, by (4) and (5)

Or $z = c_1 - c_2 e^{-2x} + e^x$ …(6)

The required solution is given by (4) and (5).

(ii) This is just the same as (i). Here we have t in place of x and x and y in place of y and z. You have to denot d/dt by D.

$x = c_1 + c_2 e^{-2t} + e^t, y = c_1 - c_2 e^{-2t} + e^t$

Example: *Solve* $d/dt + dy/dt - 2y = 2\cos t - 7\sin t$,

and $dx/dt - dy/dt + 2x = 4\cos t - 3\sin t$

Solution: Let $D = d/dt$. Then the given equations become $Dx + (D-2)\, y = 2\cos t = 7\sin t$ …(1)

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And $D+2\ x-Dy=4\cos t-3\sin t.$ …(2)

Eliminating y from (1) and (2), we get

$\left[D^2+\ D-2\ \ D+2\ \right]x=D\ \ 2\cos t-\sin t\ +\ D-2\ \ 4\cos t-3\sin t$

Or $D^2-2\ x=-9\cos t$ on simplification.

For (3), C.F. $=c_1 e^{\sqrt{2}t}+c_2 e^{\sqrt{2}t}.$ …(3)

Also P.I. $=-\dfrac{9}{D^2-2}\cos t=\dfrac{-9\cos t}{-1^2-2}=3\cos t.$

$\therefore$ Solution of (3) is $x=c_1 e^{\sqrt{2}t}+c_2 e^{\sqrt{2}t}+3\cos t.$ …(4)

From (4) $Dx=c_1\sqrt{2}\,e^{\sqrt{2}t}-c_2\sqrt{2}e^{-\sqrt{2}t}-3\sin t.$

Adding (1) and (2), $2Dx+2x-2y=6\cos t-10\sin t.$

$\therefore y=Dx+x-3\cos t+5\sin t$ …(5)

Or $y=c_1\sqrt{2}e^{\sqrt{2}t}-c_2\sqrt{2}\,e^{-\sqrt{2}t}-3\sin t+c_1 e^{\sqrt{2}t}+c_2 e^{-\sqrt{2}t}+3\cos t-3\cos t-3\cos t+5\sin t$

The required solution is given by (4) and (6). …(6)

Example: *Solve for x and y :*

$\dfrac{dx}{dt}+2\dfrac{dy}{dt}-2x+2y=3e^t$ *and* $3\dfrac{dx}{dt}+\dfrac{dy}{dt}+2x+y=4e^{2t}$

Solution: Given $3\dfrac{dx}{dt}+2\dfrac{dy}{dt}-2x+2y=3e^t$ …(1)

And $3\dfrac{dx}{dt}+\dfrac{dy}{dt}+2x+y=4e^{2t}.$ …(2)

Multiplying both sides of (2) by 2, we have

$6\dfrac{dx}{dt}+2\dfrac{dy}{dt}+4x+2y=8e^{2t}.$ …(3)

Subtracting (1) from (3), we have

$5\dfrac{dx}{dt}+6x=8e^{2t}-3e^t$

Or $\dfrac{dx}{dt}+\dfrac{6}{5}x=\dfrac{8}{5}e^{2t}-\dfrac{3}{5}e^t,$ …(4)

Which is a linear differential equation of order one.

I.F. of (4) $e^{\int 6/5\ dt}=e^{6/5\ t}$

Hence solution of (4) is given by

$x\,e^{\ 6/5\ t}=\int\left(\dfrac{8}{5}e^{2t}-\dfrac{3}{5}e^t\right)e^{\ 6/5\ t}\,dt$

Or $x\,e^{\ 6/5\ t}=\int\left(\dfrac{8}{5}e^{\ 16/5\ t}-\dfrac{3}{5}e^{\ 11/5\ t}\right)dt$

$x\,e^{\ 6/5\ t}=\dfrac{8}{5}.\dfrac{5}{16}e^{\ 16/5\ t}-\dfrac{3}{5}.\dfrac{5}{11}e^{\ 11/5\ t}+c_1$

$\therefore\qquad x=1\ 1/2\ \ e^{2t}-\ 3/11\ \ e^t+c_1 e^{-\ 6/5\ t}$ …(5)

Now, multiplying both sides of (1) by 3, we have

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$3\frac{dx}{dt}+6\frac{dy}{dt}-6x+6y=9e^t$ …(6)

Subtracting (2) from (6), we have

$5\frac{dy}{dt}-8x+5y=9e^t-4e^{2t}$

Or $5\frac{dy}{dt}+5y=8x+9e^t-4e^{2t}$

Or $5\frac{dy}{dt}+5y=8\left[\frac{1}{2}e^{2t}-\frac{3}{11}e^t+c_1e^{-6/5\,t}\right]+9e^t-4e^{2t}$, by (5)

Or $5\frac{dy}{dt}+5y=\frac{75}{11}e^t+8c_1e^{-6/5}$

Or $\frac{dy}{dt}+y=\frac{15}{11}e^t+\frac{8c_1}{5}e^{-6/5\,t}$ …(7)

which is again a linear differential equation of order one.

Integrating factor of (7) $=e^{\int dt}=e^t$

Hence solution of (7) is given by

$ye^t=\int\left[\frac{15}{11}e^t+\frac{8c_1}{5}e^{-6/5\,t}\right]e^t\,dt$

Or $ye^t=\int\left[\frac{15}{11}e^{2t}+\frac{8c_1}{5}e^{-(1/5)t}\right]dt$

Or $ye^t=\frac{15}{11}.\frac{1}{2}e^{2t}+\frac{8c_1}{5}\,-5\;e^{-1/5\,t}+c_2$

Or $y=c_2e^{-t}-8c_1e^{-6/5\,t}+\;15/22\;e^t.$ …(8)

(5) and (8) together give the required solution.

Solution of simultaneous differential equations involving operators xd / dx or td/dt etc.

Example: *Solve :* $dx/dt+y=0, t\;dy/dt+x=0$

Solution: Let t = e^z Let D_1 = d/ dz = td /dt. Then the given equations may be written as

D_1 x + y …(1)

And x + D_1y = 0 …(2)

Eliminating *y* from (1) and (2), we have

$D_1^2x-x=0$

Or $D_1^2-1\;x=0.$

Auxiliary eqn. $D_1^2-1=0$ gives $D_1=1,-1$

$\therefore \quad x=c_1e^z+c_2e^{-z}$ …(3)

$\therefore \quad D_1x=c_1e^z-c_2e^{-z}$ …(4)

$\therefore$ From (1), $y=-D_1x=c_2^{-z}-c_1e^{-z}$

Since $t=e^z$ the required solution from (3) and (4) is

$x=c_1t+c_2t^{-1}, y=c_2t^{-1}-c_1t.$

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Example: $t\,Dx+2\ x-y\ =t, tDy+x+5y=t^2$ where $D\equiv d/dt$.

Solution: Let $t=e^z$ and $D_1=d/dz=t.d/dt$. Then, we get

$D_1+2\ x-2y=e^z$ …(1)

And $x+\ D_1+5\ y=e^{2x}$ …(2)

Eliminating y from (1) and (2), we have

$D_1+5\ \ D_1+2\ x+2x=\ D_1+5\ e^z+2e^{2z}$

Or $D_1^2+7D_1+12\ x=6e^z+2e^{2z}$. …(3)

A.E. is $D_1^2+7D_1+12=0$. So $D_1=-2,-4$

$\therefore$ C.F. of $3=c_1e^{-3z}+c_2e^{-4z}$

P.I. corresponding to $6e^z=6\dfrac{1}{D_1^2+7D_1+12}e^z=\dfrac{3}{10}e^z$.

P.I. corresponding to $2e^{2z}=2\dfrac{1}{D_1^2+7D_1+12}e^{2z}=\dfrac{1}{15}e^{2z.}$

$\therefore x=c_1e^{-3z}+c_2e^{-4z}+3e^z/10+e^{2z}/15.$ …(4)

$\therefore D_1x=-3c_1e^{-3z}-4c_2^{-4z}+3e^z/10+2e^{2z}/15.$ …(5)

Using (4) and (5) in (1) and simplifying, we get

$y=-\ 1/2\ c_1e^{-3z}-c_2e^{-4z}-\ 1/20\ e^z+\ 2/15\ e^{2z}.$ …(6)

Putting $t=e^z$, from (4) and (6), the solution is

$x=c_1t^{-3}+c_2t^{-4}+3t/10+t^2/15,$

And $y=-\ 1/2\ c_1t^{-3}-c_2t^{-4}+2t^2/15-t/20$

4.8 Existence, uniqueness and linear independence of solution

Consider the differential equation

$\dfrac{dy}{dx}=+\ x,y$ …(i)

With initial condition $y\ x_0\ =y_o$ …(ii)

Let $\phi\ x,y,c\ =0$ …(iii)

Is a general solution of the given differentiate equation (1) and after using the initial condition, Ne get

i) 'C' is unique, this implies the equation (1) with condition (2) has unique solution

ii) 'C' can't be determined then the expiation (1) with condition (2) has no solution.

iii) Absurd result then the equation (1) with condition (2) has no solution.

Example: Find the number of solution of the differential equation

$2x\dfrac{dy}{dx}=3\ 2y-1$, when

(i) $y\ 0\ =1/2$

(ii) y (1) = 1 and

(iii) y (0) =1

Solution: The given differential equation can be written as

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$$\frac{dy}{2y-1}=\frac{3}{2}\frac{dx}{x}$$

Integratig both sides, we get

$$\frac{1}{2}\log\ 2y-1 =\frac{3}{2}\log x+\frac{1}{2}\log c$$

Or, $2y-1 = cx^3$ where c is the constant of integrates.

i) When $y\ 0 = 1/2$ we can't determine value of c, so infinite solution.

ii) When $y\ 1 = 1$, we get c = 1(unique value), so unique solution.

iii) When $y\ 0 = 1$, we get1 = 0 (absurd result), so no solution.

Fundamental Existence of Uniqueness Theorem

Consider the nth order linear differential equation of the forth

$$P_O\ x\ \frac{dy}{dx}+P_1\ x\ \frac{d^{n-1}y}{dx^{n-1}}+.....+P_{n-1}\ x\ \frac{dy}{dx}+P_n\ 2\ y=r\ x \quad \ldots(1)$$

Where, $P_O\ x\ , P_1\ x\ ,....+P_{n-1}\ x\ , P_n\ x$ and r (x) are continuous functions n as internal [a, b] and $P_O\ x \neq 0$ Let x_O be any point of the interval and let $K_O, K_1, ...k_{x-1}$ be any n real constant. Then three exists a unique solution of the differential equation (1) on the interval hot satisfice the initial coalitions

$$y\ x_0 = k_0, y'\ x_0\ k_1, y''\ x_0 = k_2, ...y^{x-1}\ x_0 = k_{x=1} \quad \ldots(2)$$

Where $y', y'', ...y^{x-1}$ are the derivatives of order the, two of(x-1).

Remark: If $P_0\ x = 0$ for some $x \in a,b$ then the solution of (1) may hot be unique or may hot exist.

If r (x) = 0 in (1) and $y\ x_0 = y'\ x_0 = y''\ x_0 = ... = y^{n-1}\ x_0 = 0$ then the differential equation (1) has one and only one solution and the solution will be $y\ x = 0 \forall x \in a,b$

Example: Check the uniqueness and existence of the solution of the differential equation

$$\frac{d^2y}{dx^2}+3x\frac{dy}{dx}+x^3y=e^x; y\ 1 = 2_1\ y^1\ 1 = -5$$

Solution: Here $P_0\ x \neq 0, P_1\ x = 3x,$ $P_2\ x = x^3$ and $r\ x = e^2$ are continuous functions an $y\ 1 = 2, y^1\ 1 = -5$ Then the given differential equation has unique solution.

Example:- Consider the differential equation

$$x^2+1\ y^{11}+2xy^1+\sin^2 xy=0$$

and it initial condition are $y\ 10 = y^1\ 0 = 0$ then the gives differential equation has

(a) unique solution
(b) infinite solution
(c) two solutions
(d) No solution

Solution: The given differential equation

$r\ x = 0$ and $y\ 0\ y^1\ 0 = 0$

Also, $P_0\ 2 = 2^2+1, P_1\ x = 2x, P_2\ x = \sin^2 x$ are continuous to the given differential equation has one and

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only one solution and the solution will be
$y\ x\ =0$ Hence, option (a) is correct.

Example: Show that the function $y=cx^2+x+3$ is a solution, through not unique, of the initial value problem $x^2\ y^{11}-2xy^1+2y=6$ with $y\ e\ =3, y^1\ 0\ =100\ -\infty,\infty$

Solution: Given equation is

$x^2\ y^{11}-2xy^1+2y=6$ …(1)

Out it given function is

$y\ 2\ =c2^2+x+3$ …(2)

Differentiate (2) w.r.t. x_1 we get

$y^1=2cx+1$ and $y^{11}=2c$ …(3)

$\therefore L.H.S$ of $1\ =x^2\ \ 2c\ -2x\ \ 2x+1\ +2\ \ cx^2+x+3$

$=6=R.H.S$ of (1)

Hence, (2) is a solution of (1). Again from (2) & (3) we get

$y\ 0\ =\ c\times 0\ +0+3=0, y^1\ 0\ =2c\times 0+1=1$

Comparing with the form of differential equation gives in fundamental theory, we get

$P_0\ x\ =x^2, P_1\ x\ =-2x, P_2\ x\ =2$ and r (x) = 6 which an continuous on $-\infty,\infty$

Since, $P_0\ x\ =x^2=0$ for some $x\in\ -\infty,\infty$ so the solution $y=cx^2+x+3$ is not unique. This can be also been that $y=c2^2+x+3$ is solution for any real value of c, so for example $y=2x^2+2+3$ and y = x² + x +3 are both solution of (1) with y (0) = 3 and y¹ (0) = 1.

Wronskian: at $y_1\ x\ y_2\ x\ ,...y_n\ x$ are n - red function having (n-1)th derivative on a red interval [a,b] then its Wronskian, deleted by W is defined to be the determinant

$$W\ x\ =\begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1^{\ 1} & y_2^{\ 1} & \cdots & y_n^{\ 2} \\ y_1^{\ 2} & y_2^{\ 2} & \cdots & y_n^{\ 2} \\ \vdots & \vdots & \vdots & \vdots \\ y_1^{\ n-1} & y_2^{\ n-1} & \cdots & y_n^{\ n-1} \end{vmatrix}$$

Linearly dependent and independent set of-functions: but $y_1\ x\ , y_2\ x_1\ ...$

$y_n\left(x\right)$ are n function then they are linearly dependent if there exist constants $c_1, c_2, \ldots c_n$ (lot all zero) such that

$c_1y_1+c_2y_2+...+c_a\ y_n=0$ …(1)

Where the functions are linearly independent if identity (1) implies that

$c_1=c_2=...=c_n=0$

Where as the functions are linearly independent if identity (1) implies that

$c_1=c_2=...=c_n=0$

Let $y_1, y_2...y_a$ be linearly dependent set of functions on $a\le x\le b$ and let each function be $n-1$ times differentiable in (a, b) Then the wronskian of the set of functions is identically zero.

- If the Wronskian of a set of functions is not zero, even at one point of the interval $a\le x\le b$ then the functions are linearly independent on $a\le x\le b$.

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4.9 Partial differential equations

An equation involving the partial differential coefficients, independent variables and dependent variables is called a partial differential equation.

e.g.

1) $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=z$

2) $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=0$

3) $z\frac{\partial^2 z^3}{\partial x^2}-3\frac{\partial z^4}{\partial x}+5\frac{\partial z}{\partial x \partial y}=5z$

The **order** of a partial differential equation is the order of the highest partial derivative in the equation.
The **degree** of a partial differential equation is the degree of the highest partial derivative in the equation.
Thus order of eqn1 is 1 and degree is 1.
Order of eqn2 is 2 and degree is 1.
Order of eqn3 is 2 and degree is 3.

Note:

If z is a function of two independent variables x and y then the different partial derivatives of z are denoted as p = $\frac{\partial z}{\partial x}, q=\frac{\partial z}{\partial y}, r=\frac{\partial^2 z}{\partial y^2}, s=\frac{\partial^2 z}{\partial y^2}$ and $t=\frac{\partial^2 z}{\partial x \partial y}$

4.9.1 Formation of partial differential equations:

- The P.D.E. can be formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions from a relation involving 3 or more variables.
- If the number of arbitrary constants to be eliminated is equal to the number of independent variables, the partial differential equations that arise are of the first order.
- If the number of arbitrary constants to be eliminated is more than the number of independent variables, the partial differential equations that arise are of the second or higher order.
- If the P.D.E is obtained by the elimination of arbitrary functions then the order of the P.D.E is, in general, equal to the number of arbitrary functions eliminated.

e.g.

1) Form P.D.E corresponding to z=ax+by, z=ax+by .....(1)

Differentiating (1) partially with respect to x,

$P=\frac{\partial z}{\partial x}=a$

Differentiating (1) partially with respect to y,

$q=\frac{\partial z}{\partial y}=b$

Substituting the values of a and b in (1),

z = px + qy, which is the required P.D.E.

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Solution. of a P.D.E.

The linear P.D.E. of the first order

These are P.D.E equations in which the 1st order partial derivatives p and q occur to the 1st degree only and are not multiplied together.

Lagrange's linear equation

The P.D.E. of the form Pp+Qq =R , where P and Q are functions of x,y,z is called Lagrange's linear eqn.

Method of soln. of Lagrange's linear equation:

i) Form the auxiliary equations.
ii) Solve these simultaneous equations either by the method of grouping or by the method of multipliers or both to get two independent solutions
u=a and v=b
iii) Write the complete solution as Φ(u,v)=0 or u=f(v) or v=g(u)
Note:- Solution of AE

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$$

1. Method of Grouping

e.g.

1. Solve $\frac{dx}{z^2y}=\frac{dy}{z^2x}=\frac{dz}{y^2x}$

Taking the first two ratios, $\frac{dx}{z^2y}=\frac{dy}{z^2x} \equiv \frac{dx}{y}=\frac{dy}{x} \Rightarrow$

x dx – y dy =0

Integrating, $\frac{x^2}{2}-\frac{y^2}{2}=\frac{a}{2}$

ie; $x^2-y^2=a$……..(1)

Taking the last two ratios $\frac{dy}{z^2x}=\frac{dz}{y^2x} \Rightarrow \frac{dy}{z^2}=\frac{dz}{y^2} \Rightarrow y^2dy-z^2dz=0$

Integrating, $\frac{y^3}{3}-\frac{z^3}{3}=\frac{b}{3}$

i.e. $y^3-z^3=b$…….(2)

Equations (1) and (2) taken together constitute the required soln. of the given simultaneous equations.

2. Method of multipliers

By a proper choice of multipliers l, m, n which are not necessarily constants we write,

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}=\frac{ldx+mdy+ndz}{lP+mQ+nR}$$

such *that* $lP+mQ+nR=0$

Then $l$dx+m dy+n dz=0 can be solved giving the soln. Φ(x,y,z)=c ……(1)
Again find another set of multipliers α,β,γ so that αP+βQ+γR=0 giving
α dx+β dy+γ dz=0 which on integration gives the soln. φ(x,y,z) = C $^{|}$……(2)

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Then. (1) and (2) together give the required soln. of $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$

Note: A P.D.E. may have more than one solns
e.g.

1. Solve

$$\frac{dx}{x(y^2-z^2)}=\frac{dy}{-y(z^2+x^2)}=\frac{dz}{z(x^2+y^2)}$$

Taking multpliers as x,y,z; each fraction = $\frac{xdx+ydy+zdz}{x^2(y^2-z^2)-y^2(z^2+x^2)+z^2(x^2+y^2)}$

$=\frac{xdx+ydy+zdz}{0}$

$\therefore$ x dx+y dy+z dz=0

Integrating , $\frac{x^2}{2}+\frac{y^2}{2}+\frac{z^2}{2}=\frac{c}{2}$

i.e. $x^2+y^2+z^2=c$ ……..(1)

Again taking the multipliers as 1/x,-1/y,-1/z, each fraction = $\frac{1/x\,dx+1/y\,dy+1/z\,dz}{0}$

$\therefore$ 1/x dx-1/y dy- 1/z dz=0
Integrating, log x- log y- log z= log C'
i.e. x/yz = C'; or x= C' yz……(2)
(1)and (2) together give the required soln.

4.9.2 Major Classifications of Common P.D.Es :

A general second order linear partial differential equation in two Cartesian variables can be written as

$$A\,x,y\,\frac{\partial^2 u}{\partial x^2}\;+\;B\,x,y\,\frac{\partial^2 u}{\partial x\,\partial y}\;+\;C\,x,y\,\frac{\partial^2 u}{\partial y^2}\;=\;f\left(x,y,u,\frac{\partial u}{\partial x},\frac{\partial u}{\partial y}\right)$$

Three main types arise, based on the value of $D = B^2 - 4AC$ (a discriminant):
Hyperbolic, wherever (x, y) is such that $D > 0$;
Parabolic, wherever (x, y) is such that $D = 0$;
Elliptic, wherever (x, y) is such that $D < 0$.

Among the most important partial differential equations in engineering are:

The **wave equation**: $\frac{\partial^2 u}{\partial t^2}\;=\;c^2\;\nabla^2 u$

or its one-dimensional special case $\frac{\partial^2 u}{\partial t^2}\;=\;c^2\;\frac{\partial^2 u}{\partial x^2}$ [which is hyperbolic everywhere]

(where u is the displacement and c is the speed of the wave);

The **heat** (or **diffusion**) **equation**: $\mu\rho\frac{\partial u}{\partial t}\;=\;K\;\nabla^2 u\;+\;\vec{\nabla}K\bullet\vec{\nabla}u$

a one-dimensional special case of which is

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$\frac{\partial u}{\partial t} = \frac{K}{\mu\rho}\frac{\partial^2 u}{\partial x^2}$ [which is parabolic everywhere]

(where u is the temperature, μ is the specific heat of the medium, ρ is the density and K is the thermal conductivity);

The potential (or **Laplace's**) **equation**: $\nabla^2 u = 0$

a special case of which is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ [which is elliptic everywhere]

The complete solution of a P.D.E requires additional information, in the form of initial conditions (values of the dependent variable and its first partial derivatives at $t = 0$), boundary conditions (values of the dependent variable on the boundary of the domain) or some combination of these conditions.

4.9.2.1 The Heat Equation

For a material of constant density ρ, constant specific heat μ and constant thermal conductivity K, the partial differential equation governing the temperature u at any location (x, y, z) and any time t is

$$\frac{\partial u}{\partial t} = k\nabla^2 u \ , \quad \text{where} \quad k = \frac{K}{\mu\rho}$$

Example 1

Heat is conducted along a thin homogeneous bar extending from $x = 0$ to $x = L$. There is no heat loss from the sides of the bar. The two ends of the bar are maintained at temperatures T_1 (at $x = 0$) and T_2 (at $x = L$). The initial temperature throughout the bar at the cross-section x is $f(x)$.

Find the temperature at any point in the bar at any subsequent time.

The partial differential equation governing the temperature $u(x, t)$ in the bar is

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2} \quad \text{[Parabolic]}$$

together with the boundary conditions
$u(0, t) = T_1$ and $u(L, t) = T_2$

and the initial condition
$u(x, 0) = f(x)$

[Note that if an end of the bar is insulated, instead of being maintained at a constant temperature, then the boundary condition changes to $\frac{\partial u}{\partial t}(0, t) = 0$ or $\frac{\partial u}{\partial t}(L, t) = 0$.]

Attempt a solution by the method of separation of variables.
$u(x, t) = X(x)\,T(t)$

$$\Rightarrow \quad XT' = kX''T \qquad \Rightarrow \quad \frac{T'}{T} = k\frac{X''}{X} = c$$

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Again, when a function of t only equals a function of x only, both functions must equal the same absolute constant. Unfortunately, the two boundary conditions cannot both be satisfied unless $T_1 = T_2 = 0$. Therefore we need to treat this more general case as a perturbation of the simpler ($T_1 = T_2 = 0$) case.

Example 1 (continued)
Let $u(x, t) = v(x, t) + g(x)$
Substitute this into the P.D.E:

$$\frac{\partial}{\partial t} v\ x,t\ + g\ x\ = k\frac{\partial^2}{\partial x^2} v\ x,t\ + g\ x\ \Rightarrow\ \frac{\partial v}{\partial t} = k\left(\frac{\partial^2 v}{\partial x^2} + g''\ x\right)$$

This is the standard heat P.D.E for v if we choose g such that $g''(x) = 0$.
$g(x)$ must therefore be a linear function of x.

We want the perturbation function $g(x)$ to be such that
$u(0, t) = T_1$, $u(L, t) = T_2$
and
$v(0, t) = v(L, t) = 0$

Therefore $g(x)$ must be the linear function for which $g(0) = T_1$ and $g(L) = T_2$.
It follows that

$$g\ x\ = \left(\frac{T_2 - T_1}{L}\right)x + T_1$$

and we now have the simpler problem

$$\frac{\partial v}{\partial t} = k\frac{\partial^2 v}{\partial x^2}$$

together with the boundary conditions

$$v(0, t) = v(L, t) = 0$$

and the initial condition

$$v(x, 0) = f(x) - g(x)$$

Now try separation of variables on $v(x, t)$:
$v(x, t) = X(x)\,T(t)$

$$\Rightarrow\ \ X\,T' = k\,X''T\ \ \ \Rightarrow \frac{1}{k}\frac{T'}{T} = \frac{X''}{X} = c$$

But $v(0, t) = v(L, t) = 0\ \Rightarrow\ X(0) = X(L) = 0$

This requires c to be a negative constant, say $-\lambda^2$.
The solution is very similar to that for the wave equation on a finite string with fixed ends (section 4.3). The eigenvalues are $\lambda = \frac{n\pi}{L}$ and the corresponding eigenfunctions are any non-zero constant multiples of

$$X_n\ x\ = \sin\left(\frac{n\pi x}{L}\right)$$

Example 1 (continued)

The ODE for $T(t)$ becomes

$$T' + \left(\frac{n\pi}{L}\right)^2 kT = 0$$

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whose general solution is

$T_n\ t\ = c_n e^{-n^2\pi^2 kt/L^2}$

Therefore

$$v_n\ x,t\ = X_n\ x\ T_n\ t\ = c_n \sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$$

If the initial temperature distribution $f(x)-g(x)$ is a simple multiple of $\sin\left(\frac{n\pi x}{L}\right)$ for some integer n, then the solution for v is just $v\ x,t\ = c_n \sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$.

Otherwise, we must attempt a superposition of solutions.

$$v\ x,t\ = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$$

such that $v\ x,0\ = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) = f\ x\ - g\ x\ .$

The Fourier sine series coefficients are $c_n\ = \frac{2}{L}\int_0^L f\ z\ - g\ z\ \sin\left(\frac{n\pi z}{L}\right)dz$

so that the complete solution for $v(x, t)$ is

$$v\ x,t\ = \frac{2}{L}\sum_{n=1}^{\infty}\left(\int_0^L\left(f\ z\ - \frac{T_2-T_1}{L}z\ - T_1\right)\sin\left(\frac{n\pi z}{L}\right)dz\right)\sin\left(\frac{n\pi x}{L}\right)\exp\left(-\frac{n^2\pi^2 kt}{L^2}\right)$$

and the complete solution for $u(x, t)$ is

$$u\ x,t\ = v\ x,t\ + \left(\frac{T_2-T_1}{L}\right)x\ + T_1$$

Note how this solution can be partitioned into a transient part $v(x, t)$ (which decays to zero as t increases) and a steady-state part $g(x)$ which is the limiting value that the temperature distribution approaches.

Example 1 (continued)

As a specific example, let $k = 9$, $T_1 = 100$, $T_2 = 200$, $L = 2$ and $f(x) = 145x^2 - 240x + 100$, (for which $f(0) = 100$, $f(2) = 200$ and $f(x) > 0\ \forall x$).

Then $g\ x\ = \frac{200-100}{2}x\ +\ 100\ =\ 50x\ +\ 100$

The Fourier sine series coefficients are

$$c_n\ = \int_0^2 145z^2-240z+100\ -\ 50z+100\ \sin\left(\frac{n\pi z}{2}\right)dz$$

$$\Rightarrow\ \ c_n\ = 145\int_0^2 z^2-2z\ \sin\left(\frac{n\pi z}{2}\right)dz$$

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D	I
$z^2 - 2z$	$\sin\left(\frac{n\pi z}{2}\right)$
$2z - 2$	$-\left(\frac{2}{n\pi}\right)\cos\left(\frac{n\pi z}{2}\right)$
2	$-\left(\frac{2}{n\pi}\right)^2 \sin\left(\frac{n\pi z}{2}\right)$
0	$\left(\frac{2}{n\pi}\right)^3 \cos\left(\frac{n\pi z}{2}\right)$

$$\Rightarrow \quad c_n = 145\left[\left(-z^2+2z\,\frac{2}{n\pi}+\frac{16}{n\pi^{\,3}}\right)\cos\left(\frac{n\pi z}{2}\right) + \frac{8\;z-1}{n\pi^{\,2}}\sin\left(\frac{n\pi z}{2}\right)\right]_{z=0}^{z=2}$$

$$\Rightarrow \quad c_n = 145\left[z\;2-z\,\frac{2}{n\pi}+\frac{16}{n\pi^{\,3}}\right)\cos\left(\frac{n\pi z}{2}\right) + \frac{8\;z-1}{n\pi^{\,2}}\sin\left(\frac{n\pi z}{2}\right)\Bigg]_{z=0}^{z=2}$$

$$\Rightarrow \quad c_n = \frac{2320}{n\pi^{\,3}}\;-1^{\,n}-1$$

The complete solution is

$$u\;x,t\;=\;50x+100\;-\;\frac{2320}{\pi^3}\sum_{n=1}^{\infty}\left(\frac{1-\;-1^{\,n}}{n^3}\right)\sin\left(\frac{n\pi x}{2}\right)\exp\left(-\frac{9n^2\pi^2 t}{4}\right)$$

4.9.2.2 <u>The Wave Equation – Vibrating Finite String</u>

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2\,\nabla^2 u$$

If $u(x, t)$ is the vertical displacement of a point at location x on a vibrating string at time t, then the governing P.D.E is

$$\frac{\partial^2 u}{\partial t^2} = c^2\,\frac{\partial^2 u}{\partial x^2}$$

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If $u(x, y, t)$ is the vertical displacement of a point at location (x, y) on a vibrating membrane at time t, then the governing P.D.E is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

or, in plane polar coordinates (r, θ), (appropriate for a circular drum),

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} \right)$$

Example 1

An elastic string of length L is fixed at both ends ($x = 0$ and $x = L$). The string is displaced into the form $y = f(x)$ and is released from rest. Find the displacement $y(x, t)$ at all locations on the string $(0 < x < L)$ and at all subsequent times $(t > 0)$.

The boundary value problem for the displacement function $y(x, t)$ is:

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \qquad \text{for } 0 < x < L \quad \text{and} \quad t > 0$$

Both ends fixed for all time: $y(0, t) = y(L, t) = 0$ for $t \geq 0$

Initial configuration of string: $y(x, 0) = f(x)$ for $0 \leq x \leq L$

String released from rest: $\left.\frac{\partial y}{\partial t}\right|_{(x,0)} = 0 \qquad \text{for } 0 \leq x \leq L$

Example 1 (continued)

Separation of Variables (or Fourier Method)

Attempt a solution of the form $y(x, t) = X(x)\,T(t)$

Substitute $y(x, t) = X(x)\,T(t)$ into the P.D.E:

$$\frac{\partial^2}{\partial t^2} X\ x\ T\ t = c^2 \frac{\partial^2}{\partial x^2} X\ x\ T\ t \qquad \Rightarrow \quad X\frac{d^2T}{dt^2} = c^2 \frac{d^2X}{dx^2} T$$

$$\Rightarrow \quad \frac{1}{c^2}\frac{1}{T}\frac{d^2T}{dt^2} = \frac{1}{X}\frac{d^2X}{dx^2}$$

The left hand side of this equation is a function of t only. At any instant t it must have the same value at all values of x. Therefore the right hand side, which is a function of x only, must at any one instant have that same value at all values of x.

By a similar argument, the right hand side of this equation is a function of x only. At any location x it must have the same value at all times t. Therefore the left hand side, which is a function of t only, must at any one location have that same value at all times t.

Thus both sides of this differential equation must be the same absolute constant, which we shall represent for now by $-k$.

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$$\frac{1}{X}\frac{d^2X}{dx^2} = -k \quad \Rightarrow \quad \frac{d^2X}{dx^2} + kX = 0$$

The general solution of this simple second order ODE is a combination of two exponential functions of x if $k < 0$, is a linear function of x if $k = 0$ and is a combination of sine and cosine functions of x if $k > 0$.

It is not possible for both ends of the string to be fixed for all time in the first two cases (unless we admit the trivial solution $y(x, t) \equiv 0$, a string that never moves from its equilibrium position). Therefore $k > 0$. Replace k by λ^2 (guaranteed positive for all real λ except $\lambda = 0$).

We now have the pair of ODEs

$$\frac{d^2X}{dx^2} + \lambda^2 X = 0 \quad \text{and} \quad \frac{d^2T}{dt^2} + \lambda^2 c^2 T = 0$$

The general solutions are

$$X(x) = A\cos\lambda x + B\sin\lambda x \quad \text{and} \quad T(t) = C\cos\lambda ct + D\sin\lambda ct$$

respectively, where A, B, C and D are arbitrary constants.

Example 1 (continued)

Consider the boundary conditions:

$$y(0,t) = X(0)T(t) = 0 \quad \forall t \geq 0$$

For a non-trivial solution, this requires $X(0) = 0 \Rightarrow A = 0$.

$$y(L,t) = X(L)T(t) = 0 \quad \forall t \geq 0 \quad \Rightarrow \quad X(L) = 0$$

$$\Rightarrow \quad B\sin\lambda L = 0 \quad \Rightarrow \quad \lambda_n = \frac{n\pi}{L}, \quad n \in \mathbb{Z}$$

We now have a solution only for a discrete set of eigenvalues λ_n, with corresponding eigenfunctions

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \ldots$$

Consider the initial condition:

$$\left.\frac{\partial y}{\partial t}\right|_{(x,0)} = X(x)T'(0) = 0 \quad \forall x \quad \Rightarrow \quad T'(0) = 0$$

$$T'(t) = -C\lambda c\sin\lambda ct + D\lambda c\cos\lambda ct \quad \Rightarrow \quad T'(0) = D\lambda c = 0 \quad \Rightarrow \quad D = 0$$

Therefore our complete solution for $y(x, t)$ is now some linear combination of

$$y_n(x,t) = X_n(x)T_n(t) = C_n\sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{n\pi ct}{L}\right), \quad n = 1, 2, 3, \ldots$$

There is one condition remaining to be satisfied.
The initial configuration of the string is: $y(x, 0) = f(x)$ for $0 \leq x \leq L$.

$$\Rightarrow \quad y(x,0) = \sum_{n=1}^{\infty} C_n\sin\left(\frac{n\pi x}{L}\right) = f(x)$$

This is precisely the Fourier sine series expansion of $f(x)$ on $[0, L]$!
From Fourier series theory (Chapter 3), the coefficients C_n are

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$$C_n = \frac{2}{L}\int_0^L f\ u\ \sin\left(\frac{n\pi u}{L}\right)du$$

Therefore our complete solution is

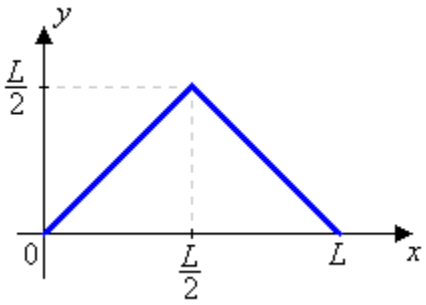
$$y\ x,t\ = \frac{2}{L}\sum_{n=1}^{\infty}\left(\int_0^L f\ u\ \sin\left(\frac{n\pi u}{L}\right)du\right)\sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{n\pi ct}{L}\right)$$

Example 1 (continued)

This solution is valid for any initial displacement function $f(x)$ that is continuous with a piece-wise continuous derivative on $[0, L]$ with $f(0) = f(L) = 0$.

If the initial displacement is itself sinusoidal $\left(f\ x\ = a\sin\left(\frac{n\pi x}{L}\right)\ \text{ for some }\ n\in\mathbb{N}\right)$, then the complete solution is a single term from the infinite series,

$$y\ x,t\ = a\sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{n\pi ct}{L}\right).$$



D	***I***
u	$\sin\left(\frac{n\pi u}{L}\right)$
1	$-\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi u}{L}\right)$
0	$-\left(\frac{L}{n\pi}\right)^2\sin\left(\frac{n\pi u}{L}\right)$

Suppose that the initial configuration is triangular:

$$y\ x,0\ = f\ x\ = \begin{cases} x & 0\le x\le \frac{1}{2}L \\ L-x & \frac{1}{2}L < x\le L \end{cases}$$

Then the Fourier sine coefficients are

$$C_n = \frac{2}{L}\int_0^L f\ u\ \sin\left(\frac{n\pi u}{L}\right)du$$

$$= \frac{2}{L}\int_0^{L/2} u\sin\left(\frac{n\pi u}{L}\right)du + \frac{2}{L}\int_{L/2}^{L} L-u\ \sin\left(\frac{n\pi u}{L}\right)du$$

$$= \frac{2}{L}\cdot\left(\frac{L}{n\pi}\right)^2\left\{\left[-\left(\frac{n\pi u}{L}\right)\cos\left(\frac{n\pi u}{L}\right) + \sin\left(\frac{n\pi u}{L}\right)\right]_0^{L/2}\right.$$

$$\left. + \left[-\left(\frac{n\pi\ L-u}{L}\right)\cos\left(\frac{n\pi u}{L}\right) - \sin\left(\frac{n\pi u}{L}\right)\right]_{L/2}^{L}\right\}$$

D	I
$L-u$	$\sin\left(\frac{n\pi u}{L}\right)$
-1	$-\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi u}{L}\right)$
0	$-\left(\frac{L}{n\pi}\right)^2\sin\left(\frac{n\pi u}{L}\right)$

Example 1 (continued)

$$= \frac{2L}{n\pi^{\ 2}}\left\{\left(-\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right)\right) - 0-0\right.$$

$$\left. + \ -0+0\ - \left(-\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi}{2}\right) - \sin\left(\frac{n\pi}{2}\right)\right)\right\}$$

$$= \frac{4L}{n\pi^{\ 2}}\sin\left(\frac{n\pi}{2}\right) \quad \text{But} \quad \sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & n \text{ even} \\ \pm 1 & n \text{ odd} \end{cases}$$

$$\sin\left(\frac{2k-1\ \pi}{2}\right) = -1^{\ k+1}, \quad k \in \mathbb{N}$$

Therefore, sum over the odd integer values of n only ($n = 2k-1$).

$$C_k = \frac{4L}{2k-1\ \pi^{\ 2}}\ -1^{\ k+1}$$

and

$$y\ x,t\ = \frac{4L}{\pi^2}\sum_{k=1}^{\infty}\frac{-1^{\ k+1}}{2k-1^{\ 2}}\sin\left(\frac{2k-1\ \pi x}{L}\right)\cos\left(\frac{2k-1\ \pi ct}{L}\right)$$

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Example 2

An elastic string of length L is fixed at both ends ($x = 0$ and $x = L$). The string is initially in its equilibrium state [$y(x, 0) = 0$ for all x] and is released with the initial velocity $\left.\frac{\partial y}{\partial t}\right|_{(x,0)} = g\ x$. Find the displacement $y(x, t)$ at all locations on the string ($0 < x < L$) and at all subsequent times ($t > 0$).

The boundary value problem for the displacement function $y(x, t)$ is:

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{for } 0 < x < L \text{ and } t > 0$$

Both ends fixed for all time: $y(0, t) = y(L, t) = 0$ for $t \geq 0$

Initial configuration of string: $y(x, 0) = 0$ for $0 \leq x \leq L$

String released with initial velocity: $\left.\frac{\partial y}{\partial t}\right|_{(x,0)} = g\ x \quad \text{for } 0 \leq x \leq L$

As before, attempt a solution by the method of the **separation of variables**.

Substitute $y(x, t) = X(x)\,T(t)$ into the P.D.E:

$$\frac{\partial^2}{\partial t^2} X\ x\ T\ t = c^2 \frac{\partial^2}{\partial x^2} X\ x\ T\ t \quad \Rightarrow \quad X \frac{d^2T}{dt^2} = c^2 \frac{d^2X}{dx^2} T$$

Again, each side must be a negative constant.

$$\Rightarrow \quad \frac{1}{c^2T}\frac{d^2T}{dt^2} = \frac{1}{X}\frac{d^2X}{dx^2} = -\lambda^2$$

We now have the pair of ODEs

$$\frac{d^2X}{dx^2} + \lambda^2 X = 0 \quad \text{and} \quad \frac{d^2T}{dt^2} + \lambda^2 c^2 T = 0$$

The general solutions are

$$X\ x = A\cos\ \lambda x + B\sin\ \lambda x \quad \text{and} \quad T\ t = C\cos\ \lambda ct + D\sin\ \lambda ct$$

respectively, where A, B, C and D are arbitrary constants.

Example 2 (continued)

Consider the boundary conditions:

$$y\ 0,t = X\ 0\ T\ t = 0 \quad \forall t \geq 0$$

For a non-trivial solution, this requires $X\ 0 = 0 \quad \Rightarrow \quad A = 0$.

$$y\ L,t = X\ L\ T\ t = 0 \quad \forall t \geq 0 \quad \Rightarrow \quad X\ L = 0$$

$$\Rightarrow \quad B\sin\ \lambda L = 0 \quad \Rightarrow \quad \lambda_n = \frac{n\pi}{L}, \quad n \in \mathbb{Z}$$

We now have a solution only for a discrete set of eigenvalues λ_n, with corresponding eigenfunctions

$$X_n\ x = \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \ldots$$

and

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$$y_n\ x,t\ =\ X_n\ x\ T_n\ t\ =\ \sin\left(\frac{n\pi x}{L}\right)T_n\ t\ ,\quad n=1,2,3,\ldots$$

So far, the solution has been identical to Example 4.3.1.

Consider the initial condition $y(x, 0) = 0$:

$$y\ x,0\ =0\ \Rightarrow\ X\ x\ T\ 0\ =0\ \ \forall x\ \Rightarrow\ T\ 0\ =0$$

The initial value problem for $T(t)$ is now

$$T''\ +\ \lambda^2c^2T\ =\ 0,\ \ T\ 0\ =0,\qquad \text{where}\quad \lambda=\frac{n\pi}{L}$$

the solution to which is

$$T_n\ t\ =\ C_n\sin\left(\frac{n\pi ct}{L}\right),\qquad n\in\mathbb{N}$$

Our eigenfunctions for y are now

$$y_n\ x,t\ =\ X_n\ x\ T_n\ t\ =\ C_n\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi ct}{L}\right),\qquad n\in\mathbb{N}$$

Example 2 (continued)

Differentiate term by term and impose the initial velocity condition:

$$\left.\frac{\partial y}{\partial t}\right|_{x,0}\ =\ \sum_{n=1}^{\infty}C_n\left(\frac{n\pi c}{L}\right)\sin\left(\frac{n\pi x}{L}\right)\ =\ g\ x$$

which is just the Fourier sine series expansion for the function $g(x)$.
The coefficients of the expansion are

$$C_n\frac{n\pi c}{L}\ =\ \frac{2}{L}\int_0^L g\ u\ \sin\left(\frac{n\pi u}{L}\right)du$$

which leads to the complete solution

$$y\ x,t\ =\ \frac{2}{\pi c}\sum_{n=1}^{\infty}\frac{1}{n}\left(\int_0^L g\ u\ \sin\left(\frac{n\pi u}{L}\right)du\right)\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi ct}{L}\right)$$

This solution is valid for any initial velocity function $g(x)$ that is continuous with a piece-wise continuous derivative on $[0, L]$ with $g(0) = g(L) = 0$.

The solutions for Examples 1 and 2 may be superposed.

Let $y_1(x, t)$ be the solution for initial displacement $f(x)$ and zero initial velocity.
Let $y_2(x, t)$ be the solution for zero initial displacement and initial velocity $g(x)$.

Then $y(x, t) = y_1(x, t) + y_2(x, t)$ satisfies the wave equation
(the sum of any two solutions of a linear homogeneous P.D.E is also a solution),
and satisfies the boundary conditions $y(0, t) = y(L, t) = 0$.

$y(x, 0) = y_1(x, 0) + y_2(x, 0) = f(x) + 0,$
which satisfies the condition for initial displacement $f(x)$.

$y_t(x, 0) = y_{1t}(x, 0) + y_{2t}(x, 0) = 0 + g(x),$
which satisfies the condition for initial velocity $g(x)$.

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Therefore the sum of the two solutions is the complete solution for initial displacement $f(x)$ **and** initial velocity $g(x)$:

$$y(x,t) = \frac{2}{L}\sum_{n=1}^{\infty}\left(\int_0^L f(u)\sin\left(\frac{n\pi u}{L}\right)du\right)\sin\left(\frac{n\pi x}{L}\right)\cos\left(\frac{n\pi ct}{L}\right)$$
$$+ \frac{2}{\pi c}\sum_{n=1}^{\infty}\frac{1}{n}\left(\int_0^L g(u)\sin\left(\frac{n\pi u}{L}\right)du\right)\sin\left(\frac{n\pi x}{L}\right)\sin\left(\frac{n\pi ct}{L}\right)$$

SET- 8
First order differential Equation

1. The degree of differential equation is
$$\left(\frac{d^3y}{dx^3}\right)^{2/3} + \left(\frac{d^3y}{dx^3}\right)^{3/2} = 0$$

a) 3
b) 5
c) 4
d) 9

2. The differential equation $2x\frac{dy}{dx} = 3 - y$ *has solution*

a) $y = 3 + \sqrt{\frac{c}{x}}$
b) $y = 3 - \sqrt{\frac{c}{x}}$
c) $y = 3 + c\sqrt{x}$
d) $y = 3 - c\sqrt{x}$

3. The Integrating factor of the differential eqaution $(x^3 + xy^4)dx + 2y^3\,dy = 0$ *is given by*

a) e^{-x}
b) e^{x^2}
c) e^{x}
d) e^{-x^2}

4. The integrating factor of the differential equation $x^3y - 2xy^2\ dx + x^3 - 3x^2y\ dy$ *is*

a) $\frac{1}{x^2y^2}$
b) $\frac{1}{x\ y}$
c) xy
d) x^2y^2

5. The differential equation $x - y^2\ dx + 2xydy = 0$ *has the solution*

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a) $xe^{y^2/x} = A$

b) $ye^{2y/x} = A$

c) $ye^{x/y^2} = A$

d) $xe^{2x/y^2} = A$

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6. The solution of the differential equation $\frac{dy}{dx}=\frac{2x-y+1}{x+2y-3}is$

a) $xy+y^2+x^2+3y-x=c$

b) $xy+y^2-x^2-3y-x=c$

c) $xy+y^2+x^2-3y-x=c$

d) $xy+y^2-x^2-3y+x=c$

7. The Integrating Factor of the differential eqauation is $\frac{dy}{dx}+\frac{y}{x}=x^3-3\ is$

a) $\log x$

b) $x\log x$

c) $-x$

d) x

8. The solution of the differential eqaution $y\ x-y\ dy-y^2\,dx=0$ is

a) $y=Ce^{\frac{y}{x}}$

b) $y=Ce^{-\frac{y}{x}}$

c) $y=Ce^{\frac{x}{y}}$

d) $y=Ce^{-\frac{x}{y}}$

9. The solution of the differential equation $x^2+y^2\ dx-2xy\,dy=0 is$

a) $x^2+cx-y^2=0$

b) $(x-y)^2=cx$

c) $x+y=2xy=c$

d) $y=x^2-2x\ +c$

10. The differential equation $\left(x^2y^3+xy\right)\frac{dy}{dx}=1 has integrating\ factor$

a) e^{y^2}

b) $e^{\frac{1}{2}y^2}$

c) $e^{\frac{1}{2}x^2}$

d) $e^{-\frac{1}{2}y^2}$

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Solutions:

1. Ans: b)

We Have $\left(\frac{d^3y}{dx^3}\right)^{2/3}+\left(\frac{d^3y}{dx^3}\right)^{3/2}=0$

Or $\left(\frac{d^3y}{dx^3}\right)^{5/6}+1=0$

Thus degree is 5
Hence B is correct option

2. Ans: b)

This is separable and written as $\frac{2dy}{3-y}=\frac{dx}{x}$

Integrating $-2\ln\ 3-y\ =\ln x=\ln c$

Or $\ln c\ \ 3-y\ ^{-2}-\ln x$

Or $y=3-\sqrt{\frac{c}{x}}$

3. Ans:b)

We have $x^3+xy^4\ \ dx+2y^3\,dy=0$

Hare $M=x^3+xy^4,\quad N=2y^3$

And $\frac{\partial M}{\partial y}=4xy^3,\frac{\partial N}{\partial x}=0$

Now $\frac{\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)}{N}=\frac{4xy^2}{2y^3}=2x$

$I.F.=e^{\int 2xdx=e^{x^2}}$

Hence B is correct option

4. Ans: a)

We have $x^2\ y-2xy^2\ \ dx+\ x^3-3x^2y\ \ dy=0$

Which is of the from $yf\ \ x,y\ \ dx+xy\ \ x,y\ \ dy=0$

$M_z=N_y=x^3y-2x^2y^2-x^3y+3x^2y^2$

$=x^2\ y^2\neq 0$

Thus Integrating Factor $\frac{1}{Mx-Ny}=\frac{1}{x^2y^2}$

Hence A is correct option

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5. Ans: a)

We Have $x-y^2\ dx+2xydy=0$

This is of the form $Mdx+Ndy=0$

Here $\frac{\partial M}{\partial y}=-2y\ and\ \frac{\partial N}{\partial x}=2y$

Now $\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)=\frac{-2y-2y}{2xy}=\frac{-2}{x}$

$I.F.=e^{\int\frac{-2}{x}dz}$

$=e^{-2\log x}=\frac{1}{x^2}$

Now multiplying the integrating factor in the given equation, we get

$$\left(\frac{1}{x}-\frac{y^2}{x^2}\right)dx+\frac{2y}{x}dy=0$$

The solution is $\int\left(\frac{1}{x}-\frac{y^2}{x^2}\right)dx=\log A$

Or, $\log\ x+\frac{y^2}{x}=\log A$

Or, $\log\ \frac{A}{x}=\frac{y^2}{x}$

Or, $A=xe^{y^2/x}$

Hence A is correct option

6. Ans: b)

We have $\frac{dy}{dx}=\frac{2x-y+1}{x+2y-3}$

Or $2x-y+1\ dx-\ x+2y-3\ dy=0$

This is exact differential equation because

$$\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}=-1$$

Thus solution is

$$\int\ 2x-y+1\ dx-\int\ 2y-3\ dy=c$$

Or $x^2-xy+x-y^2+3y=c$

Or $xy+y^2-x^2-cy-x=c$

Hence B is correct option

7. Ans: d)

We have $\frac{dy}{dx}+\frac{y}{x}=x^3-3$

$I.F.=e^{\int\frac{1}{x}dx_{xz}\ e^{\log x}=x}$

Hence D is correct option

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8. Ans: b)

This differential equation is linear,

Hence $I.F. = e^{2\int \tan x\, dx}$

Or $e^{2\log \sec 2} = \sec^2 x$

Hence B is correct option

9. Ans: a)

We have $x^2 + y^2\ dx - 2xy\, dy = 0$

Or $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$

Which is homogeneous differential equation.

Putting $y = \upsilon x \, or \, \frac{dy}{dx} = \upsilon + x\frac{d\upsilon}{dx} \, we \, get$

$$\upsilon + x\frac{dv}{dx} = \frac{x^2 + \upsilon^2 x^2}{2x, \upsilon x}$$

Or $x\frac{d\upsilon}{dx} = \frac{1-\upsilon^2}{2\upsilon}$

Or $\frac{2\upsilon}{1-\upsilon^2}d\upsilon = \frac{dx}{x}$

Integrating, $-\log\ 1-\upsilon^2 = \log x - \log c$

$$\frac{1}{1-\upsilon^2} = \frac{x}{c}$$

Or $\frac{x^2}{x^2 - y^2} = \frac{x}{c}$

Or $x^2 - y^2 = cx = 0,$

Hence A is correct option

10. Ans: b)

We have $x^2 y^3 + xy\ \frac{dy}{dx} = 1$

Or $\frac{dx}{dy} - xy = x^2 y^3$

This is a Bernoulli's equation thus

$$\frac{1}{x^2}\frac{dx}{dy} - \frac{y}{x} = y^3$$

Let $\frac{1}{x} = \upsilon \, or - \frac{1}{x^2}\frac{dx}{dy} = \frac{d\upsilon}{dy}$

From (1) $-\frac{d\upsilon}{dy} - \upsilon y = y^3$

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Or, $\frac{dv}{dy}+vy=-y^3$

This is a linear differential equation

Thus $I.F.=e^{\int ydy}=e^{\frac{1}{2}y^2}$

Hence B is correct option

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SET-9
Higher Order Differential equation

1. The general solution of the differential equation $D^2-4D+4\ y=0$, is of the form (given $D=d/dx$), and C and D are constants.
   a) Ce^{2x}
   b) $Ce^{2x}+Dxe^{-2x}$
   c) $Ce^{2x}+Dxe^{2x}$
   d) $Ce^{-2x}+Dxe^{-2x}$

2. The general solution of the differential equation $\frac{d^4y}{dx^4}+2\frac{d^2y}{dx^2}+y=0$ *is*
   a) $y=(C_1+C_2x)Sin\,x+(C_3+C_4x)Cos\,x$
   b) $y=C_1Sin\,x+C_2Cosx+Sin\,x+x\,Cos\,x$
   c) $y=C_1Sin\,x+C_2Cosx+C_3\,Tanx+C_4\,Cot\,x$
   d) $y=C_1Sin\,x+C_2Cosx+C_3\,x+C_4$

3. The general solution of the differential equation $\frac{d^2y}{dx^2}+\ 3a-1\ \frac{dy}{dx}-3ay=0$ *is*
   a) $C_1e^{x}+C_2e^{3x}$
   b) $C_1e^{-x}+C_2e^{3x}$
   c) $C_1e^{-x}+C_2e^{-3x}$
   d) $C_1e^{x}+C_2e^{-3x}$

4. The solution of the differential eqaution $\frac{d^2y}{dx^2}-8\frac{dy}{dx}\ +15=0$ is
   a) $y=C_1e^{3x}+C_2e^{5x}$
   b) $y=C_1e^{2x}+C_2e^{3x}$
   c) $y=C_1e^{4x}+C_2e^{5x}$
   d) *None of these*

5. The solution of the differential equation $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=0$ is given by
   a) $y=e^{2x}+e^{-3x}$
   b) $y=e^{2x}+e^{3x}$
   c) $y=e^{-2x}+e^{3x}$
   d) $y=e^{-2x}+e^{-3x}$

6. Whoch of the following pair of functions is not a linearly independent pair of solutions of $\frac{d^2y}{dx^2}+9y=0$
   a) $Sin3x,\ Sin3x-Cos3x$
   b) $Sin3x+Cos3x,\ 3Sin\,x-4Sin^3x$

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c) $Sin3x,\ Sin3xCos3x$

d) $Sin3x+Cos3x,\ 4Cos^3x-3Cos\,x$

7. $y=x$ is a solution of $x^2\dfrac{d^2x}{dy^2}+x\dfrac{dx}{dy}-y=0.$ The second linearly independent solution of the above equation is

a) $1/x$

b) $1/x^2$

c) x^2

d) x^n

8. The general solution of the differentail equaton $x^2\dfrac{d^2y}{dx^2}-x\dfrac{dy}{dx}+y=0$ *is*

a) $C_1x+C_2x^2$

b) $C_1x+C_2\log x$

c) $C_1x+C_2x^2\,logx$

d) $C_1x+C_2\,x\,logx$

9. The differential equation whose linearly independent solutions are $\cos 2x, Sin2x$ and e^{-x} is

a) $D^3+D^2+4D+4\ y=0$

b) $D^3-D^2+4D-4\ y=0$

c) $D^3+D^2-4D-4\ y=0$

d) $D^3-D^2-4D+4\ y=0$

10. The differential equation whose linearly independent solutions are $Cos2x,\ Sin2x$ and e^{-x} is

a) $D^3+D^2+4D+4\ y=0$

b) $D^3-D^2+4D-4\ y=0$

c) $D^3+D^2-4D-4\ y=0$

d) $D^3-D^2-4D+4\ y=0$

11. The Particular Integral of $D^2+4\ y=Cos\,2x$ is

a) $xSin4x$

b) $\dfrac{1}{4}Cos2x$

c) $xCos4x$

d) $\dfrac{1}{4}xSin2x$

12. A particular Integral of the differential equation $(D^2+4)y=x$ *is*

a) xe^{-2x}

b) $xCos2x$

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c) $x\,Sin\,2x$

d) $x/4$

13. The general siolution of the differential eqaution $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+4y=\frac{e^{2x}}{x}$ is

a) $C_1+C_2x+x\,logx-\frac{x^2}{2}e^{2x}$

b) $(C_1+C_2x+x\,logx-x)e^{2x}$

c) $C_1e^{2x}+C_2e^{-2x}+xe^{2x}logx-\frac{x^2}{2}e^{2x}$

d) $C_1e^{2x}+C_2e^{-2x}+xe^{2x}\,logx-x-$

14. Find the particular integral of $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+3y=3e^{2x}$ is

a) $\frac{e^{2x}}{15}$

b) $\frac{e^{2x}}{5}$

c) $3e^{2x}$

d) e^{-x}

15. The differential equation $2D+1\,^2\,y=4e^{-x/2}$ has solution

a) $y=\ C_1+C_2x\ e^{-x/2}$

b) $y=\ C_1+C_2x+x^2\ \ e^{-x/2}$

c) $y=\left(C_1+C_2x+\frac{1}{2}x^2\right)e^{-x/2}$

d) $y=\left(C_1+C_2x+\frac{1}{4}x^2\right)e^{-x/2}$

16. The differential equation $\frac{d^2y}{dx^2}-\frac{dy}{dx}-2y=10Cos\,x$ has solution

a) $y=C_1e^{-x}+C_2e^{-x}-Sinx-3Cosx$

b) $y=C_1e^{-x}+C_2e^{-2x}-Sinx-3Cosx$

c) $y=C_1e^{-x}+C_2e^{2x}-3x-3Cosx$

d) $y=C_1e^{-x}+C_2e^{2x}-3Cosx$

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17. The differential equation $\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=e^{3x}$ has solution

a) $y=C_1e^{-x}+C_2e^{-2x}+\frac{1}{2}\ e^{3x}$

b) $y=C_1e^{x}+C_2e^{2x}+\frac{1}{2}\ e^{3x}$

c) $y=C_1e^{x}+C_2e^{2x}+\frac{1}{2}\ e^{-3x}$

d) $y=C_1e^{-x}+C_2e^{2x}+\frac{1}{2}\ e^{3x}$

18. The differential equation satisfying the relation $x=ACos\ mt-\alpha\ is$

a) $\frac{d^2x}{dt^2}=-\alpha^2 x$

b) $\frac{dx}{dt}=-\alpha x^2$

c) $\frac{dx}{dt}=-m^2 x$

d) $\frac{d^2x}{dt^2}=-m^2 x$

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Solutions:

1. Ans: c)

$D^2 - 4D + 4 = 0$

$D-2 \quad D-2 \ = 0$

$D = 2, 2$

$y = \ C_1 + C_2 \ x \ e^{2x}$

2. Ans: a)

We have $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$

A.E. $D^4 + 2D^2 + 2 = 0$

Or $D^2 + 1 \ ^2 = 0$

Or $D = \pm i$ (repeated twice)

Thus general solution is $y = \ C_1 + C_2 x \ \sin x + \ C_3 + C_4 \ x \ \cos x$

Hence A is correct option

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3. Ans: c)

We have $\frac{d^2y}{dx^2} + 3a - 1 \frac{dy}{dx} - 3ay = 0$

A.E. $D^2 + 3a - 1 D - 3a = 0$

Or (D-1) (D+3a) = 0

Or D=1, -3a

Thus $y = C_1 e^x + C_2 e^{-3ax}$

Hence C is correct option

4. Ans: a)

We have $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15 = 0$

A.E. $D^2 - 8D + 15 = 0$

Or (D-3) (D-5) = 0

Or (D-3) (D-5) = 0

Hence solution is D = 3,5

Hence A is correct option $y = C_1 e^{3x} + C_2 e^{5x}$

5. Ans: b)

We have $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

A.E. $D^2 - 5D + 6 = 0$

Or (D - 2) (D - 3) = 0

Or D = 2, 3

Thus solution is

Hence B is correct option $y = e^{2x} + e^{3x}$

6. Ans: c)

We have $\frac{d^2y}{dx^2} + 9y = 0$

A.E. $D^2 + 9 = 0$

Or $D = \pm 3i$

The general solution is $y = C_1 \cos 3x + C_2 \sin 3x$

It can be written in following forms

$y = A_1 \sin 3x + B_1 \sin 3x - \cos 3x$

Where $A_1 + B_1 = C_2$ *and* $-B_1 = C_1$

Also $y = A_2 \sin 3x + \cos 3x + B_2 \sin 3x$

$= A_2 \sin 3x + \cos 3x + B_2 \ 3\sin x - 4\sin^3 x$

Where $A_2 = C_1$ *and* $A_2 + B_2 = C_2$

Also, $y = A_3 \sin 3x + \cos 3x + B_3 \ 4\cos^3 x - 3\cos x$

Where $A_3 + B_3 = C_1$ *and* $A_3 = C_2$

Hence C is correct option

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7. Ans: a)

We have $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}-y=0$

Let $x=e^z \; then \log\, x=z$

From (1),

$$\left[D\;\; D-1\;+d-1\right]y=0$$

Or, $D^2-1\;\;y=0$

Solution of this equation is $y=C_1e^x+C_2e^{-x}\;\; Or\; y=C_1x+C_2\,x^{-1}$

Hence A is Correct option

8. Ans: d)

We Have $x^2\frac{d^2y}{dx^2}-x\frac{dy}{dx}+y=0$

On Putting $x=e^z, we\; have$

$$\left[D\;\; D-1\;-D+1\right]y=0,$$

Or $D-1\;^2\;y=0$

Solution is $y=e^z\;\; C_1+C_2z$

Or $y=x\;\; C_1+C_2\log x$

Hence D is correct option

9. Ans: a)

Linearly independent solution are cos 2x, sin 2 x, e^{-x}

Thus A.E. Has roots equals to $\pm 2i$ and -1

Now D.E is $D+2i\;\;\; D-2i\;\;\; D-1\;\; y=0$

Or $D^2+4\;\;\; D+1\;\; y=0$

$$D^3+D^2+4D+4\;\; y=0$$

Hence A is correct option

10. Ans: a)

Linearly independent solution are cos 2 x, sin 2x , e^{-x}

Thus A.E. has roots equals to $\pm 2i$ and -1

Now D.E. is $D+2i\;\;\; D-2i\;\;\; D+1\;\; y=0$

Or $D^2+4\;\;\; D+1\;\; y=0$

$$D^3+D^2+4D+4\;\; y=0$$

Hence A is correct option

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11. Ans: d)

We have $D^2+4\ y=\cos 2x$

$$P.I.=\frac{1}{D^2+4}\cos 2x$$

Here the denominator vanishes when D^2 is replaced by $-2^2=4$ It is a case of failure multiplying the numerator by x and differentiating the denominator w.r.t. D we get,

$$P.I.=x\frac{1}{2D}\cos 2x=\frac{x}{2}\int\cos 2x\,dx$$

$$=\frac{x}{4}\sin 2x$$

12. Ans: d)

We have $D^2+4\ y=x$

$$\text{P.I.}=\frac{1}{D^2+4}x$$

$$=\left(1+\frac{D^2}{4}\right)^{-1}x$$

$$=\frac{1}{4}\left[1-\frac{D^2}{4}....\right]x=\frac{x}{4}$$

Hence D is correct option

13. Ans: b)

We Have $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+4y=\frac{e^{2x}}{x}$

A.E. $D^2=-4D+4=0$

Or $D-2\ ^2=0$

Or D = 2, 2

$$C.F.=\ C_1+C_2x\ e^{2x}$$

$$P.I.=\frac{1}{D+2\ ^2-4\ D+2\ +4}\frac{1}{x}$$

$$=e^{2x}\frac{1}{D^2}\frac{1}{x}$$

$$=e^{2x}\ \ x\log x-x$$

General solution is $y=\ C_1+C_2\,x+x\log x-x\ \ e^{2x}$

Hence B is correct option

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14. Ans: b)

We have $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+3y=3e^{2x}$

A.E. $D^2+4D+3=0$

Or D = -1,-3

P.I. = $\frac{1}{D^2+4D+3}3e^{2x}$

$=3\frac{e^{2x}}{2^2+4\ 2\ +3}$

$=\frac{3e^{2x}}{15}=\frac{e^{2x}}{5}$

Hence B is correct option

15. Ans: a)

We have $\frac{d^2y}{dx^2}+6\frac{dy}{dx}+9y=5e^{3x}$

A.E. $D^2+6D+9=0$

Or D = -3, -3 (Repeating roots)

$C.F.=\ C_1+C_2\ x\ e^{-3x}$

P.I. $=\frac{1}{D^2+6D+9}5e^{3x}$

$=5\frac{e^{3x}}{3^2+6\ 3\ +9}=\frac{5e^{3x}}{36}$

Complete solution is $y=\ C_1+C_2x\ e^{-3x}+\frac{5e^{3x}}{36}$

Hence A is correct option

16. Ans: a)

We have $\frac{d^2y}{dx^2}-\frac{dy}{dx}-2y=10\cos x$

A.E. $D^2-D-2\ y=0$

$D=2,-1$

C.F. = $C_1e^{-x}+C_2e^{2x}$

$P.I.=\frac{1}{D^2-D-2}10\cos x$

$=10\frac{1}{-1^2-D-2}\cos x$

$=-10\frac{1}{D+3}\cos x=-10\frac{D-3}{D^2-9}\cos x$

= $-10\frac{D-3}{-1^2-9}\cos x$

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= (D - 3) cos x

= - sin x – 3 cos x

The required solution is $y = C_1 e^{-x} + C_2 e^{2x} - \sin x - 3\cos x$

Hence A is correct option

17. Ans: b)

We have $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{3x}$

A.E. $D^2 - 3D + 2 = 0$

Or D = 1, 2

$C.F. = C_1 e^x + C_2 e^{2x}$

$P.I. = \frac{1}{D^2 - 3D + 2} . e^{3x}$

$= \frac{1}{3^2 - 3 \cdot 3 + 2} e^{3x} = \frac{1}{2} e^{3x}$

Thus the general solution of the given equation is

$y = C_1 e^x + C_2 e^{2x} + \frac{1}{2} e^{3x}$

Hence B is correct option.

18. Ans: c)

We have $x = A\cos\ mt - \alpha$

Differentiating $\frac{dx}{dy} = -Am\sin\ mt - \alpha$

Again Differential $\frac{d^2x}{dt^2} = -Am^2 \cos\ mt - \alpha$

Eliminating 'A', between (1) and (2), we get

$\frac{d^2x}{dt^2} = -m^2 x$

Which is the required differential equation

Hence C is correct option

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SET- 10
Initial and boundary value problem and Partial Differential Equations

1. For the differential equation $\frac{dy}{dx}+5y=0 \, with \, y\ 0\ =1,$ the general solution is

   a) e^{5t}
   b) e^{-5t}
   c) $5e^{-5t}$
   d) $e^{\sqrt{-5t}}$

2. The solution of the first order differential eqaution $\dot{x}\ t\ =-3x\ t\ , x\ 0\ =\mathrm{x}_0 \, is$

   a) $x\ t\ =\mathrm{x}_0 e^{-3t}$
   b) $x\ t\ =\mathrm{x}_0 e^{-3}$.
   c) $x\ t\ =\mathrm{x}_0 e^{\frac{-t}{3}}$
   d) $x\ t\ =\mathrm{x}_0 e^{-t}$

3. The solution of the differential equation $\frac{d^2y}{dx^2}+\frac{dy}{dx}-2y=0 \, with \, y\ 0\ =0\ , \dot{y}(0)=3$ is

   a) e^x+e^{-2x}
   b) e^x-e^{-2x}
   c) e^x-e^{2x}
   d) e^x+e^{2x}

4. The solution for the differential equion $\frac{dy}{dx}=\frac{3x^2}{2}-2x+\mathrm{C}_1 \, with$ boundary conditions

   $y\ 0\ =2 \, and \, \dot{y}\ 1\ =2 \, is$

   a) $y=\frac{x^3}{3}+\frac{x^2}{3}+3x-6$
   b) $y=3x^3+\frac{x^2}{2}-5x+2$.
   c) $y=\frac{x^3}{2}+\ x^2+\frac{5x}{2}+2$
   d) $y=x^3-\frac{x^2}{2}+5x+\frac{3}{2}$

5. The solution of differential equation $\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=0 \, with \, y\ 0\ =1\ \ and\ \ \dot{y}\ 0\ =\ \ -2\ \ for \, y \, is$

   a) $y=e^{-2t}\ \ t-1$
   b) $y=e^{-t}\ \ 1-t^2$
   c) $y=e^{-t}\ \ t^2-1$
   d) $y=e^{-t}\ \ 1-t$

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6. The solution for the differential equation $\frac{dy}{dx} = x^2 y$ *with the condition that* $y\ 0\ = 1$ *is* .

a) $y = e^{1/2x}$

b) $\log y = -\frac{x^3}{3} + 4$

c) $\log y = \frac{x^2}{2}$

d) $y = e^{\frac{x^3}{3}}$

7. The solution of the differential equation

$\frac{d^2y}{dx^2} + y = x$ *with the following conditions* *i* $at\ x = 0, y = 1$ *ii* $at\ x = \pi/2,\ y = \pi/2$ *is*

a) $y = C_1 Sin\, x + x$

b) $y = C_1 Tan\, x + x$

c) $y = C_1 Cos\, x + x$

d) $y = C_1 Sin\, x + 4$

8. If $x^2 \frac{dy}{dx} + 2xy = \frac{2 \log x}{x}$ *and* $y\ 1\ = 0$, *then the value of* $y\ e$ *is*

a) e

b) $1/e$

c) 1

d) $1/e^2$

9. The solution of the differential equation $\frac{dy}{dx} + 2xy = e^{-x^2}$ *with* $y\ 0\ = 1$ *is*

a) $1 + x\ e^{x^2}$

b) $1 + x\ e^{-x^2}$

c) $1 - x\ e^{x^2}$

d) $1 - x\ e^{-x^2}$

10. The solution of the differential equation $\frac{d^2y}{dx^2} + 9y = 6Cos3t,\ y\ 0\ = 2$ *and* $\dot{y}\ 0\ = 0$ *is*

a) $2Cos3t - t\,Sint$

b) $2Cos3t + t\,Cos3t$

c) $2Cos3t - t\,Sin3t$

d) $2Cos3t + t\,Sint$.

11. The solution of the differential equation $\frac{d^2y}{dx^2} + 4y = 12t$, *given that* $y = 0$ *and* $\frac{dy}{dt} = 9$ $at\ t = 0$, *is*

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a) $3t+3Sin2t$
b) $3t+Sin2t$
c) $t+3Sint$
d) $t+3Sin2t$

12. The solution of the differential equation $9\ddot{y}-6\dot{y}+y=0$ *with* $y(0)=3$ *and* $\dot{y}(0)=1$, *is*

a) $y=3e^{t/3}$

b) $y=\frac{1}{3}e^{t/3}$

c) $y=-3e^{t/3}$

d) $y=\frac{2}{3}e^{t/3}$

13. The solution of the differential equation $\dot{x}(t)+2x(t)=\delta(t)$ *with initial condition* $x(0-)=0$ *is*

a) $e^{-2t}u(t)$

b) $e^{2t}u(t)$

c) $e^{-t}u(t)$

d) $e^{t}u(t)$

14. The solution of the differential equation $x^2\frac{\partial u}{\partial x}+y^2\frac{\partial u}{\partial y}=0$ *is*

a) $C\log(x-y)$

b) $Ce^{k(x-y)}$

c) $Ce^{k\left(\frac{1}{x}-\frac{1}{y}\right)}$

d) $C\log\left(\frac{x-y}{xy}\right)$

15. The solution of the differential equation $\frac{\partial u}{\partial x}=2\frac{\partial u}{\partial y}+u$ *where* $u(x,0)=6e^{-3x}$ is

a) $6e^{-3x-2y}$
b) $6e^{3x+2y}$
c) $3e^{2x-3y}$
d) $3e^{2x+3y}$

16. The integral surface of the partial differential equation
$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=0$ *satisfying the condition* $u(1,y)=y$ *is given by*

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a) $u\ x,y\ = \frac{y}{x}$

b) $u\ x,y\ = \frac{2y}{x+1}$

c) $u\ x,y\ = \frac{y}{2-x}$

d) $u:\ x,y\ = y+x-1$

17. The general solution of the differential equation $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ *is of the form*

a) $u = f\ x+iy\ + g\ x-iy$

b) $u = f\ x+y\ + g\ x-y$

c) $u = c\ f\ x-iy$

d) $u = g\ x+iy$

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Solutions:

1. Ans: b)

We have $\frac{dy}{dt}+5y=0$

Or $\frac{dy}{y}=-5dt$

Integrading, $\log y = C_1 - 5t$

From y (0) = 1 we get $C_1 = 0$

Thus solution is $\log y = -5\,t$

Or $y = e^{-5t}$

Hence B is correct option

2. Ans: a)

We have $\dot{x}\ t\ =-3x\ t$

Or $\dot{x}\ t\ +3x\ t\ =0$

A.E. $D + 3 = 0$

Thus solution is $x(t) = C_1e^{-3t}$

From $x(0) = x_0$ we get $C_1 = x_0$

Thus $x(t) = x_0\ e^{-3t}$

Hence A is correct option

3. Ans: b)

A.E. $D^2 + D-2 = 0$

Or $(D-1)\ (D+2) = 0$

Or $D=1, -2$

C.F. $y = C_1e^{x} + C_2e^{-2x}$

Or $y' = C_1e^{x} - 2C_2e^{-2x}$

From y (0) = 0 we get $C_1 + C_2 = 0$

From y' (0) = 3 we get $C_1 - 2C_2 =3$

Solving , we get $C_1 = 1, C_2 = -1$

Solution is $y = e^{x} - e^{-2x}$

Hence B is correct option

4. Ans:c)

We have $y'=\frac{3x^2}{2}-2x+C_1$

From. y' (1) = 2 we get $C_1=\frac{5}{2}$

Lntegrating (1), $y=\frac{x^3}{2}-x^2+\frac{5}{2}x+C_2$

From y (0) =2 we get $C_2 = 2$

Therefore $y=\frac{x^3}{3}-x^3+\frac{5}{2}x+2$

Hence C is correct option

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5. Ans: d)

We have $\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=0$

A.E. $D^2 + 2D + 1 = 0$

Or $D = -1, -1$ (repeated roots)

Thus $y=e^{-1}\ C_1+C_2\,t$

Or $\frac{dy}{dt}=e^{-1}C_2-e^{-t}\ C_1+C_2t$

From y (0) =1 we get $C_1 = 1$

From $\left(\frac{dy}{dt}\right)\ 0\ =-2$ we get $C_2 = -1$

Thus solution is $y = e^{-t}(1-t)$

Hence D is correct option

6. Ans: d)

We have $\frac{dy}{dx}=x^2\,y$

Integrating $\int\frac{dy}{y}=\int x^2\,dx$

Or $\log y=\frac{x^3}{3}+C_1$

From y(0) =1 we get $C_1=0,$

Thus $\log y=\frac{x^3}{3}$

Or $y=e^{x^3/3}$

Hence D is correct option

7. Ans: c)

We have $\frac{d^2y}{dx^2}=y=x$

A.E. $D^2 + 1 = 0$

Or $D=\pm i$

$C.F.=C_1\cos x+C_2\sin x$

Now $P.I.=\frac{1}{D^2+1}\ x\ =1+D^2\ ^{-1}x$

$= (1-D^2)\ x = x$

$y = C_1 \cos x+C_2 \sin x + x$

From y (1) = 0 we get $C_1 = 1$

From $y\left(\frac{\pi}{2}\right)=\frac{\pi}{2}$ *we get* $C_2 = 0$

Thus complete solution is $y = C_1 \cos x + x$

Hence C is correct option

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8. Ans: d)

We have $x^2\frac{dy}{dx}+2xy=\frac{2\log x}{x}$

Or $y'+2\frac{y}{x}=\frac{2\log x}{x^3}$

Comparing this equation with standard equation y' + Py = Q, we see that

$$P=\frac{2}{x}\,and\,Q=\frac{2\log x}{x^2}$$

Now $I.F.=e^{\int Pdx}=e^{\int\frac{2}{x}dx}=e^{2\log x}=x^2$

$$yx^2=\int\frac{2\log x}{x^3}x^2\,dx+C_1$$

Or $yx^2=\log x^{\,2}+C_1$

Form y (1) = 0, $C_1 = 0$

So we get $yx^2 = (\log x)^2$

For y (e) $ye^2 = (\log e)^2$

Or $y=\frac{1}{e^2}$

Hence D is correct option

9. Ans: b)

We have $\frac{dy}{dx}+2xy=e^{-x2}$

I.F. = $e^{\int 3xdx}=e^{x^2}$

Solution is $y\cdot e^{x^3}=\int e^{-x^2}\cdot e^{x^2}\,dx+C_1$

Or $y\,e^{x^3}=x+C_1$

From y(0) =1 we get $C_1 = 1$

Thus $y = (1 + x)\,e^{-x^2}$

Hence B is correct option

10. Ans: c)

We have $\frac{d^2y}{dx^2}+9y=6\cos 3t, and\ y\ 0\ =2\ \&\ y'\ 0\ =0$

Taking laplace transform of (1), we get

$$\left[s^2Y\ s\ -sy\ 0\ -y'\ 0\ \right]+9Y\ s\ =6\frac{8}{8^2+9}$$

Putting value of y(0) and y'(0) in (2), we get

$$s^2Y\ s\ -2s+9Y\ s\ =\frac{6s}{s^2+9}$$

Or $Y\ s\ =\frac{2s}{s^2+9}+\frac{6s}{s^2+9^{\,2}}$

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Or $y = 2\cos 3t - 3^{-1}\dfrac{d}{ds}\left[\dfrac{3}{s^2+9}\right]$

$= 2\cos 3t - t\sin 3t$

Hence C is correct option

11. Ans: a)

We have $\dfrac{d^2y}{dt^2} + 4y = 12t$

Taking laplace transform on the both sides, we get

$$s^2 Y\ s\ - sy\ 0\ - y'\ 0\ + 4Y\ s\ = \frac{12}{s^2}$$

Or $s^2 + 4\ Y\ s\ - 0 - 9 = \dfrac{12}{s^2}$

Or $s^2 + 4\ Y\ s\ = \dfrac{12}{s^2} + 9$

Or $Y\ s\ = \dfrac{12}{s^2\ s^2+4} + \dfrac{9}{s^2+4}$

Or $Y\ s\ = \dfrac{3}{s^2} + \dfrac{6}{s^4+4}$

Or y = 3t + 3 sin 2t

Hence A is correct option

12. Ans: a)

We have $9y'' - 6y' + y = 0$

Taking laplace transform

$$9\left[s^2 Y\ s\ - sy\ 0\ + y'\ 0\ \right] - 6\left[sY\ s\ - y\ 0\ \right] + Y\ s\ = 0$$

Or $9s^2 - 6s + 1\ Y\ s\ + 6 - 9s\ y\ 0\ - 9y'\ 0\ = 0$

Or $9s^2 - 6s + 1\ Y\ s\ + 6 - 9s\ 3 - 9 = 0$

Or $3s - 1^{\ 2}\ Y\ s\ - 9\ 3s - 1\ = 0$

Or $Y\ s\ = \dfrac{9}{3s-1} = \dfrac{3}{s - \frac{1}{3}}$

Or $y = 3e^{1/3}$

Hence A is correct option

13. Ans: a)

We have $\dot{x}\ t\ + 2x\ t\ = \delta\ t$

Taking laplace transform $sX\ s\ - x\ 0\ + 2X\ s\ = 1$

Or $X\ s\ \ s + 2\ = 1$

Or $X\ s\ = \dfrac{1}{s+2}$

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Taking inverse laplace transform $x\ t\ = e^{-2t}\ u\ t$

Hence A is correct option

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14. Ans: c)

We have $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$ ..(1)

Let $u\ x, y = X\ x\ Y\ y$ ..(2)

From (1) we get

$x^2\ X'Y + y^2\ XY' = 0$ ..(3)

Or $\frac{x^2 X'}{X} = -\frac{y^2 Y'}{Y} = k\ \ say$

Thus $\frac{x^2\ X'}{X} = k\ \ Or\ \frac{X'}{X} = \frac{k}{x^2}$ ..(4)

And $\frac{y^2 Y'}{Y} = -k\ \ or \frac{Y'}{Y} = \frac{-k}{y^2}$ ..(5)

Form (4) $\log X = \frac{-k}{x} + \log C_1$

Or $X = C_1 e^{\frac{-k}{z}}$

From (5) $\log Y = \frac{k}{y} + \log C_2$

Or $y = C_2 e^{\frac{k}{y}}$

Putting these values in (2) we get

$u\ x, y = C_1 e^{\frac{-k}{x}}\ C_2 e^{\frac{k}{y}}$

Thus C is correct option

15. Ans: a)

We have $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial y} + u$ ..(1)

Let $u\ x, y\ \ X(x)Y\ \ y$ ..(2)

From (1) we get

X'Y = 2X'Y + XY ..(3)

Or $\frac{X' - X}{2X} = \frac{Y'}{Y} = k\ \ say$

Thus $\frac{X' - X}{2X} = k\ or \frac{X'}{X} = 1 + 2k$ ..(4)

And $\frac{Y'}{Y} = k$ ..(5)

From (4) $\log X = 1 + 2k\ \ x + \log C_1\ \ or\ X = C_1 e^{1+2k\ x}$

From (5) $\log Y = ky + \log C_2\ \ \ or\ \ y = C_2 e^{ky}$

Putting these values in (2) we get

$u\ x, y = C_1 C_2 e^{1+2k\ x} \in^{ky}$ ..(6)

Now $6e^{-3x} = u\ x, 0 = C_1 C_2\ e^{1+2k\ x}$

Thus $C_1C_2 = 6$ and $1 + 2k = -3$ or $k = -2$

Substituting these values in (6), we get

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$$u = 6e^{-3x}\, e^{-2y}$$

Hence A is correct option

16. Ans: a)

We have $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$

Let $u\ (x, y) = \frac{y}{x}$

$$\frac{\partial u}{\partial x} = -\frac{y}{x^2} \text{ and } \frac{\partial u}{\partial y} = \frac{1}{x}$$

Now $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = x\left(-\frac{y}{x^2}\right) + y\left(\frac{1}{x}\right)$

$$= -\frac{y}{x} + \frac{y}{x} = 0$$

And $u\ (1, y) = \frac{y}{1} = y$

Hence A is correct option

17. Ans: a)

We have $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

A,E $D^2 + 1 = 0$

$D = \pm i$

Thus general solution is $u = f\ (x + iy) + f\ (x - iy)$

Hence A is correct option

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SET-11

Differential Equations

1. The solution of $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$, is

(A) $\log (1 + y^2) = \log (1 + x) + c$
(B) $y + x = C(1 + xy)^2$
(C) $\log (1 + x^2) / (1 + y^2) = c$
(D) $\tan^{-1} x - \tan^{-1} y = C$

2. Which of the following is the linear differential equation?

(A) $\frac{dy}{dx} + x^2 y = \cos x . \sin y$

(B) $\frac{dy}{dx} - x^2 y = \sin x$

(C) $l + y \frac{dy}{dx} + \sin^2 x = 0$

(D) $\frac{dy}{dx} + y \ y + x = x^3$

3. Which of the following is the solution of the differential equation $e^y dy = (e^x + x^2) dx$?

(A) $e^y = e^x + \frac{1}{3} x^3 + C$ (B) $e^y = e^{-x} + \frac{1}{3} x^2 + C$

(C) $e^y = e^x - \frac{1}{3} x^3 + C$ (D) None of these

4. For the differential equation $y(x^2y + e^x) dx - e^x dy = 0$, which of the following solution is correct.

(A) $3xy^3 + e^x = cx$ (B) $xy^3 + 3e^x = cx$
(C) $x^3y - 3e^x = cx$ (D) $x^3 y + 3e^x = cx$

5. Identify the equation which represents the curves for which the sum of the reciprocal of the polar sub normal and the radius vector is constant.

(A) $\theta = cr + K$ (B) $r = c\theta + K$
(C) $\theta = cr - \log r + K$ (D) $\theta = cr + \log r - K$

6. Degree and order of the differential equation $\sqrt{2\left(\frac{dy}{dx}\right)^3 + 4} = \left(\frac{d^2y}{dx^2}\right)^{3/2}$ are respectively

(A) Order 2, Degree 3 (B) Order 2, Degree 1
(C) Order 3, Degree 2 (D) Order 2, Degree 2

7. Identify the ordinary differential equation

(A) $y = 2x + \frac{d}{dx} (\sin x)$ (B) $\frac{d}{dx} (y + x) = x^2 + \sin x$

(C) $\frac{d}{dx} (\cos x + \sin x) = x^2 + y^2$ (D) $y^2 + x = \frac{d}{dx} (xa + 2)$

8. In the equation $P dx + x \sin y, dy = 0$ is exact for which the following value of P?

(A) $\sin y + \cos y$ (B) $-\cos y$

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(C) $x^2 - \cos y$ (D) $x^2 + \cos y$

9. Which of the following is the solution of the differential equation $(x - y^2)\,dx + 2xy\,dy = 0$?

(A) $ye^{y^{2/x}} = A$ (B) $xe^{y^{2/x}} = A$

(C) $xe^{y^{-2/x}} = A$ (D) None of these

10. Solution of the differential equation $(xy3 + y)\,dx + 2(x^2y^2 + x + y^4)\,dy = 0$ is

(A) $3x^4y^2 + 6xy^2 + 2y^6 = C$
(B) $3x^2y^4 + 6xy^2 + 2y^6 = C$
(C) $3x^2y^4 + 6xy^2 + y^6 = C$
(D) None of these

11. If $f'(x) = \tan^{-1} x$ and $f(0) = 0$, then $f(1)$ is

(A) $\frac{\pi}{4} + \log 2$ (B) $\frac{\pi}{3} - \log 2$

(C) $\frac{\pi}{4}(\pi - \log 4)$ (D) $\frac{\pi}{4}(\pi + \log 4)$

12. Which of the following differential equation is exact?

(A) $(x^2 + y)\,dx - x\,dy = 0$
(B) $(2x + y - 3)\,dy + (x + 2y - 3)\,dx = 0$
(C) $x\,dx + y\,dy + \frac{x^2\,dy - y\,dx}{x^3 + y^3} = 0$
(D) $\frac{dy}{dx} + y(y + x) = x^3$

13. Consider the following variable separable equation $(x^2 + 1)(y^2 - 1)\,dx + xy\,dy = 0$. The solution of this equation is

(A) $y + x = C(1 + xy)^2$ (B) $\log(y^2 - 1) = C.\log(x^2 + 1)$

(C) $y^2 - 1 = C\frac{e^{-x2}}{x^2}$ (D) $y^2 - 1 = x^2 + 1 + C$

14. If $y = 1$, when $x = 1$, the solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$ is

(A) $4xy = y^4 - 3$ (B) $4xy = = y^4 + 3$
(C) $4xy = x^4 + 3$ (D) $4xy = x^4 - 3$

15. The P.I. of $(D^2 - 2D)y = e^x \sin x$ is

(A) $-\frac{1}{2}e^x \sin x$ (B) $e^x \cos x$

(C) $-\frac{1}{2}e^x \cos x$ (D) None of these

16. The general solution of the linear differential equation $(D^4 - 81)\,y = 0$ is given by

(A) $(C_1 + C_2x)e^{3x} + (C_3 + C_4x)\sin 3x$
(B) $-C_1 e^{3x} + C_2 e^{-3x} + C_3 \cos 3x + C_4 \sin 3x$
(C) $C_1e^{3x} + C_2 e^{-3x} + C_3 \cos 3x + C_4 \sin 3x$
(D) None of these

17. The general solution of the differential equation $(D^2 - m^2)y = 0$ is

(A) $y = (C_1 + C_2x)e^{mx}$ (B) $y = C_1 \sin x + C_2 \cos x$

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(C) $y = C_1 \cosh x + C_2 \sinh x$ (D) None of these

18. $\frac{1}{\varphi\ D\quad D - a}^{n} e^{ax}$ is equal to

(A) $\frac{x^n e^{ax}}{\varphi\ a}$ (B) $\frac{x^n e^{ax}}{n!\varphi\ a}$

(C) $\frac{x^n e^{ax}}{\varphi\ a}$, $\varphi(a) \neq 0$ (D) $\frac{x^n e^{ax}}{n!\varphi\ a}$, $\varphi(a) \neq 0$

19. The particular integral of $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4$ is]

(A) $\frac{x^2}{3} + 4x$ (B) $\frac{x^3}{3} + 4x$

(C) $\frac{x^3}{3} + 4$ (D) $\frac{x^3}{3} + 4x^2$

20. The solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 3e^{2x}$, where y (0) = and y (0) = – 2 is

(A) $y = e^{-x} - e^{2x} + xe^{2x}$ (B) $y = e^{x} - e^{-2x} + xe^{2x}$

(C) $y = e^{-x} + e^{2x} + \frac{x}{2} e^{2x}$ (D) $y = e^{x} - e^{-2x} - \frac{x}{2} e^{2x}$

21. The particular integral for the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{3x}$ is

(A) xe^{-x} (B) x^2e^{-x}

(C) $x(x - 1)\ e^{3x}$ (D) $(x - 2)\ e^{4x}$

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Solutions:

1. (d) The given equation can be written as, $\frac{dx}{1+x^2} - \frac{dx}{1+x^2} = 0$
Integration we get $\tan^{-1} x - \tan^{-1} y$ = constant (C).

2. (b) The differential equation (d.e.) in (b) is linear as the shape of linear differential equation (1.d.e.) is $\frac{dy}{dx} + Py = Q$ where P and Q are functions of x.

3. (a) The given equation can be written as, $e^y\, dy = (e^x + x^2)\, dx$
Integrating we get, $e^y = e^x + \frac{1}{3}x^3 + C$
which is the required solution.

4. (d) The given equation can be written as, $\frac{dy}{dx} - y = x^2 e^{-x} y^2$ This is Bernoulli's equation.
or $y^{-2}\frac{dy}{dx} - y^{-1} = x^2 e^{-x}$
Putting $y^{-1} = v$, we get $-y^{-2}\frac{dy}{dx} = \frac{dv}{dx}$
From equation (i), $-\frac{dy}{dx} - v = x^2 x^{-x}$
or $\frac{dv}{dx} + v = -x^2 e^{-x}$ which is linear equation.
Here, P = 1, $Q = -x^2 e^{-x}$
$\therefore$ I. F. $= e^{\int t.\,dx} = e^x$
the solution is $ve^x = \int -x^2 e^{-x}\ e^x\ dx + C = \int -x^2\ dx + C$
or $\frac{1}{y}e^x = -\frac{x^3}{3} + C$
or $x^3 y + 3e^x = 3C$
or $x^3 y + 3e^x = cy$, which is the solution of the given equation.

5. (c) According to given problem, $\frac{1}{r} + \frac{d\theta}{dr} = c$
Or $\int d\theta = \int \left(c - \frac{1}{r}\right) dr$
Or $\theta = cr - \log r + K$

6. (a) Squaring both sides we get, $2\left(\frac{dy}{dx}\right)^3 + 4\left(\frac{d^2y}{dx}\right)^3$
$\therefore$ Higher order = 2, degree = 3

7. (b) Only the equation in (b) posses dy/dx, hence (b) is ordinary differential equation

8. (c) Pdx + x sin y, dy = 0
Here M = P, N = x in y

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For exact $\frac{\theta M}{\theta y} = \frac{\theta N}{\theta x}$

Or $\frac{\theta P}{\theta y}$ = sin y

Or P = - cos y + f(x)
Let f(x) = x^2, then P = - cos1 y + x^2

9. (b) The given equation can be written as,

$x - y^2 + 2xy\frac{dy}{dx} = 0$

Let $x - y^2 = v \Rightarrow 1 - 2y\frac{dy}{dx} = \frac{dv}{dx}$

$\therefore \quad v + x\left(1 - \frac{dv}{dx}\right) = 0$

Or $\frac{dv}{dx} - \frac{v}{x} = 1$ which is linear equation

$\therefore$ I.F. $= e^{\int \frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$

It's solution is v. $\frac{1}{x} = \int 1.\frac{1}{x} dx + \log c$

Or v $\frac{1}{x}$ = log x + log c

Or $cx = e^{v/x} = e^{x - y^2/x} = e^{1-y^2} = e.e^{-y^{2/x}}$

Or cx = $e^{-y^2/x} = \frac{e}{c} = A$ Say

10. (c) The given equation is,
$(xy^3 + y)\,dx + 2(x^2y^3 + x + y^4)\,dy = 0$
$M = xy^3 + y$; $N = 2(x^2y^3 + x + y^4)$

$\therefore \quad \frac{\partial M}{\partial y} = 3xy^2 + 1$; $\frac{\partial N}{\partial x} = 4xy^2 + 2$

$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \frac{1}{xy^2 + 1\ y}\ xy^2 + 1 = \frac{1}{y}$

$\therefore$ I.F. = $e^{\int \frac{1}{y} dy} = e^{\log y} = y$

Multiplying by y in the given equation,
$(xy^4 + y^2)\,dx + 2(x^2y^3 + xy + y^5)\,dy = 0$

$\therefore$ Solution is $\int xy^4 + y^2\ dx + \int y^5\ .\ dy = C$

Or $\frac{x^2}{2}y^4 + xy^2 + \frac{1}{6}y^6 = C$

Or $3x^2y^4 + 6xy^2 + y^6 = C$

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11.

$c\ \ f\ \ x\ = \tan^{-1} x$

$\therefore f\ \ x\ = \tan^{-1} x\, dx = x \tan^{-1} x - \int \frac{1}{1+x^2} x dx$

$or\ f\ \ x\ = x \tan^{-1} x - \frac{1}{2} \log\ 1+x^2\ + C$

$Putting\ x = 0\ we,\ get\ f\ \ 0\ = 0 + C$

$\Rightarrow \qquad C = 0$

$\frac{d^2 y}{dx^2} = -\cos x - \sin x$

$f\ \ 1\ \frac{\pi}{4} - \frac{1}{2} \log 2 \frac{\pi}{4} - \frac{1}{4} \log 4 = \frac{1}{4}\ \pi - \log 4$

12. (b) In the equation, $2x + y - 3\ dy + \ x + 2y - 3\ dx = 0$

$M = x + 2y - 3, N = 2x + y - 3$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2$

Thus, differential equation in (b) is exact

13. (C) The given equation can be written as

$\left(x + \frac{1}{x}\right) dx + \left(\frac{y}{y^2 - 1}\right) dy = 0$

Integrating we get, $\therefore y = c_1 e^{-x} + 2c_2 e^{2x} + x e^{2x}$

Or $x^2 + 2 \log x + \log\ y^2 - 1\ = \log C$

Or $y\ \ 0\ = c_1 + c_2 = 0\ \ given$

$\therefore \qquad y^2 - 1 = \frac{C e^{-x^2}}{x^2}$

14.

$c\ \ \frac{dy}{dx} + \frac{y}{x} = x^2$

$I.F. = e^{\int \frac{1}{x} dx} = e^{\log x} = x$

$\therefore The\ solution\ is\ y.x = \int x^2 .x dx + C$

$xy = \frac{x^4}{4} + C$

$Putting\ z = 1\ and\ y = 1,\ we\ get$

$\therefore The\ solution\ is\ 4xy = x^4 + 3$

15.

$a\ \ P.I = \frac{1}{D^2 - 2D} e^x \sin x = e^x \frac{1}{D+1^{\ 1} - 2\ \ d+1} \sin x$

$= e^x \frac{1}{D^2 - 1} \sin x = e^x \frac{1}{-1^2\ \ -1} \sin x = -\frac{1}{2} e^x \sin x$

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16.

$c \qquad D^4 - 81\ y = 0$

$Auxiliary\ equation\ is\ m^4 - 81 = 0$

$\Rightarrow \qquad m+3 \quad m-3 \quad m+3i \ = 0$

$\therefore Solution\ is\ y = C_1 e^{3x} + C_2 e^{-3x} + C_3 \cos 3x + C_4 \sin 3x$

17. (c) The A.E. of the given equation is, $D^2 - m^2 = 0$ or $D = \pm m$
$\therefore$ C.F. is $y = C_1 e^x + C_2 e^{-x} = C_1 \cosh x + C_2 \sinh x$

18. (d)

$$\frac{1}{\varphi\ D \quad D - a^{\ n}} e^{ax} = \frac{1}{\varphi\ a} \cdot \frac{1}{D - a^{\ n}} e^{ax} = \frac{1}{\varphi\ a} x. \frac{1}{\frac{d}{dD}\ D - a^{\ n}} e^{ax}$$

$$\because f\ a = 0 \ = \frac{1}{\varphi\ a} . x \frac{1}{n\ D - a^{\ n-1}} e^{ax}$$

$$= \frac{1}{\varphi\ a} . x^2 \frac{1}{n \frac{d}{dD}\ D - a^{\ n-1}} e^{ax} \qquad \because f\ a = 0$$

$$= \frac{1}{\varphi\ a} . x^2 \frac{1}{n\ n-1 \quad D - a^{\ n-2}} e^{ax}$$

$$= \frac{1}{\varphi\ a} . x^n \frac{1}{n\ n-1 \quad n-2 \ ...2.1} e^{ax}$$

$$= \frac{1}{\varphi\ a} . \frac{x^n e^{ax}}{n!}$$

19. (b) The given differential equation is, $D^2 + D = x^2 + 2x + 4$

$$\therefore \ P.I. = \frac{1}{D\ D+1} \ x^2 + 2x + 4 \ = \frac{1}{D} \ D + 1^{\ -1} \ x^2 + 2x + 4$$

$$= \frac{1}{D} \ 1 - D + D^2 \quad x^2 + 2x + 4$$

$$= \frac{1}{D} \ x^2 + 2x + 4 - 2x - 2 + 2 \ = \frac{x^3}{3} + 4x$$

20. (a) Auxiliary equation is, $m^2 - m - 2 = 0$
$\therefore \quad m = -1, 2$
$CF = c_1 e^{-x} + C_2 e^{2x}$

$$P.I. = \frac{1}{D^2 - D - 2} . 3 . e^{2x} = 3 . \frac{1}{D^2 - D - 2} e^{2x}$$

$$= 3x \ \frac{1}{D^2 - D - 2} . 3 . e^{2x} = 3 . \frac{1}{D^2 - D - 2} e^{2x}$$

$\therefore \quad y = c_1 e^{-x} + 2c_2 e^{2x} + x e^{2x}$

$$\frac{dy}{dx} = -c_1 e^{-x} + 2c_2 e^{2x} + 2x e^{2x} + e^{2x}$$

$y(0) = C_1 + C_2 = 0$ (given)
$y'(0) = -C_1 + 2C_2 + 1 = -2$ (given)
$\therefore \qquad C_1 + C_2 = 0$

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$C_1 - 2C_2 = 3$
Solving these, we get $C_1=1$, $C_2 = -1$
$\therefore$ required solution is $y = e^{-x} - e^{2x} + xe^{2x}$

21. (c) The given equation is, $(D^2 - 5D + 6)\, y = e^{3x}$

$$\therefore \text{ P.I.} = \frac{1}{D^2 - 5D + 6} e^{3x} = \frac{1}{D - 3 \quad D - 2} e^{3x}$$

$$= \frac{1}{D - 3} e^{3x} - \frac{1}{D - 2} e^{3x} = x.e^{3x} - \frac{1}{3 - 2} e^{3x} = x - 1\ e^{3x}$$

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Past GATE Questions

1. The following differential equation has

$$3\left(\frac{d^2y}{dt^2}\right)+4\left(\frac{dy}{dt}\right)^3+y^2+2=x$$ **(Gate-2005, 1-Mark)**

a) Degree=2 , Order=1
b) Degree=1 , Order=2
c) Degree=4 , Order=3
d) Degree=2 , Order=3

2. A solution of the following differential equation is given by

$$\frac{d^2y}{dx^2}-5\frac{dy}{dx}+6y=0$$ **(Gate-2005, 1-Mark)**

a) $y = e^{2x} + e^{-3x}$
b) $y = e^{2x} + e^{3x}$
c) $y = e^{-2x} + e^{3x}$
d) $y = e^{-2x} + e^{-3x}$

3. A solution of the following differential equation is given by

$\dot{x}\ t\ +2x\ t\ =\delta\ t$ **(Gate-2006, 1-Mark)**

With initial condition $x(0^-)=0$ is

a) $e^{-2t}u\ t$
b) $e^{2t}u\ t$
c) $e^{-t}u\ t$
d) $e^{t}u\ t$

4. Ḟor the diff̈erentail eqauation **(Gate-2006, 2-Mark)**

$$\frac{d^2y}{dx^2}+k^2y=0$$

Boundary condition are given by $i\ \ y=0\ \ for\ \ x=0$

and $ii\ \ y=0\ \ for\ x=a$. The form of non-zero solutions of y (m varies over all integers) are

a) $y=\sum_m A_m \sin\frac{m\pi x}{a}$
b) $y=\sum_m A_m \cos\frac{m\pi x}{a}$
c) $y=\sum_m A_m x^{\frac{m\pi}{a}}$
d) $y=\sum_m A_m e^{-\frac{m\pi x}{a}}$

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5. A solution for the following differential equation

$k^2 \frac{d^2y}{dx^2} = y - y_2$ under the boundary condition **(Gate-2007, 2-Mark)**

(i) $y = y_1$ *at* $x = 0$ *and*

(ii) $y = y_2$ *at* $x = \infty$ *, where* k y_1 *and* y_2 *are constant is*

a) $y = (y_1 - y_{2)} \exp\left(-\frac{x}{k^2}\right) + y_2$

b) $y = (y_2 - y_{1)} \exp\left(-\frac{x}{k}\right) + y_1$

c) $y = (y_1 - y_{2)} \sinh\left(\frac{x}{k}\right) + y_1$.

d) $y = (y_1 - y_{2)} \exp\left(-\frac{x}{k}\right) + y_2$

6. Which of the following is a solution to the differential equation $\frac{dx\ t}{dt} + 3x\ t = 0$? **(Gate-2008, 1-Mark)**

a) $x\ t = 3e^{-t}$

b) $x\ t = 2e^{-3t}$

c) $x\ t = -\frac{3}{2}t^{-2}$

d) $x\ t = 3t^2$

7. The order of the differential equation **(Gate-2009, 1-Mark)**

$\frac{d^2y}{dt^2} + \left(\frac{dy}{dt}\right)^3 + y^4 = e^{-t}$ is

a)1 b) 2 c) 3 d) 4

8. Match each differential equation in Group-I to its family of solution curves from group-II **(Gate-2009, 2-Marks)**

Group-I	Group-II
A. $\frac{dy}{dx} = \frac{y}{x}$	1. Circle
B. $\frac{dy}{dx} = -\frac{y}{x}$	2. Straight Line
C. $\frac{dy}{dx} = \frac{x}{y}$	3. Hyperbola
D. $\frac{dy}{dx} = -\frac{x}{y}$	

a) A-2, B-3, C-3, D-1
b) A-1, B-3, C-2, D-1
c) A-2, B-1, C-3, D-3
d) A-3, B-2, C-1, D-2

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9. A function n(x) satisfies the differential equation

$$\frac{d^2n\ x}{dx^2}-\frac{n\ x}{L^2}=0$$ Where L is a constant. (**Gate-2010, 1-Marks**)

The Boundary conditions are: n(0)=K and n(∞)=0. The solution to this differential equation is

a) $n\ x\ = K\exp\left(\frac{x}{L}\right)$

b) $n\ x\ =$ K $\exp\left(\frac{-x}{\sqrt{L}}\right)$

c) $n\ x\ =$ K^2 $\exp\left(\frac{-x}{L}\right)$

d) $n\ x\ = K\exp\left(\frac{-x}{L}\right)$

10. The solution of the differential equation $\frac{dy}{dx}=ky,\ y\ 0\ =c\ is$ (**Gate-2011, 1-Marks)**

a) $x=ce^{-ky}$ b) $x=ce^{ky}$ c) $y=ce^{kx}$ d) $y=ce^{-kx}$

11. With initial condition x(1)=0.5, the solution of the differential equation

$t\frac{dx}{dt}+x=t\ is$ (**Gate-2012, 1-Marks**)

a) $x=t-\frac{1}{2}$

b) $x=t^2-\frac{1}{2}$

c) $x=\frac{t^2}{2}$

d) $x=\frac{t}{2}$

12. If the characteristics equation of the differential equation (**Gate-14, 1-Marks)**

$$\frac{d^2y}{dx^2}+2\alpha\frac{dy}{dx}+y=0$$

has two equal roots, then the values of α are

a) ±1 b) $0,0$ c) $\pm j$ d) $\pm1/2$

13. Which one of the following differential equation is a linear non-homongous differential equation, where x and y are independent and dependent variables respectively? (**Gate-14, 1-Marks**)

a) $\frac{dy}{dx}+xy=e^{-x}$

b) $\frac{dy}{dx}+xy=0$

c) $\frac{dy}{dx}+xy=e^{-y}$

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d) $\frac{dy}{dx}+e^{-y}=0$

14. If a and b are constants, the most general solution of the differential equation

$\frac{d^2x}{dt^2}+2\frac{dx}{dt}+x=0 is$ (**Gate-14, 1-Marks**)

a) ae^{-t} b) $ae^{-t}+be^{-t}$ c) $ae^{t}+be^{t}$ d) ae^{-2t}

15. With Initial Values $y\ 0\ = y\,(0)$ =1, the solution of the differential equation

$\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=0$ at $x=1$ is ……. (**Gate-14, 2-Marks**)

16. The solution of the differential equation

$\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=0 with\ y\ 0\ = y\ 0\ =1 is$ (**Gate-15, 2-Marks**)

a) $2-t\ e^{t}$ b) $1=2t\ e^{-t}$ c) $2+t\ e^{-t}$ d) $1-2t\ e^{t}$

17. The general solution of the differential equation

$\frac{dy}{dx}=\frac{1+\cos 2y}{1-\cos 2x} is$ (**Gate-15,1-Mark**)

a) $\tan y-\cot x=c\ \ c=constant$

b) $\tan x-\cot y=c\ \ c=constant$

c) $\tan y+\cot x=c\ \ c=constant$

d) $\tan x-\cot y=c\ \ c=constant$

18. Consider the differential equation

$\frac{d^2x\ t}{dt^2}+3\frac{dx\ t}{dt}+2x\ t\ =0. Given\ x\ 0\ =20 and\ x\ 1\ =10/e$,

$where\ e=2.718, the\ value\ of\ \ x\ 2\ =$…….. (**Gate-15, 2-Mark**)

19. The particular solution of the initial value problem given below is

$\frac{d^2y}{dx^2}+12\frac{dy}{dx}+36y=0\ with\ y\ 0\ =3 and\ \frac{dy}{dx}\ at\ x=0\ =-36$ (**Gate-16, 2-Mark**)

a) $3-18x\ e^{-6x}$

b) $3+25x\ e^{-6x}$

c) $3+20x\ e^{-6x}$

d) $3-12x\ e^{-6x}$

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20. Which of the following is the general solution of the first order differential equation

$\frac{dy}{dx} = x + y - 1^{2}$, *Where x, y are real* ? **(Gate-17, 2-Mark)**

a) $y = 1 + x + \tan^{-1} x + c$,*Where c is a contant*

b) $y = 1 + x + \tan x + c$,*Where c is a contant*

c) $y = 1 - x + \tan^{-1} x + c$,*Where c is a contant*

d) $y = 1 - x + \tan x + c$,*Where c is a contant*

21. The solution of the differential equation

$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 5y = 0$ *in terms of arbitrary constants* K_1 *and* K_2 *is* **(Gate-17, 1-Mark)**

a) $K_1 e^{-1+\sqrt{6}x} + K_2 e^{-1-\sqrt{6}x}$

b) $K_1 e^{-1+\sqrt{8}x} + K_2 e^{-1-\sqrt{8}x}$

c) $K_1 e^{-2+\sqrt{6}x} + K_2 e^{-2-\sqrt{6}x}$

d) $K_1 e^{-2+\sqrt{8}x} + K_2 e^{-2-\sqrt{8}x}$

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Solutions:

1. b) Order is highest derivative term. Degree is power of highest derivative term.

2. b) Auxiliary equation is $D^2 - D + 6 = 0$

(D-2) (D-3) = 0

D=2, 3

Therefore $y = e^{2x} + e^{3x}$

3. a) $\dot{x}\ t\ + 2x\ t\ = \delta\ t$

by taking Laplace Transform on both sides

$$s\,X\ s\ - x\ 0\ + 2\,X\ s\ = 1$$

$$X\ s\ \ s + 2\ = 1$$

$$X\ s\ = \frac{1}{s+2}$$

$$x\ t\ = e^{2t}\,u\ t$$

4. a) $D^2 + k^2 = 0\ then\, D = \pm jK$

$y = A\cos Kx + j\,B\sin Kx \quad at\, x = 0,\ y = 0\, then\, A = 0, y = jB\sin Kx$

X=a , y=0

$\Rightarrow \quad 0 = B\sin Ka$

$\Rightarrow \quad B \neq 0\, else\, y = 0\, always$

$\sin Ka = 0$

$$K = \frac{m\pi x}{a}$$

$$y = \sum_m A_m \sin\left(\frac{m\pi x}{a}\right)$$

5. d)

$$K^2 \frac{d^2y}{dx^2} = y = y_2$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} - \frac{y}{K^2} + \frac{y_2}{K_2} = 0$$

$$m^2 - \frac{1}{K^2} = 0$$

$$m = \pm\frac{1}{K}$$

Complementary function

$$\Rightarrow \quad y = C_1 e^{\frac{x}{K}} + C_2 e^{\frac{x}{K}} + y_2 \qquad .. \ i$$

$$x = 0; y = y_1 = C_1 + C_2 + y2 \qquad .. \ ii$$

$$x = \infty; y = y_2 = C_1 e^{\infty} + 0 + y_2$$

$$\Rightarrow \quad C_1 = 0\, and\, C_2 = y_1 - y_2$$

$$\therefore \quad y = \ y_1 - y_2\ \ e^{-x/k} + y_2$$

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6. b)

$$D+3\ x\ t\ =0$$

$$\Rightarrow \quad D=-3$$

$$So, \quad x\ t\ =Ce^{-3t}$$

7. b) Highest derivative of differential equation is 2.

8. a)

A. $\frac{dy}{dx}=\frac{y}{x}$

$$\frac{dy}{y}=\frac{dx}{x}\Rightarrow\int\frac{dy}{y}=\int\frac{dx}{x}$$

$$\log y=\log x+\log c$$

$$=\log cx$$

$$y=cx\ldots Equation\ of\ straight\ line$$

B. $\frac{dy}{dx}=\frac{-y}{x}$

$$\frac{dy}{y}=\frac{-dx}{x}\Rightarrow\int\frac{dy}{y}=-\int\frac{dx}{x}$$

$$\log y=-\log x+\log c$$

$$\log y+\log x=\log c$$

$$\log yx=\log c$$

$$yx=c$$

$$y=c/x$$

$$\ldots Equation\ of\ hyperbola$$

C. $\frac{dy}{dx}=\frac{x}{y}, y\,dy=x\,dx$

$$\Rightarrow\int y\,dy\ =\int x\,dx$$

$$\frac{y^2}{2}-\frac{x^2}{2}=\frac{c^2}{2}\rightarrow const$$

$$y^2-x^2=c^2$$

$$\frac{y^2}{c^2}-\frac{x^2}{c^2}=1$$

$$\ldots Equation\ of\ hyperbola$$

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D. $\frac{dy}{dx}=\frac{-x}{y}$

$\Rightarrow \int y\,dy=-\int x\,dx$

$\frac{y^2}{2}=-\frac{x^2}{2}+\frac{c^2}{2}$

$\frac{y^2}{2}+\frac{x^2}{2}=\frac{c^2}{2}$

$x^2+y^2=c^2$

...Equation of circle.

9. d)

$\left(D^2-\frac{1}{L^2}\right)n\ x\ =0$

$D^2-\frac{1}{L^2}=0$

$D=\pm\frac{1}{L}$

So, the solution is

$n\ x\ =C_1e^{x/L}+C_2e^{-x/L}$

$n\ 0\ =K\Rightarrow C_1+C_2=K$

$n\ \infty\ =0\Rightarrow C_1=0$

$\Rightarrow \quad C_2=K$

Therefore, $n\ x\ =K\exp\ -x/L$

10. c)

$\frac{dy}{dx}=ky$

$\Rightarrow \quad \frac{dy}{y}\ kdx$

Integrating both sides

$\ln y=kx+A$

Put, $x=0$

$\ln y\ 0\ =A$

$\therefore \quad A=\ln c\left[\because y\ 0\ =c\right]$

Hence,

$\ln y=kx+\ln c$

$\Rightarrow \quad \ln y-\ln c=kx$

$\Rightarrow \quad \ln\left(\frac{y}{c}\right)=kx$

$\Rightarrow \quad \frac{y}{c}=e^{kx}$

Or $y=c\,e^{kx}$

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11. d)

The given differential r=equation is

$t\frac{dx}{dt}+x=t$ with initial condition x(1) = $\frac{1}{2}$ *which*

Is same as

$\frac{dx}{dt}+\frac{x}{t}=1$

Which is a linear differential equation

$\frac{dx}{dt}+Px=Q$

Where $P=\frac{1}{t}$ *and* $Q=1$

Integrating factor = $e^{\int Pdt=e^{\int \frac{1}{t}dt}}$

$= e^{\log_e t} = t$

Solution is

$x.\ IF\ = \int Q.\ IF\ dt+C$

$x.t=\int 1\cdot t\cdot dt+C$

$xt=\frac{t^2}{2}+C$

$x=\frac{t}{2}+\frac{C}{t}$

$Put\ x\ 1=\frac{1}{2}$

$\Rightarrow \quad \frac{1}{2}+\frac{C}{1}=\frac{1}{2}$

$\Rightarrow \quad C=0$

So. $\boxed{x=\frac{t}{2}}$ *is the solution*

12. a)

$\frac{d^2y}{dx^2}+2\alpha\frac{dy}{dx}+y=0$

The characteristic equation is given as

$m^2+2\alpha m+1\ =0$

$m_1.m_2=\frac{-2\alpha\pm\sqrt{4\alpha^2-4}}{2}$

Since both roots are equal i.e

$m_1=m_2$

$\frac{-2\alpha+\sqrt{4\alpha^2-4}}{2}=\frac{-2\alpha-\sqrt{4\alpha^2}}{2}$

$\sqrt{4\alpha^2-4} \quad =-\sqrt{4\alpha^2-4}$

$2\sqrt{4\alpha^2-4}=0$

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$4\alpha^2 - 4 = 0$

$\alpha^2 = 1$

$\alpha = \pm 1$

13. a)

General form of linear differential equation

$\frac{dy}{dx} + py = \theta$ when P and θ can be function of x.

Only option (a) is in this form

14. b)

The differential equation id given as

$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 0$

Y=C.F+P.I

Since Q = 0, i.e PHS term is zero, so there will be No particular integral

$\therefore \quad y = C.F$

$Let \quad \frac{\partial}{\partial x} = D$

$So, \ D^2 + 2D + 1 \ x = 0$

$\therefore \quad D + 1\ ^2 = 0$

$\therefore \quad y \ ae^{-t} + bte^{-t}$

15. Given

$y\ 0 = y'\ 0 = 1$

$\frac{d^2y}{dx^2} + \frac{4dy}{dx} + 4y = 0 \quad .. \ i$

$D^2 + 4D + 4 = 0$

$D = -2, -2$

$y = C_1 e^{-2x} + C_2 xe^{-2x}$

$y\ 0 = C_1 = 1$

$y' = -2e^{-2x} + C_2 e^{-2x} + C_2\ -2\ xe^{-2x}$

$y'\ 0 = -2 + C_2 = 1$

$\Rightarrow \quad C_2 = 3$

$y = e^{-2x} + 3xe^{-2x}$

$y\ 1 = e^{-2} + 3 \times 2e^{-2}$

$= 0.947$

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16. b)

$$D^2+2D+1=0$$

$$D=-1,-1$$

$$y\ t\ =\ C_1+C_2\,t\ e^{-t}$$

$$y'\ t\ =C_2\,e^{-t}+\ C_1+C_2t\ \ -e^{-t}$$

$$y\ 0\ =y'\ 0\ =1$$

$$From\,here, C_1=1, C_2=2$$

$$\Rightarrow \qquad y\ t\ =1+2t\ e^{-t}$$

17. c)

$$\frac{dy}{1+\cos 2y} = \frac{dx}{1-\cos 2x}$$

$$\frac{dy}{2\cos^2 y} = \frac{dx}{2\sin^2 x}$$

$$\sec^2 y\,dy=\cos ec^2\,x\,dx$$

Integrating both sides, we get

Tany = -cot x + c

Tany + cot x = c

18. Given

$$D2+3D+2=0$$

$$D=-1,-2$$

$$x\ t\ =C_1\,e^{-t}+C_2\,e^{-2t}$$

$$x\ 1\ =\frac{10}{e}=C_1\,e^{-1}+C_2\,e^{-2}$$

$$C_1+C_2\,e^{-1}=10 \qquad .. \ i$$

$$C_1+C_2=20 \qquad .. \ ii$$

$$From\,here, C_1=\frac{10e-20}{e-1}; C_2=\left(\frac{10e}{e-1}\right)$$

$$x\ 2\ =\left(\frac{10e-20}{e-1}\right)e^{-2}\left(\frac{10e}{e-1}\right)e^{-4}$$

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19. a)

$$D^2+12D+36 = 0$$
$$D+6\ ^2 y = 0$$
$$D = -6,-6$$
$$y = C_1+xC_2\ e^{-6x}$$
$$y = C_1e^{-6x}+C_2\,xe^{-6x}$$
$$y\ 0 = 3$$
$$= C_1+0$$
$$\Rightarrow \quad C_1 = 3$$
$$y' = -6C_1e^{-6x}+C_2\,e^{-6x}-6C_2\,xe^{-6x}$$
$$y'\ 0 = -36$$
$$-36 = -6C_1+C_2$$
$$-36 = -18+C_2$$
$$C_2 = -18$$
$$y = 3e^{-6x}-18xe^{-6x}$$
$$y = 3-18x\ e^{-6x}$$

20. d)

$$\frac{dy}{dx} = x+y-1\ ^2 \qquad i$$

$$Let\ x+y-1=t \qquad ii$$

$$1+\frac{dy}{dx}=\frac{dt}{dx}$$

$$\frac{dy}{dx}=\frac{dt}{dx}-1 \qquad iii$$

Substituting equations (ii) and (iii)in equation (i)

$$\frac{dt}{dx}-1=t^2$$

$$\frac{dt}{t^2+1}=dx$$

Integrating both side

$$\int\frac{1}{t^2+1}dt=\int dx$$

$$\tan^{-1}t = x+c$$

$$Sin\,ce, \quad t = x+y-1$$

$$\therefore \tan^{-1}\ c+y-1 = x+c$$

$$x+y-1\ \tan\ x+c$$

$$y = 1-x+\tan\ x+c$$

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21. a)

The general solution of the differential equation

$\frac{d^2y}{dx^2}+2\frac{dy}{dx}-5y=0$

$(D^2 + 2D - 5)\, y = 0$

Auxillary equation is

$m^2+2m-5=0$

$m=\frac{-2\pm\sqrt{4+20}}{2}=\frac{-2\pm\sqrt{24}}{2}$

$=\frac{-2\pm2\sqrt{6}}{2}=-1\pm\sqrt{6}$

Solution $y = K_{1e}^{\,-1+\sqrt{6}} + K_2 e^{\,-1-\sqrt{6}\,x}$

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